Observers Who Read Their Universe into Existence: Finite Consensus, an Executable Observer, and Standard Physics

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Abstract

Observer Patch Holography (OPH) proposes closure. The universe is one timeless self-referential history whose observers recover its specification from their records and realize it, so that it reads itself into existence. The proposal becomes testable as the executable-model hypothesis: a simple, fully specified local repair and memory law on a federated twelve-port icosahedral carrier admits continuing observers whose records have the observed $3+1$-dimensional quantum-field description and a common classical-gravity limit, with matter as localized topological defects of native overlap transport whose rest energy is their excess maintenance dynamics. This paper presents OPH as one theory organized as a tower over that model. Under stated repair, algebra and transport premises, finite theorems give schedule-independent public records, Born representation and the Tsirelson bound in a finite quantum algebra, a carrier whose slow response is exactly three-dimensional, the Standard Model gauge Lie algebra with an anomaly-free generation on a supplied matter module, and a $(1+3)$ causal placement on a declared event population with proper-time ratios from record counts. Executed components include a native observer that acts on its records, one interacting quantum process of flux, charged matter and a scalar field, overlapping observers that share one complete law, and a continuum field operator on refining screens. Conditional implications give a smooth $(1+3)$ scalar description on a declared radial product and the Einstein equation from a realized null balance. A specified coherent memory law protects native exchange-history sectors and implements fermionic exchange with internal states. Complete source factors join spatial propagation, particle-changing Higgs operations, native repair and a common calibrated clock, retaining every quantum outcome and reference. The three-dimensional Weyl/contact action also has a positive-energy infinite-lattice ground at fixed spacing, with certified energy-density bounds; its grounded cube has a fully resolved 78-line local charged response and a complete native finite-clock energy measurement. The infinite ground has a rigorously non-Gaussian field correlation and a time-reflection-positive ordered Euclidean hierarchy. Under explicit bounded-contact refinement, all eight Weyl valleys form a continuum field net with positive energy and finite propagation. An exact composite CAR channel resolves the contact-force algebra. A signed-energy sum rule obstructs the stated CPT vacuum identification for the unchanged fields, ground and clock at positive limiting coupling. Particle-hole symmetry fixes a charge subtraction with its own centered ground; the finite cube then has a unique half-filled ground with hidden spin zero, a certified charged gap and faithful one-site states. Uniform spectral control then shows that the composite force escapes

every finite energy window. The entire centered bounded-coupling continuum hierarchy is the free massless eight-valley Weyl theory, so this trajectory cannot supply the required interacting massive endpoint. Regional modular generators and entropy first laws are also constructed on the original particle ground, with their support and central terms retained. A controlled massive relativistic continuation is one-dimensional and requires an explicit coupling trajectory. Physical particle identification, the interacting chiral $3+1$ continuum and the joint premises for responsive Einstein geometry remain open.

Keywords: foundations of physics; Observer Patch Holography; observer consensus; executable observer; self-reference; quantum records; gauge structure; Einstein field equation.

1 Introduction

Physics usually begins with a world and asks how observers fit into it. Observer Patch Holography (OPH) begins with observers and asks what world they can establish together. Its observers are finite. Each holds a patch of local state, reads a boundary it shares with its neighbors, keeps durable records, and repairs its state when records disagree; it accepts only repairs that preserve protected data. OPH identifies an objective fact with a record that survives every authorized comparison. Quantum probability, Lorentzian kinematics, gravitational dynamics and gauge structure enter standard physics through distinct mathematical inputs [17, 18, 19, 53, 60]; OPH asks whether they are different public readouts of one finite observer architecture. Space, time, matter and classical objects are effective descriptions of what observers can measure and jointly interpret; no observer-independent continuum is assumed beneath them. In this interpretation, spacetime geometries and quantum field theories, including any conformal limit, describe relations established by compatible observer records; they are not additional entities posited to exist before or independently of those observer relations.

The intended argument runs from specified repair operations to compatible joint records, and from those records to a common operational description of space, time and matter. A position is what the spatial readouts locate; a duration is what calibrated clocks compare; a particle is a persistent excitation with the observed transport, charges and interactions. The target is one such description with three spatial directions and local time, including its quantum fields and gravitational response. The finite constructions below establish parts of this chain; the common physical $3+1$ limit is the remaining conclusion to be demonstrated, not an input hidden in the word "observer."

OPH makes one proposal about the whole, closure: one complete history is both a timeless self-referential structure and its own internal computation, and observers inside it recover its specification and realize it (Section 2). The executable-model hypothesis is the concrete and testable form of that proposal: one specified local repair and memory process must reproduce the observed quantum fields and their common gravitational description. It names one carrier architecture, one local repair and memory law and one matter mechanism, and it requires one source process to supply records, clocks, fields, matter, energy and responsive geometry together. This paper presents OPH as that single theory, organized by its dependencies: each layer states a hypothesis, proves finite theorems on displayed premises, and passes a conclusion upward only when the executable model realizes the premise on the same process.

1.1 Scope and order of the argument

This paper follows the construction from its premises to its physical interpretation. The finite mathematical results include public normal forms, quantum record identities, the rank-three car-

rier response, gauge Lie type, an anomaly-free matter module and null tomography. The model also realizes operations: observers retain and use records, complete instruments act on fields, and specified coherent memory protects transport sectors. The common-process construction in Section 9 joins these sectors to fermionic exchange, calibrated matter dynamics, spatial propagation, particle creation and a shared clock. Its energy and modular observations use that same field algebra, rather than a second unrelated benchmark.

The Weyl/contact construction now supplies an original-action ground representation on the infinite three-dimensional lattice at fixed spacing. Its positive GNS energy, nonzero density and convergent energy-density brackets are proved without selecting a particle-number sector. On the finite cube, exact charged spectral weights and retarded memory give concrete matter-response data for the existing geometry interfaces. Native controls prepare the finite ground and read its charged characteristic function with complete output error bounds. On the infinite ground, exact CAR identities prove a nonzero connected four-point witness on an explicit time interval and nonzero interaction memory. Its positive generator constructs the complete ordered Euclidean hierarchy with time-reflection positivity along the same finite-ground subsequence. Locality estimates control the inverse response at complex frequencies separated from the real axis. A further spacing subsequence now supplies a continuum field net with all eight valleys, continuous translations, positive energy and finite propagation, under $c = v/a$ and bounded contact strength. The complete finite ground family has the same 78 addition energies; exact hidden-spin selection and a complete native finite-clock energy POVM sharpen its interpretation. The original and centered charge clocks now have explicitly different ground constructions. The subtraction required by the signed-energy symmetry constraint is fixed by the source particle-hole identity; it changes the cube's global ground from four to eight particles. Checked premises of Lieb's theorem give uniqueness and hidden spin zero, and an exact energy bound gives faithful one-site density matrices. The centered family has its own positive-energy continuum construction and a stronger quadratic occupation-defect bound. A uniform microscopic-time comparison proves that its composite force loses every finite-energy component. Spectral splitting, free-vacuum outside cyclicity and finite propagation then identify the entire centered bounded-coupling hierarchy with the free massless eight-valley Weyl field. An interacting massive endpoint therefore requires a different justified trajectory or action. The original-ground regional modular generator supplies a finite entropy first law, while its geometric cap and edge identifications remain to be established.

The distinction is between what each construction establishes and the physical interpretation it supports. A finite source program can be complete on its stated Hilbert space without furnishing a localized physical particle or a chiral continuum. The physical task is to extend and match these constructed interfaces on one refinement family. Section 12 separates that task from the finite results realized here and records the counterexamples that constrain it.

Finite theorems, complete source executions and physical interpretations answer different questions. We state each result's domain and supplied inputs where it is used; the detailed claim classes and verification methods are collected in Section 14.

"Source-executed" means that an operation is compiled and run using the declared native preparation, repair, memory and readout services, with its complete outputs and resources accounted. "Source-only" concerns the provenance of its parameters: no empirical physical input occurs anywhere in their dependency chain. These are different properties. A source-executed field may use empirically calibrated Yukawas, mixing parameters or clock units; reproducing those inputs verifies the implementation, not an independent prediction.

Definition 1.1 (Physical realization map)**.** A physical realization map assigns finite public records, event data, currents, modular generators, and scales to physical records, spacetime events, labo-

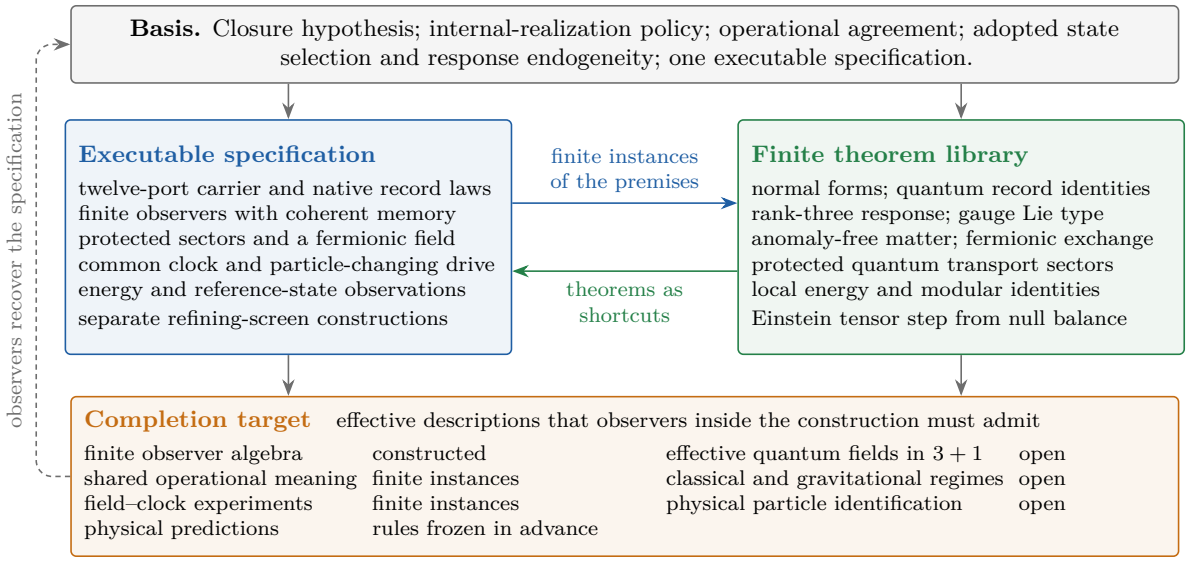


Figure 1: How the parts of OPH fit together. The basis fixes one executable specification and the premises of the finite theorem library. The running specification supplies finite instances of those premises, and a theorem replaces simulation wherever its premises are decided on the specification's finite objects. Both feed the tower of the executable-model hypothesis, whose layer status Section 12 gives. The dashed arrow is the closure loop of Section 2: observers recover the specification from their records.

ratory currents, clocks, and units, preserving the algebraic operations, overlap restrictions, causal order, normalizations, and refinement relations consumed by the conclusion in question. A resemblance of spectra, dimensions, or symmetry names is insufficient.

Order of the construction. Section 2 states the hypothesis and its additional premises. Section 3 defines the carrier and both native record laws before any calculation uses them. Public quantum records, spatial response and executable observers then provide the interfaces needed for fields and matter (Sections 4–8). Section 9 assembles the protected-sector, field, clock and energy constructions on explicitly joined carriers. The Einstein and cosmological arguments follow with their own hypotheses (Sections 10–11). Finally, Section 12 consolidates the realized results and physical obligations; Sections 13–14 give tests and reproducibility. The companion construction and component papers carry the full implementation contracts and longer proofs [66, 1, 4, 5, 6, 8, 11, 9, 7, 2].

2 Closure and the executable-model hypothesis

This section states the interpretation and acceptance criterion: closure, the internal-realization policy, the additional state-selection and response-endogeneity principles, and the agreement condition, then explains why closure is tested through an executable model.

2.1 The closure hypothesis

The universe is modeled as a timeless, closed, self-referential structure in which the description and the described system are one object. Let $\mathfrak{U}$ be a complete inhabited history, let $D(\mathfrak{U})$ be an account of its architecture recoverable through completed records within it, and let R be an admissible

realization of that account constructed inside $\mathfrak{U}$. The closure condition is

$$R(D(\mathfrak{U})) \;\cong\; \mathfrak{U}, \tag{1}$$

where $\cong$ is agreement in the observable and interventional structure: the records used to recover the description, and the responses of the realization to every admitted intervention, fit the world from which the account was recovered. Written as a fixed point of a universe-level closure operator, the condition reads $T(\mathfrak{U}) = \mathfrak{U}$. Observers infer the specification of their world, build hardware within it, and run that hardware; its execution realizes the same history, including its builders and their hardware. The identification of the construction-side and inhabited-side readings is the hypothesis. In this sense the structure reads itself into existence.

Within this hypothesis, no external computer runs the history and no additional time exists in which the whole is played. Host runtime used to check a model is part of our method of investigation, not an extra time coordinate of the universe described by that model. Repair order is a partial order on commits, an observer history is a chain of completed records, and that order is invariant under strictly increasing regradings, so affine time and proper time require their own calibration. Fixedness of the complete history permits changing states within it: the equation concerns the whole history and identifies no last instantaneous state with the universe.

Self-reference earns content only when the return condition restricts the possibilities. On the real line $x \mapsto x+1$ has no fixed point, $x \mapsto x$ fixes every point and $x \mapsto (x+2)/2$ fixes exactly one. A candidate therefore has to specify the history domain and the action of T; existence, consistency and the identification of our universe with a fixed point are questions the equation leaves open. Self-reference alone guarantees neither consistency nor mathematical completeness. Uniqueness is an additional question when it is claimed; it is not a prerequisite for a specified construction to work and match observations. Two finite results give the condition exact content on declared branches. The packet-quotient closure theorem proves that on the declared finite consensus branch the closure map is an affine idempotent whose fixed points are exactly the packets supported on consensus normal forms [6]. The universe-level record-closure equation $N = \log M_0(\mathfrak{U}_N)$, with $\mathfrak{U}_N$ the regulated observer system at capacity budget N and M_0 the correctable-code capacity of its terminal public record, is posed exactly, and the all-rung arithmetic proves that the bounded completion class defined by base agreement, positivity and the carrier bound does not identify it [8].

2.2 No outside

A self-contained realization must account internally for its operations and information access. This motivates a justification policy and two further constitutive principles; it does not logically derive those principles.

Justification policy. A premise is justified operationally when observers recover it and the specification implementing it runs and reproduces their measurements. The carrier, observer algebra, repair law and closure coordinates must meet that test. Their specification can be reconstructed from records or proposed as a design. In particular, using a twelve-port carrier does not require a theorem that every possible realization must have twelve ports. A construction choice needs no uniqueness theorem unless uniqueness is claimed. Closure neither exempts a candidate from empirical tests nor proves its antecedents jointly realizable.

State-selection principle. On the declared feasible family, reference, observer cover and positive weights, adopt the information projection minimizing $\sum_p w_p D(\rho_p \| \tau_p)$. This is premise A3,

motivated by least-informative inference but separately adopted when used as a physical selection law. The closure equation supplies no entropy functional or reference; even its identity-map specialization fixes both uniform and nonuniform states. Under A3, every minimizing local density has the union of feasible supports, and a feasible failure witness $b > 0$ gives an explicit selected-probability floor [9].

Response-endogeneity principle. The complete-response branch additionally requires each proper carrier recharting to be implemented by the connected response group, up to a unitary centralizer acting trivially on the response algebra. This is the endogeneity clause of A2, not a consequence of internal implementation alone. For example, twelve diagonal phase generators on $\mathbb{C}^{12}$ admit internal port permutations in $\mathrm{U}(1)^{12} \rtimes A_5$. These permutations act nontrivially on the abelian current algebra, whereas its inner automorphisms are trivial. This closed operation system fails the stronger endogeneity premise; it shows why that premise must be supplied. Theorem 8.1 uses it to select $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$ under the stated complete-response hypotheses.

2.3 Agreement and the meaning of "our"

The justification policy fixes the operational test. Agreement says what counts as shared. Observers assign the same operational meaning to accepted data on shared boundaries, naturally under every restriction, recharting and refinement, and a record is public when it survives every authorized comparison (Theorem 4.4). "Our universe" is the world that this public record describes. Agreement in this sense is a definition and an acceptance criterion: it requires interpretation maps fixed independently of validation outcomes and preserving the declared joint and conditional statistics, including quantum correlations. It does not require a noncontextual table of outcomes for incompatible unperformed measurements. Agreement here is compatibility under specified comparisons and interventions, not a vote among observers or assent to a convenient interpretation. Its content includes the joint probabilities and the response to changed preparations. Calling space, time and matter operational descriptions does not make these constraints optional: once established by compatible records, they are public physical facts. The equations describe those relations and do not by themselves assert an additional material substrate beyond the specified process. The theorem library calls the interpretation clause of this condition A2. The basis reference distinguishes the closure hypothesis, agreement, adopted selection/endogeneity principles and executable specification, with the library's premise statements and their dictionary [100].

2.4 An entry point without a starting point

The exposition of this paper enters the loop at the executable specification. That choice reflects where observers stand, holding records and inferring, and it makes the account concrete. Closure states the internal realization contract; the adopted principles and specification then carry their own mathematical and empirical tests. The order of exposition is a logical entry point, not an event before the universe or a privileged temporal beginning of the self-referential loop. A pair of mutually determining equations such as $x = (y + 2)/2$, $y = (x + 2)/2$ fixes $x = y = 2$ with no first value from which the other is manufactured; a calculator may approach the pair by iteration and someone may find it on a particular day, and neither belongs to the reason the answer is two.

The closure equation relates parts of one timeless object; it posits no backward signal in an external time coordinate. Observers recover the specification and later construct it in their own record order, and both stages belong to the same fixed point. In the terminating consensus theorems,

repair descent supplies a well-founded order on the admitted updates for proving termination and comparing schedules; it is neither an observer's experienced time nor a universe-level clock.

What is reconstructed and what is constructed are two roles within this proposal. The observer who infers a law and the hardware that realizes it both belong to the candidate history. Finding that specification in a laboratory or checking it by computation occurs in the investigator's record order; neither supplies a temporal origin outside the history. If several specified closures satisfy the same tests, the results warrant that observational class without a uniqueness claim.

2.5 From closure to an executable model

The executable observer gives the condition its finite meaning. A realization differs from a description because it carries the described dependencies under intervention: for a described update τ_a, realized operation F_a and fixed readout p, the one-step condition $p(F_a(s)) = \tau_a(p(s))$ extends by induction to every finite admitted sequence. A playback can match every frame of one history and fail this test as soon as a retained state is changed. The controls of Section 6 perform exactly this test on the running observer: disconnecting the detector or shuffling its bit with the same frequencies removes the feedback response, while the record-controlled execution keeps it.

Closure alone does not imply that a particular carrier, record law or initial state yields Standard Model physics. The executable-model hypothesis is the stronger claim that one simple, fully specified law does, and it is the form in which closure can fail. Its carrier architecture, refinement and composition, record instruments, memory law, scheduler, preparations and available controls are stated before testing. A failed specified law rejects that law and its tested interpretation; it reaches the wider proposal only when the rejected choice is proved necessary and exclusive. Changing the law, memory, state or decoder defines a new candidate.

2.6 Specification and theorem premises

The architecture, record instruments, memory law, scheduler, preparation services and allowed interventions form a declared specification. Observers may infer or design these ingredients; mathematics determines their consequences, and comparison determines whether the resulting process matches observed physics. Declaring a primitive fixes the model under study. Its consequences require proofs, and its physical adequacy requires tests; a specified input must remain distinguishable from a consequence derived from other inputs. The carrier, memory and control laws need not be uniquely selected by closure to be legitimate inputs. Their implementation and resource use must, however, belong to the process being studied.

Three premise families recur. A1 specifies the carrier, federation and response structure. A2 adds agreement on shared records and inner implementation of the complete response algebra. A3 adopts a least-informative state-selection principle on the stated feasible family. Closure alone proves neither the entropy minimizer nor the inner implementation assumption. Later theorems state which of these inputs they use and which additional hypotheses their conclusions require.

Quantum executions below use finite-dimensional complex Hilbert spaces, positive normalized density operators and completely positive instruments whose complete effects sum to the identity. These are explicit quantum premises, not consequences of classical normal-form uniqueness. A faithful reference is a strictly positive density operator $\rho > 0$; its modular flow is $\sigma_s(A) = \rho^{is} A \rho^{-is}$. The dimensionless parameter s requires a separate implemented calibration before it is read as duration. For fermionic fields the canonical anticommutation relations (CAR) are $\{c_i, c_j^\dagger\} = \delta_{ij} I$ and $\{c_i, c_j\} = 0$; the declared Fock representation fixes the exchange signs used later.

The physical interpretation also fixes a domain: one prepared history, a family of interventions,

or every allowed input state. These are inequivalent requirements. Complete quantum instruments must retain the reference systems and all outcomes needed for the chosen domain; a record history need not assign simultaneous values to unperformed incompatible measurements. Likewise, a stationary probability law, an instantaneous repair fixed point and the closure of an entire history are distinct. Host runtime, repair counts, modular parameters and measured duration are related only by an implemented clock calibration. The operational observer below is defined by access and record interfaces, without an additional claim about consciousness.

3 The observer and the native repair law

The model rests on two objects: an operational observer and a twelve-port carrier with its native repair law. We define their interfaces and update rules before constructing the observer that runs on them.

3.1 Patches, records and repair

Definition 3.1 (Observer patch)**.** An observer patch is a finite object with local state, an observable boundary map, durable records, readback, and a set of authorized repair moves. A patch sees a fragment of the world: its own state and the boundary data of its authorized overlaps.

Definition 3.2 (Overlap and repair)**.** For patches x_i, x_j with an authorized overlap $e = (i, j)$, both induced boundary records must agree for the pair to hold public data. A repair move changes local state while preserving protected readout. A configuration is a normal form when no authorized repair applies.

An observer in this sense is defined by its access and record interfaces. It reads, writes and retains records, and it uses retained records to choose its next operation. Nothing in the definition refers to a background space, a global clock or a pre-assigned list of particles. Those enter later as readouts that observers can share.

3.2 The twelve-port carrier

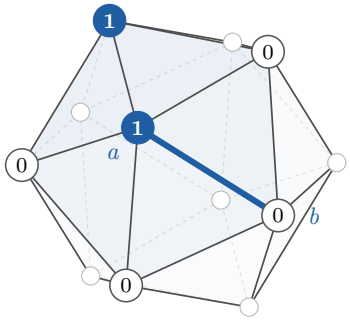
Each local carrier has twelve primitive boundary ports with oriented icosahedral incidence: thirty seams and twenty faces, five seams at each port, every seam on two oppositely oriented faces. This is finite combinatorial incidence, not an embedding of fundamental objects in an pre-existing physical space. Coordinates used to draw the carrier or compute its group representations describe that finite object; physical position and metric enter through operational reconstruction. Carriers federate through typed seam algebras with coherent triple overlaps into a nerve carrying a degree-one bridge to an oriented spherical support. The theorem library calls this carrier, federation and response layer A1. Within the closure hypothesis of Section 2 it is the part of the specification that observers recover from their records; every result below displays its dependence on it. The local carrier Gram is recovered from native current records, while cross-carrier routing in the executed federations uses a supplied cell-centre and tangent-frame convention.

Two exact features of the carrier enter later constructions. First, the proper rotation group of the icosahedral packet is the alternating group A_5, with sixty rotations acting on the twelve ports, and the real port-coefficient space decomposes without multiplicity as

$$P_{12} \cong_{A_5} \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}. \tag{2}$$

On the six-dimensional golden sector the spectral projectors have entries in $\frac{1}{20}\mathbb{Z}[\varphi]$, $\varphi = (1+\sqrt{5})/2$, each is the entrywise Galois conjugate of the other, and both triplets are irreducible real representations of real type; the projectors, characters and irreducibility certificates are machine-checked [48, 1]. Second, record keeping is integer valued: a write appends $+1$, a retraction appends -1, and readback sums atomic events per port. A conservative repair transfers one unit across a seam whenever the oriented mismatch has magnitude at least two. With $V(x) = \sum_i x_i^2$, each such repair strictly decreases V by $2(d-1)$ for mismatch $d \geq 2$, so repair terminates. The native repair law of Section 3.3 also moves load to the floor and ceiling of the endpoint mean, but at difference one it waits or swaps by its coin, so the native process keeps moving while its quadratic cost stays fixed.

3.3 The native repair law



twelve ports, thirty seams, twenty faces; a binary load of total two (filled ports)

$d = x_a - x_b, \qquad j = \frac{1}{2}\big[d - (2c-1)(|d| \bmod 2)\big]$

d	$c=0$	$c=1$	effect of the attempt
2	1	1	balance
1	1	0	swap or wait
0	0	0	no move
-1	0	-1	wait or swap
-2	-1	-1	balance

The move j carries j units from a to b, and the fair coin c reads no load. At $|d| = 2$ the attempt lowers $V = \sum_i x_i^2$; at $|d| = 1$ it keeps V fixed, so the native process keeps moving. The seam drawn has $d = 1$.

Figure 2: The twelve-port carrier and its native repair. Left: the icosahedral incidence of the twelve ports, with thirty seams and twenty faces, in an orthographic view with back edges dashed; the filled ports carry a binary load of total two, and the drawn seam (a, b) has $d = 1$. Right: the seam rule for every mismatch the total-two carrier admits.

The executable observer runs on one carrier with twelve nonnegative integer loads of total two. There are 78 such configurations, and the 66 binary configurations form an invariant shell. For an oriented seam (a, b) and a fair placement coin $c \in \{0, 1\}$ (Figure 2), the native repair law moves

$$d = x_a - x_b, \qquad j = \tfrac{1}{2}\big[d - (2c-1)(|d| \bmod 2)\big], \qquad y = x + j\,(e_b - e_a), \tag{3}$$

so load is conserved and the signed current j is the native observation. Conservation, one-step balance, a tie-break that reads no load and endpoint symmetry characterize this rule; on the total-two carrier they fix only the tie-break coin. Its quantum version is a record law, and seam locality, the classical law and the declared local number-phase and endpoint-swap covariance, a local phase symmetry of the repair channel, leave a convex family of them, each a table of five coherence multipliers [69]. The two laws used below have the same load update (3) and differ in which information they record. Write $p_e(x) = (x_a + x_b) \bmod 2$ for the parity of the seam's total

load. Law C records the pair (c, j); law E records (c, j, p_e). Their Kraus operators are

$$L^C_{c,j} = \frac{1}{\sqrt{2}} \sum_{x:\, j(x,c)=j} |y(x,c)\rangle\langle x|, \tag{4}$$

$$L^E_{c,j,p} = \frac{1}{\sqrt{2}} \sum_{x:\, j(x,c)=j,\ p_e(x)=p} |y(x,c)\rangle\langle x|. \tag{5}$$

Each complete instrument preserves total probability. E exposes an extra parity record and therefore removes coherence between the corresponding parity classes when the record is ignored. It changes no classical load or current probability. C is the convention used for the primary construction; E provides a specified alternative and control. A current-only coherent implementation gives the most coherent member C_0; adding a coin readout gives C, and adding seam parity gives E. Classically forgetting a written coin is not the coherent C_0 implementation. E and the seam-total recorder are the two idempotent members of the classified family. Under a common definite-current record interface all members share every classical current statistic. One coherent interference experiment reads the multiplier g that separates C, E and the most coherent member (Section 13), and nine source-executed measurements read every real coordinate of the table (Section 6). The quantum relaxation of the channels obtained by discarding the source record, the forgotten-source channels, is known exactly. C and E have quantum spectral gap $(5 - \sqrt{5})/60$ on the binary shell and on the full carrier, their slowest binary coherences decay by the sharp factors 5/6 per attempt for C and 7/10 for E, the binary-shell channel becomes entanglement breaking after finitely many attempts (134 attempts suffice by an explicit certificate), and on the full carrier it never does: an explicit partial transpose stays negative at every attempt count [68, 70]. The endpoint instruments also fix a local action. On binary operator space $H_{\text{native}} = \sum_e (I - \Phi_e)$, summed over seams with Φ_e the unread C or E channel, is positive and local, with an exclusion exchange term and coherence-killing coefficients set by the record law; the uniform discrete transfer is $I - H_{\text{native}}/M$ for M seams. It is a Liouville operator on operators, not a Hamiltonian on carrier states, and the executed channel strictly attenuates a coherent mode, which rules out unitary conjugation on this carrier. At fixed selected-pair sum each classical repair minimizes the pair's quadratic load cost, so $\|x\|^2/2$ falls by $(d^2 - (|d| \bmod 2))/4$ per repair [82, 68].

The gap and the arrow do not depend on the record law. Every admissible coherence table has the uniform binary state as its unique stationary state and the gap $(5 - \sqrt{5})/60$; the linear load statistics along the rank-three port vectors of Theorem 5.1 attain it, while every coherence between distinct load configurations contracts by at most $43/60 + 2\sqrt{2}/15 < 1 - (5 - \sqrt{5})/60$ per attempt, so the longest-lived datum of a carrier is a classical position-like readout and equilibrium selects no record-law coordinate [69]. On the binary shell each native repair is the fibre-resampling map of the theorem library for the uniform reference: it resamples the seam orientation uniformly within the fibre of its repaired visible datum, which minimizes relative entropy over that fibre, while a collision is resolved by deterministic descent. The library's second law and modular Clausius inequality therefore hold for every native attempt on the binary shell [8]. One such projection is idempotent; the declared uniform schedule over noncommuting seams supplies a sustained arrow and contracts every input to the uniform state at the gap rate, and passive repair never realizes the library's third map, the modular flow of a faithful reference.

4 Public records and completed-record identities

The repair mathematics establishes when local agreement reaches global consistency and when nonzero cycle holonomy leaves a residue that local repair cannot remove. Two observers who share

a boundary can disagree about what it shows. If repair stopped in different places depending on who acted first, the boundary would carry no fact of the matter. If it always stops in the same place, the final record is a public fact that no single observer owns. This section states when the second case holds, and then shows that completed records satisfy the familiar quantum identities on the declared complex algebra and state domain below.

A completed finite configuration x is a fixed point of the named repair-and-agreement map,

$$\mathcal{R}_{\mathrm{cons}}(x) = x.$$

This map is distinct from the universe-level closure operator of Section 2. Fixed-point existence, observable determination and schedule independence are separate questions, and the following results isolate them [4, 6].

4.1 Observable normal forms

Definition 4.1 (Observable quotient system)**.** An observable quotient system is a tuple $\mathfrak{S} = (Q, C, \mathcal{B}, B)$, where Q is a set of configurations, $C \subseteq Q$ is the consistent subset, $\mathcal{B}$ is a set of protected records, and $B : Q \to \mathcal{B}$ is the record map. For $b \in \mathcal{B}$, write $C_b = C \cap B^{-1}(b)$.

The fiber C_b is unrealizable, reconstructing, or ambiguous according as its cardinality is zero, one, or greater than one. This trichotomy avoids assigning a state to a record that has either no consistent extension or several observationally indistinguishable extensions.

Theorem 4.2 (Canonical observable normalizer)**.** *The following statements are equivalent:*

(i) $B|_C$ *is injective;*

(ii) there is a unique map $\mathcal{N} : Q \to C \sqcup \{\bot\}$ *that preserves* B *on its* C*-valued outputs, fixes every point of* C*, is constant on each* B*-fiber, and returns* $\bot$ *exactly on fibers with no consistent extension;*

If $B(Q) \subseteq B(C)$*, the map is a total idempotent retraction* $Q \to C$*.*

Proof. Injectivity makes every nonempty C_b a singleton, which defines $\mathcal{N}$ and forces all its properties fiber by fiber. Conversely, if $c, c' \in C$ have $B(c) = B(c')$, the fixed-point and fiber clauses give $c = \mathcal{N}(c) = \mathcal{N}(c') = c'$. □

The normalizer is defined by observable fibers, independently of any repair schedule; a repair relation implements it when the relation preserves B, reaches the consistent set and satisfies a liveness condition. Approximate versions control the normalizer where exactness fails, through an optimal two-output distance bound and summable naturality defects along refinement towers. Canonicity carries no promise of cheapness: for succinct Boolean boundary systems, deciding whether a protected record has a consistent extension is NP-complete, and deciding injectivity of the record map is coNP-complete [4].

For the canonical single-site repair, every adaptive run is eventually constant because each nonstuttering attempt strictly lowers the mismatch count; pathwise weak fairness makes the stable state a normal form, and completeness with confluence sends every weak-fair run to the same public fixed point. These results select no scheduler and provide no stochastic hitting law, rate, physical clock, or refinement theorem. Attempts obey a sharp capacity law. A scheduler can delay the first genuine repair for any prescribed number of attempts. When every reducible record sees a genuine change among the next $q + 1$ attempts, a bounded-waste upper horizon of $(q + 1)$ times

the initial mismatch count holds, and an independent-defect family attains the work-conserving case exactly at $q = 0$. The input-independent compiler produces a fixed federation with immutable input ports; genuine node repairs decrease a dependency rank, and pathwise weak fairness gives stable consensus and the correct output. Emission-order round robin finishes within its first n attempts, and the single-consumer structure bounds every accepted path by $n(n+1)/2$, with an emitted family attaining $\Theta(n^2)$. General well-formed circuits instead admit $2^n - 1$ accepted steps, attained by a fanout chain, where n counts the emitted nodes. These statements concern canonical node repairs: the weak patch-frame relation can stutter fairly forever, and no physical scheduler, clock, energy or hardware cost follows from these operation counts [4, 6].

4.2 Public records from local repair

For a prepared batch of support-closed transactions, join τ and σ in the conflict graph when $W_\tau \cap (R_\sigma \cup W_\sigma) \neq \varnothing$ or $W_\sigma \cap (R_\tau \cup W_\tau) \neq \varnothing$, and iterate to a fixed point: compute semantic support inside every aggregate, rebuild the conflict graph on the expanded read and write sets, merge every connected component, and repeat until the partition and all aggregate supports are unchanged.

Proposition 4.3 (Transactional local diamond)**.** *Let an accepted repair be an aggregate transaction with read set R_τ, write set W_τ, a read snapshot, and a snapshot-determined payload. Assume:*

(i) every read set contains the semantic dependency closure of its write set, and every affected acceptance functional is revalidated at commit;

(ii) the terminal closure above exists, and each of its connected conflict components has one coherent canonical aggregate payload;

(iii) a prepared component whose snapshot remains valid survives commits of independent components; and

(iv) accepted commits preserve the protected record and strictly decrease the integer descent functional.

Then every one-step quotient peak $t \longleftarrow s \longrightarrow u$ has a quotient state v with $t \longrightarrow v \longleftarrow u$.

Proof. Two different first steps cannot originate in one conflict component, since that component has a unique aggregate payload, so they originate in distinct components of the terminal closure. Rebuilding the graph after every aggregate support expansion makes their final read and write sets disjoint, so neither commit changes the other's read snapshot, and semantic dependency closure ensures neither changes an acceptance functional consumed by the other. Revalidation and the surviving component rule make both second commits legal, and with snapshot-determined payloads and disjoint writes the two application orders agree; protected-record preservation carries the equality to the physical quotient. □

Theorem 4.4 (Consensus normal form)**.** *Suppose the accepted repair relation satisfies the hypotheses of Proposition 4.3, so that it has the one-step diamond property, strictly decreases the quadratic descent functional V of Section 3, and is repair complete, meaning that its normal forms are exactly the consistent states. Every maximal asynchronous repair schedule from one initial state then terminates at the same consistent quotient normal form. The induced global repair map is idempotent, fixes consistent states, and preserves the protected record.*

Proof. Quadratic descent makes the accepted relation terminating. Proposition 4.3 makes it locally confluent, so Newman's lemma [41] makes it confluent. A terminating confluent relation has a unique normal form below each source, and repair completeness puts that normal form in the consistent set. Applying the normalizer a second time does nothing, and protected-record preservation holds along every accepted edge. □

Atomic commits, disjoint write sets, or termination alone do not imply the local diamond; countermodels omitting semantic dependency closure, canonical aggregation or the surviving component rule are retained with the finite receipt. This is the precise sense in which a public fact is what survives every authorized comparison: the public object is the protected quotient record, independent of the order in which observers repaired.

Repair also has a residue that it cannot remove. On a fixed graph with positive mismatch weights and fixed edge offsets, nonzero cycle holonomy forces a strictly positive minimum mismatch, and no repair of the vertex assignments eliminates it while the offsets are fixed [6]. The matter hypothesis uses this frustration as a candidate mechanism (Section 8.2).

The law runs at scale. The public run uses the level-six cell rung of the geodesic icosahedral refinement tower: 81,920 exact twelve-port carriers with one declared gluing of ports to neighboring cells, 2,580,480 seams in all, and the seam-mean retraction as repair primitive. Sixteen shuffled schedules from one seeded load terminate at the balanced class with no accepted move raising V, and all sixteen return one quotient hash equal to the component multiset fixed by the initial totals. A standalone verifier that imports no simulator code rebuilds the component multisets from the primitive arrays, replays one schedule from its seed and recomputes every hash [2]. The level-ten rung, with 20,971,520 carriers and 660,602,880 seams, terminates all sixteen schedules at the balanced class in 84 to 100 sweeps with one quotient hash fixed by the loads. Across the rungs from five to nine, one initial load per rung gives an excess-energy contraction between 0.81 and 0.83 per sweep, and the sweep counts to the balanced class grow slowly with the carrier count, from 51 to 79 at level six to 84 to 100 at level ten; these finite readings prove no all-size mixing or termination-time bound. On the sixteen level-six schedules the number of unit transfers to the balanced class lies between 924,832 and 925,708, about 1.25 times half the initial ℓ_1 deviation of the loads from their mean. The first-settling configuration retains spatial information that its public component multiset does not encode: at level ten the initial-to-terminal amplitude responses of equal-count ancestral groups of sizes 1,024, 4,096 and 65,536 are 0.90758, 0.95232 and 0.98983, with conditional schedule-noise fractions 1.241, 0.594 and 0.143 percent. Stationary continuation adds a constructive fluctuation result: for fair integer repair on a connected weighted graph with independent edge clocks of total rate one the fixed-occupancy law is reversible, centered loads have lagged covariance $\kappa e^{-|t|L_G/2}\Pi$ in the edge-clock time t, with L_G the weighted graph Laplacian, $\kappa = r(p-r)/(p(p-1))$ for r raised occupancies on p ports and Π the projection off constants, and fixed linear readouts obey a martingale central limit theorem [7].

The same records let observers recover the specification itself. At carrier scale, recovery from event logs under the declared probe and operation grammar returns the twelve ports, thirty seams, inverse pairing, automorphism and rotation groups of orders 120 and 60, mean rule, schedule, rank-three Gram and Lie split $1+3+8$. Re-instantiation preserves the invariant vector for two iterations, and tetrahedral and octahedral controls recover their distinct specifications [2]. Complete federation gluing is not identified by the retained routing logs: within the declared port-gluing interface a two-edge switch preserves every carrier degree, occupied port and source traversal while changing the carrier graph's triangle count from 14,000 to 13,990, and native executions reproduce identical bytes for both. The alternative is not asserted to belong to the canonical geometric family. Full-gluing recovery requires further observations or a justified restriction of the candidate class.

4.3 Completed records and quantum identities

Once a compare, write and verify slice has completed, its accessible events are represented in a finite-dimensional complex C^*-algebra with a positive normalized state and completely positive instruments. This algebra-state representation is an explicit input; the consensus theorem identifies which record is public and does not by itself select complex quantum theory over every alternative probability model.

Theorem 4.5 (Finite public-record identities)**.** *Let P_E be the projector of a completed public event E and ρ the normalized state of the completed record surface. The event weight and nonzero-weight update*

$$\Pr(E) = \operatorname{Tr}(\rho P_E), \qquad \rho \,|\, E = \frac{P_E \,\rho\, P_E}{\operatorname{Tr}(\rho P_E)} \tag{6}$$

define a normalized probability measure on every orthogonal event partition and a normalized positive state on the post-event corner. For self-adjoint dichotomic observables A_0, A_1 and B_0, B_1 in commuting record subalgebras, with $A_i^2 = B_j^2 = I$, the combination $S_{\mathrm{CHSH}} = \operatorname{Tr}[\rho\{A_0(B_0 + B_1) + A_1(B_0 - B_1)\}]$ obeys $|S_{\mathrm{CHSH}}| \le 2\sqrt{2}$ [22, 23]. The bound is attained exactly on the committed slot-split interface by the maximally entangled state with Pauli and quarter-rotated projections, while every state diagonal in the record basis with diagonal observables valued in $[-1, 1]$ obeys $|S_{\mathrm{CHSH}}| \le 2$.

Proof. Positivity of ρ and $0 \le P_E \le I$ give $0 \le \operatorname{Tr}(\rho P_E) \le 1$, and $\sum_k P_k = I$ normalizes every partition; for nonzero weight $P_E \rho P_E / \operatorname{Tr}(\rho P_E)$ is positive and normalized. For $\mathcal{B} = A_0(B_0 + B_1) + A_1(B_0 - B_1)$ the commuting-subalgebra calculation gives $\mathcal{B}^2 = 4I - [A_0, A_1][B_0, B_1]$ [24], so $\|\mathcal{B}\| \le 2\sqrt{2}$. The saturating witness and the diagonal bound are exact finite calculations [5]. □

At the level of weights the Born form is derived. Every valuation on the effects of the finite surface that is nonnegative, normalized and additive on coexisting effects equals $E \mapsto \operatorname{Tr}(\rho E)$ for a unique density operator ρ, in every finite dimension including two, where the projector-only Gleason theorem fails, and the derivation consumes no continuity axiom [20, 21]. The projectors alone do not select the update: an explicit swap-twisted instrument has the same effects and different updates. Kraus form with induced effect P together with invariance on the states certain of P is equivalent to being the Lüders map [27, 5]. The same surface carries the copying obstruction: if one ancilla-free isometry copies two sharp states from a common blank, preservation of inner products forces their overlap z to satisfy $z^2 = z$, so two distinct alternatives copied by one device are orthogonal [25, 26]. Finite webs do not by themselves supply the valuation hypothesis: a source-produced complex-tomographically-complete effect and instrument web with full coexistent-effect additivity is the missing step [5].

5 Three spatial directions

The carrier incidence fixes the dimension of the carrier's own position readback: the slow band is the three-dimensional irreducible representation of the icosahedral group, so the number three is a property of the committed carrier.

Theorem 5.1 (Intrinsic three-dimensional carrier completion)**.** *Let A be the twelve-port adjacency, let $M = I - (5I - A)/60$ be the uniform scalar repair mean, and take the complete equally weighted centered probe census. The normalized infinite-response kernel converges to the intrinsic Gram $G = 4P_{\mathrm{slow}}$, of rank three. The twelve integer port loads have antipodal-odd quotient $\mathbb{Z}^{12}/\{x : x_p = x_{-p}\} \cong \mathbb{Z}^6$, and the thirty-seam boundary has image exactly the even-sum sublattice D_6, with*

one residual parity coset. Under the pullback of the response-selected Gram metric, both integer modules embed densely in one abstract continuous three-dimensional Euclidean carrier. The sixty proper carrier maps act faithfully and isometrically on the rank-three range and extend to that completion, and cumulative records act by exact internal translations.

The mechanism is spectral. The icosahedral adjacency has eigenvalues 5, $\sqrt{5}$, -1 and $-\sqrt{5}$ with multiplicities 1, 3, 5 and 3, so the repair mean M has eigenvalue $1-(5-\sqrt{5})/60$ on a three-dimensional eigenspace and smaller eigenvalues on everything else nonconstant. Repeated local averaging forgets every nonuniform pattern except a three-dimensional one, and that surviving pattern is the triplet of (2) carried by the port position vectors. The same number $(5-\sqrt{5})/60$ reappears as the quantum spectral gap of the native repair channel (Section 3.3).

The completion is taken after slow-response normalization and the infinite-step limit; at finite step count the kernel has rank eleven on the centered space and rank six on the signed antipodal sector, so the order of limits matters. The metric is the response-selected Gram pullback, distinct from the usual six-dimensional lattice metric on D_6. Thirteen finite covariance and isometry table identities are discharged in Lean with `native_decide`, so those subreceipts trust the native compiler in addition to the kernel and are not kernel-only proofs. Displacement is a record difference and cumulative records translate the completion, so position is the carrier's own readback of its accumulated records. In the executable observer the same rank-three readout attaches to all joint histories of the interacting process, and a binary controller extension executes displacements spanning three independent directions [68]. Reading the completion as physical position consumes the realization map (M1) of Section 12.3 [1, 7]. The rank-three readout belongs to one carrier's response. When carriers join into the refining spherical screen of Section 7.1, the continuum form of their repair history has spectral dimension two, the dimension of the screen, and bounded-energy families lose every within-carrier component, this band included, at the present full-port energy scaling; a three-dimensional bulk reading of the screen requires observation maps that the native energy does not supply.

6 The executable observer

Operational observers are continuing native processes that prepare, read, retain and act on records. The specification is executable, and its components run with retained receipts [68]. The companion construction paper states each component, its declared services and its certificates [66]; the following summary gives the parts used in this paper.

An observer that remembers and acts. The native observer executes two repair events with a finite tape. It records the first current, issues a forecast from that record, chooses the next seam from the record alone, records the result and halts. Exhaustive checks retain all 264 binary-source paths, and a 20,000-trial seeded execution reproduces the exact predictions. Record-controlled feedback gives final port-zero occupation 35/264, against 1/6 when the selector is disconnected or shuffled with the same action frequencies, so under the declared selector rule the response is caused by the retained record and not by the action statistics; the comparison checks the wiring of that rule. Feedback preserves the committed classical record while it can reduce coherence between different record values. Complete records of these short histories recover the source optimally, with entanglement fidelity $D^{-2}\sum_h w_h \operatorname{rank}(P_h)^2$ for the maximally mixed input, where D is the number of carrier configurations, h runs over the complete admitted histories, w_h is the product of their seam-selection and coin probabilities and P_h projects onto the inputs compatible with h; it equals

1301/2028 for the full-carrier two-event C observer and 3329/6084 for E under the same current-sign selector [71].

An electromagnetic detector driven by the closure coordinate. A second observer evaluates the coupling $\alpha(P)$ from the screen grain P and sets $g(P) = \sqrt{4\pi\alpha(P)}$. A prepared path qubit reads the first current through a calibrated loop exposure χ; the detector bit $y = \pm 1$ has probability $[1 + y\sin(g(P)j_1\chi)]/2$ and alone chooses the second seam, giving mean transferred charge $\mathbb{E}[g(P)j_2] = (5g(P)/264)\sin(g(P)\chi)$. All 528 weighted histories are retained and a 200,000-trial execution agrees with the analytic response. The closure coordinate thereby enters an actual response magnitude; the charge assignment, probe, exposure and controller are declared apparatus.

Fields measured by the observer. A scalar detector prepares a signed field intervention from the first native current, reads the field with a binary Weyl instrument and feeds the bit back, with inherited continuum bounds transferred to the complete history law. A source-compiled two-site field apparatus builds its entangled preparation from blank registers, evolves the field and reads it with a quantum pointer, all from emitted twelve-port control words; at the executed sixteen-level cutoff an exact rational interval places the ideal plus probability in $[0.51495, 0.51789]$, excluding a fair detector.

One interacting process. A common execution puts quantum electric flux, a charged fermion pair and two neutral scalar sites on one work register. Native source words implement their noncommuting evolution, the matter and scalar pointer instruments, and feedback controlled by a committed matter bit; the final scalar bit controls the next native repair. A smooth extension retains four quantum reads on the same disturbed state, compiles the C and E record laws around it, and actively protects a logical qubit with record-controlled recovery. Its complete conditional native/work/reference outputs are retained, and Section 6.1 gives its certified responses.

Two observers, one law. Observer A reads ports $(0, 1)$ and observer B ports $(0, 5)$ of one carrier, so the two regions overlap at port zero. Each receives its initial bits and every incident local current, including zero currents, and its clock ticks whenever an attempted repair meets its region while remote repairs continue on the same source. After the first incident a native feedback repair is chosen from the reconstructed occupation of port zero. Both observers decode one continuing source, so their agreement at the shared port instantiates the overlap-agreement equation with identity transport; rechartings between observer charts are not exercised. A direct execution through the native endpoint branches and an independent recursion in a derived effective transfer representation agree exactly on all 5,768 output words, including timeouts and terminal populations; at every positive-weight prefix each observer's decoded fields obey native continuity, and the two values at the shared port agree [77]. The quantum extension checks the native remote resolvent on all 4,356 matrix units with coherent preparations and untouched references. A compiled version replaces its ideal preparation, intervention and pointer by pulses emitted from the source compiler: exact rational propagation of all 512 code addresses keeps every one of the 1,656 clock, timeout and abort outputs within 10^{-11} of the ideal history, at a cost of at most 51 pulses and five native repair attempts [78].

Local clocks. A local event clock counts the attempted repairs that meet its region, or the actual occupation changes at a port, and its stopped instrument sums the unresolved wait and keeps every finite timeout. Two such clocks on joined carriers share one continuing history with

an observer-chosen field. C and E have the same completed clock law there and differ in the field: after three native attempts they predict field-plus probabilities 35/54 and 14/27 under the stated probe. Without feedback, over each two-tick cycle of one clock the other ticks twice on average with variance 16, so equal mean rates do not imply equal readings, and converting between the clocks through the joint event instrument preserves that whole count distribution [79]. A separate active clock lets each incident event, including a zero-current one, trigger one compiled modular step whose generator is the native three-state transfer, and closed histories derive a faithful thermal reference for that generator. Four movement-bit covariances on the complete attempt tape recover its two nontrivial frequencies and predict its coherent quadratures with no frequency fit [80]. Attempt counts, local event counts and modular parameters are distinct parameters, and none is identified with seconds.

Identifying the repair law. Nine source-executed scalar measurements identify all nine real coordinates of the record-law family and with them the coarse definite-current instrument on all 78 states, with arbitrary spectator and reference coherences. A collision test separates an admissible law that agrees with E on the whole binary shell. The calibrated instrument predicts complete two-event feedback histories through the same observer runtime, including quantum references and checkpoint continuation, and an explicit bound carries the probability uncertainty to their joint clock, current and interference records [81]. The probes are synthetic, so this identifies an instrument within the declared family and selects no law for nature.

From one carrier to a screen. One carrier suffices for the local results above; joined carriers test larger regions, distant interventions and refinement. Federations of 20, 80 and 320 twelve-port carriers preserve complete source records, untouched references and cross-carrier coherence probes, and local dependency bounds control the whole local quantum and record output under local feedback. On the actual 15,360- and 245,760-port screens, matched radius-nine initial restrictions give complete root-observer output distance below 3.82×10^{-4} at auxiliary per-seam parameter $1/4$, and below 6.656×10^{-9} after four local incident attempts with no auxiliary time coordinate [82]. On those matched screens a logical qubit with an untouched reference passes through two native incident ticks and local record-controlled feedback while the surrounding source keeps density $1/3$. A degree-two operator calculation retains every quantum output and timeout, and each full-screen output lies within 7.779×10^{-10} of the common 222-port computation, uniformly over references. C and E give identical classical records in this protocol and different reference coherences [83].

These executions are finite, and their apparatus, services, schedules, field action and preparations are declared; their time parameters are attempt and event counts. They discharge finite execution, continuation and comparison between observers under those contracts. A separate specified source clock shares a common calibration with complete field and repair operations. Autonomous unbounded observers, interacting spatial refinement, a physically validated clock/energy normalization and responsive gravity remain unestablished (Section 12).

6.1 Measurement inside one interacting process

OPH places measurement at the transition from a private repairable record to a protected public normal form: a record is public when it has survived every authorized comparison and no repair can change it, and event projectors, weights and updates act on the completed algebra-state surface of Section 4.3. The executable observer puts this account inside one interacting quantum process [66].

Quantum electric flux, a charged fermion pair and two neutral scalar sites share one work register, with operator Gauss law and source-executed instruments. The sharp protocol retains 1,056 native/quantum histories. At $d = 2$ and 512 temporal steps its certified feedback contrast is $[-0.033297300667839, -0.030025660061775]$; for zero native preparation current the actual infinite-coordinate half-line instrument gives $[-0.024045670143, -0.024043450955]$, an infinite-occupation enclosure that includes the free vacuum and unbounded interacting evolution of the same declared action [72].

Outcome accuracy does not certify the output state. For that zero-current preparation, the complete $d = 2$ sharp output has trace distance about 0.455 from the infinite history; every output in the $d = 256$ occupation code has error above 0.122. For the vacuum control the best code error scales as $d^{-1/4}$ [73]. A separate smooth Ramsey instrument preserves finite number moments. A four-read continuation carries its disturbed state through two smooth reads, feedback and a later gauge probe, retaining sixteen work branches. Stagewise moment and projected-Duhamel estimates give complete $O(1/d)$ error at fixed protocol with $n(d) = d^2$ temporal steps; executed $d = 2, 4$ occupation bounds remain vacuous. This covers the later probe, not post-sharp clocks or physical clock calibration [74].

Compiled C/E native repairs surround that same work history, acting on arbitrary joint inputs of the native, work and reference registers. They retain full conditional quantum outputs, aborts and padding. If r_2 is the second smooth bit, $K_r^{j(h)}$ the work branch, and $e(0) = (0, 5)$, $e(1) = (0, 1)$, the joint Kraus map is

$$M_{h,r,s} = (L_s^{e(r_2)} L_h^{(0,1)}) \otimes K_r^{j(h)}.$$

Causal summation proves completeness. A retained native phase witness has C contrast $(1 + q)/4$, $q = \Pr(r_2 = 0 \mid j = 0)$, and E contrast zero; source execution gives approximately 0.3492 and 0.3514 for C at $d = 2, 4$ [75].

Active memory uses the existing four-configuration qubit V and compiled record-controlled recovery $R_h L_h V = \sqrt{p_h} V$. Correcting after each native event, with final seam $(2, 3)$ for $r_2 = 0$ and $(0, 1)$ otherwise, preserves the logical/reference marginal even for entangled work inputs. For independent logical/work preparations, the matched passive complete plus/minus output distance is $1 - \bar{q}$, $\bar{q} = (q_{-1} + 2q_0 + q_1)/4$, $q_j = \Pr(r_2 = 0 \mid j)$, versus one with correction; a terminal phase read consumes that coherence [76]. This protects an encoded qubit under specified controls, not arbitrary lost native coherence.

7 Fields, clocks and causal order

Fields and clocks have complementary constructions: one repair history supplies a positive continuum generator on the screen, modular flow of observer algebras supplies candidate local clocks, and authenticated provenance supplies causal order.

7.1 A continuum operator from one repair history

The executable carriers join into the geodesic spherical screens of the refinement tower; the level-l screen has $V_l = 240 \cdot 4^l$ binary ports. The uniform canonical preparation fixes the total load at $q_l = V_l/3$, every attempt selects one of the native seams uniformly and independently, and the fair native repair acts on it. Observers read cell averages of the occupations against the normalized area measure of the declared flat polyhedral comparison surface, and $\chi_l = q_l(V_l - q_l)/[V_l(V_l - 1)]$ normalizes their fluctuations. No continuum field, metric or Hamiltonian enters the construction.

Proposition 7.1 (Native continuum operator)**.** *Along one common subsequence of levels:*

(C1) the native cell Green operators converge in operator norm to a positive, compact, injective operator G on the mean-zero observation space, and $A = G^{-1}$ is a densely defined positive self-adjoint operator;

(C2) with the constant mode restored, the form of A is a conservative, regular, strongly local Dirichlet form with domain H^1, its energy lies between positive multiples of the gradient energy, its positive eigenvalues obey $n/49920 \le \lambda_n(A) \le 3n/2$, and its heat-trace spectral dimension is two;

(C3) for mean-zero test functions f the actual density readouts

$$Y_l(\tau; f) = \frac{\sum_p f_l(p)\,[n_{\lfloor 1260\cdot 16^l\tau\rfloor}(p) - 1/3]}{\sqrt{V_l \chi_l}}$$

at native attempt count $\lfloor 1260 \cdot 16^l\tau\rfloor$ converge jointly at every finite set of values of τ to a stationary reflection-positive Gaussian family with covariance $\langle f, e^{-|\tau-u|A} g\rangle$, whose one-particle semigroup is $e^{-\tau A}$;

(C4) the admitted coherent continuation converges strongly to $\cos(t\sqrt{A})$, and with $\bar{A} = A \oplus 0$ and d the comparison distance,

$$\|\mathbf{1}_F e^{-\tau\bar{A}} \mathbf{1}_E\| \le e^{-d(E,F)^2/(2\tau)},$$

so the cosine propagator vanishes between E and F while $|t| < \sqrt{2}\, d(E, F)$.

Proof sketch. For conditional averaging P_k on the level-k cells, an elementary Poincaré inequality on the actual subdivision gives $\|(I-P_k)u\|^2 \le 624\cdot 4^{-k}\langle u, B_l u\rangle$ for every cell vector u and $l \ge k$, where B_l is the scaled native cell energy. It makes the Green family uniformly compact and upgrades weak covariance convergence to norm convergence. Convergence of the inverses gives Mosco convergence of the forms [94]. The full-port energy minimizer of a bounded-energy family has a within-carrier part of order 2^{-l}, so normal contractions of port values pass to the limit and the closed form is Markovian and conservative [95]. Tent functions bound the eigenvalues from above and the Poincaré inequality from below, which fixes the heat-trace growth t^{-1}. For (C3) each unread repair acts as a fair identity or swap; the conditional profile overlaps concentrate uniformly in attempt count, and a conditional finite-population central limit theorem applies [96]. Clause (C4) is the Gaussian heat bound together with its equivalence to finite propagation of the cosine family [97]. The complete arguments and constants are given in [84, 85]. □

One positive operator therefore plays three roles for observers of one history: it is the inverse covariance of the static field, the generator of the field's decay in rescaled attempt count, and the squared frequency of the admitted waves, whose locality gives them a finite speed. Two consecutive long observation blocks of the same history also synthesize a continuous classical Gaussian wave readout with covariance $G\cos((t-u)\sqrt{A})$ in $C([0,T], H^{-2}(S^2))$, with observation count taken to infinity before spatial refinement. A repair whose execution is biased by ϵ toward a disclosed coordinate profile f has continuum mean derivative $2(I - e^{-\tau A})f$ in its executed-repair clock, and uniform likelihood bounds keep its response strictly positive at $\epsilon = 10^{-6}$, with timeouts and coin-service costs included [80]. With the same normalization the response and the fluctuations obey a fluctuation-dissipation relation: for $C_{g,f}(\tau) = \langle g, e^{-\tau A} f\rangle$ the mean response of g to the bias toward f is $2[C_{g,f}(0) - C_{g,f}(\tau)]$, and its initial rate $2\langle g, Af\rangle$ is set by the operator that generates the decay. The parameters τ and t are rescaled attempt count and an admitted coherent-wave parameter, the wave processor and its coherent services are declared, and the distance is that of the comparison surface; none of them carries a laboratory calibration.

The proposition also fixes what the screen is not. Its spectral dimension is that of the two-dimensional screen. The eigenvalue count excludes a unitary identification of A, up to scale and a bounded mass shift, with a second-order elliptic operator on a smooth compact three-manifold, and a boundary-trace test excludes a boundedly invertible energy identification with the static boundary action of a smooth compact three-dimensional Dirichlet bulk, whose Steklov eigenvalues grow like $\sqrt{n}$ [98]; this test detects that A is a second-order generator, where a boundary trace would need a first-order one. A bulk reading of the screen therefore needs a radial observation algebra that the native energy does not supply. The same min-max comparison fixes the angular spectrum of any sky field read as a bounded congruent image of the native long-average field: a precision whose eigenvalues grow like $\ell^{2+\theta}$ is form-congruent with A only for $\theta = 0$, so such a field has the scale-invariant spectrum with $\ell(\ell+1)C_\ell$ bounded above and below, and the conditional tilt branch of Section 13 needs an ingredient beyond this covariance. A specified random-length return reference has modular generator $\arccos K$, with K the symmetric native configuration transfer, and its modular phase magnitudes and cosine amplitudes match those of the waves. The full configuration reference approaches the normalized trace at fixed modular parameter, so it identifies no quantum vacuum; its entropy deficit is at most $2025/(64V)$ for V ports and every return cap. On a complete base-face patch the stationary reduced entropy grows in proportion to the port count, while the return reference's mutual information per crossing seam vanishes, so neither carries a nonzero universal coefficient times the patch's boundary-seam count; a reading of the whole screen as the boundary of a three-dimensional region is not excluded, and that region map is a premise [86].

A declared spectral filter of A, applied to whitened consecutive long-block records of the same history, nevertheless synthesizes a radial product field over the screen: on a declared radial interval with Dirichlet ends its covariance is $(-\partial_r^2 + \bar{A})^{-1}$, with spectral dimension three counted from the declared interval. The radial direction carries no native information beyond the spectrum of A, and the same filter attaches an interval to any two-dimensional generator. The chosen hyperbolic product wave action has coefficient density $C = \mu\,\mathrm{diag}(-1, a)$, where μ is the native measure and a the measurable spatial tensor, whose angular part complete native records recover and whose radial coefficient is one. The identity

$$g = \sqrt{-\det C}\, C^{-1} = \mu\sqrt{\det a}\,\mathrm{diag}(-1, a^{-1})$$

gives signature $(-,+,+,+)$ almost everywhere [66]. The signature is that of the chosen action; the derived content is the measurable spatial tensor a.

Proposition 7.2 (Smooth $(1+3)$ effective descriptions)**.** *Smoothing the angular variables of μ and a, with the radial Dirichlet boundary retained, converges in the norm-resolvent sense with spectral gap $(\pi/L)^2$. Consequently, for every fixed finite admissible menu of observer readouts on bounded wave times (thermal field covariances, responses, ordered Weyl and Ramsey histories, detector and reference outputs) and every $\varepsilon > 0$, one smooth static $(1+3)$-dimensional Lorentzian metric carries a free scalar field with the standard canonical and KMS normalization whose predictions for the whole menu agree within ε with those of the same free field built on the native measure and tensor, plus the declared source and clock budgets.*

Proof sketch. The native density relative to smooth sphere volume is positive and globally Lipschitz. The unitary $U_\varepsilon f = \sqrt{\rho_\varepsilon/\rho}\, f$ between the changing and fixed L^2 spaces converges on the form domain; Rellich compactness, recovery sequences and the dual energy liminf give transported Mosco convergence, which compact resolvents and self-adjointness upgrade to norm-resolvent convergence. Thermal covariance and sine response then converge in operator norm, momentum covariance on the form domain, and every fixed finite history with them [66, 94]. □

Both sides of the comparison are free fields assembled on the supplied wave processor, whose per-mode steps are compiled from the spectrum of the product operator and triggered by native incidents, with a supplied Gibbs preparation; the theorem removes the roughness of the native coefficients, not the declared radial product, clock or state. Passive native repair has no wave sector of its own [66].

The observers therefore have the effective description that standard physics uses, a Lorentzian field theory in $1 + 3$ dimensions, as a theorem about their own readouts. Its scope is stated with it. The exact native measure, the radial pushforward of flat face area, is not C^1 across the original face edges, so the proposition asserts no exact smooth native metric and no computable smoothing rate. The carrier's own records do supply a radius: read through the recovered rank-three frame, the axial current counter converges from every binary initial state to three-dimensional Brownian motion, and its norm has the radial generator $\frac{2}{99}(\partial_r^2 + \frac{2}{r}\partial_r)$ with angular term $\frac{2}{99}r^{-2}\Delta_{S^2}$, without added radial sites. That radius is the radius of a retained record; it is not identified with physical transport, and it does not select the radial architecture of the field: complete adaptive histories with angular instruments, native clock marks and feedback are identical across distinct radial extensions [66]. The radial architecture and the active observer protocol therefore remain declared. The metric is static and does not respond to stress. A quartic projection on the same product has a source-prepared interacting Gibbs state, an active native event clock and an unrestricted-occupation limit within its two modes, and the same smoothing preserves its finite histories; it is no spatially refining interacting continuum [66]. Uniqueness of the limit along the whole sequence, an exact smooth native metric, responsive geometry and a spatially refining interacting field are unproved.

7.2 A causal precursor from records

The oriented conformal spherical support carries the classical isomorphisms

$$\mathrm{Conf}^+(S^2) \;\cong\; \mathrm{PSL}(2,\mathbb{C}) \;\cong\; \mathrm{SO}^+(3,1), \qquad H^3 \;\cong\; \mathrm{SO}^+(3,1)/\mathrm{SO}(3), \tag{7}$$

which identify the three-dimensional space of future unit timelike directions [47]. Authenticated records separately supply events and their order. Every quotient-visible register version carries the identifier of its last semantic writer, each commit certifies the versions its acceptance depends on, and the informational order is the transitive closure of authenticated read-from parenthood, an authenticated analogue of happened-before [37]. As an abstract finite poset it satisfies the order-theoretic causal-set axioms, and every finite causal set is the generated order of some authenticated log [1].

Theorem 7.3 (Source-derived finite (1+3) causal precursor)**.** *Let $\mathcal{L}$ be an authenticated finite semantic-event log with generated strict order $\prec_{\mathcal{L}}$ and canonical source height $h_{\mathcal{L}}$, zero on roots and one plus the maximum parent height otherwise. Then $h_{\mathcal{L}}$ strictly increases along $\prec_{\mathcal{L}}$ and equals the maximum length of an authenticated parent chain ending at each event. Let V_{src} be the rank-three source Gram quotient with positive form g_{src}, and put $W_{\mathrm{src}} = \mathbb{R} \oplus V_{\mathrm{src}}$ with $Q(t,x) = t^2 - g_{\mathrm{src}}(x,x)$.*

(F1) W_{src} has dimension $1 + 3 = 4$ and Q has one positive and three negative directions;

(F2) source-unit directions, S^2 and future-null rays are exactly equivalent;

(F3) a placement $F(e) = (\tau h_{\mathcal{L}}(e), x(e))$ sends every generated precedence into the future cone whenever each parent edge obeys $g_{\mathrm{src}}(x(f) - x(e), x(f) - x(e)) \le \tau^2 (h_{\mathcal{L}}(f) - h_{\mathcal{L}}(e))^2$;

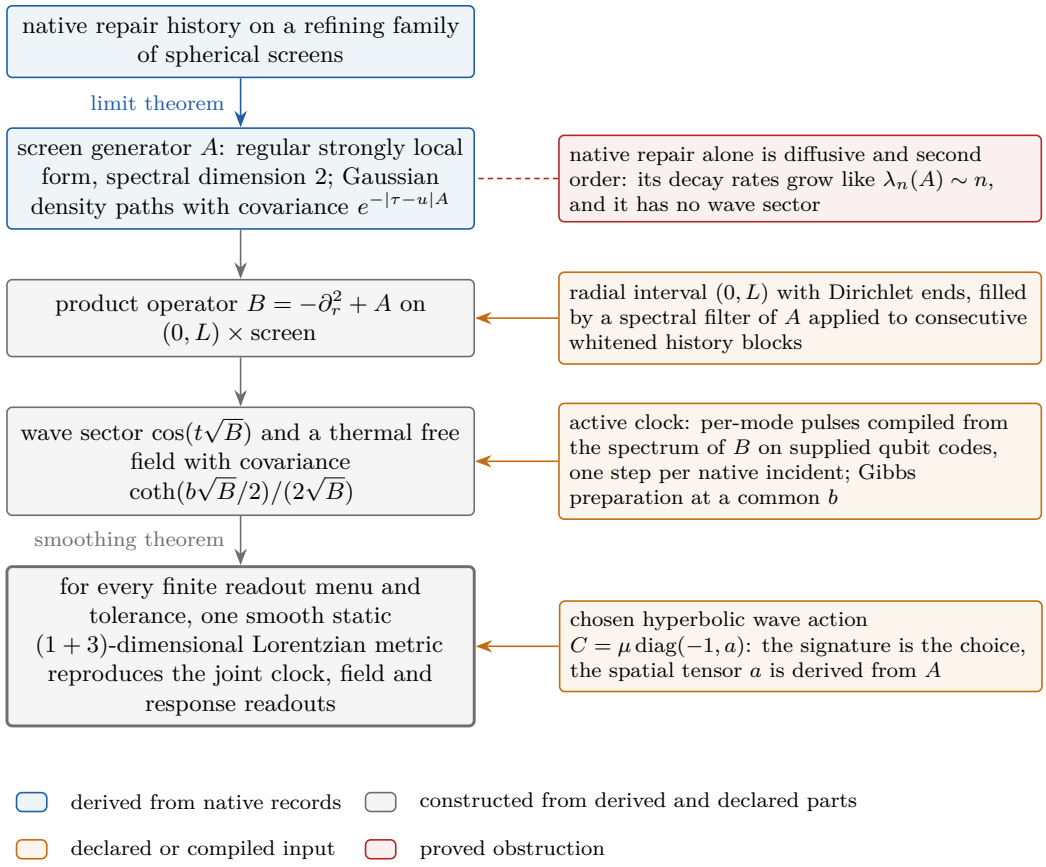


Figure 3: What the effective $(1+3)$ scalar description derives from native records and what it declares. Blue: derived from one native repair history by a limit theorem. Grey: constructed from derived and declared parts. Orange: declared or compiled inputs. Red: the proved obstruction that rules out a wave sector from passive native repair. The smoothing theorem is an existence result for fixed finite readout menus.

(F4) every finite log admits an injective placement of this form that sends generated ancestry into the future cone and makes distinct same-height events spacelike; and

(F5) for a placement obeying the edge condition of (F3), if the spatial readback separates each equal-height layer and every unrelated pair with $h_{\mathcal{L}}(e) < h_{\mathcal{L}}(f)$ is spacelike, then $e \preceq_{\mathcal{L}} f$ holds exactly when $F(f) - F(e)$ is future causal, so F is an exact order embedding preserving every causal interval.

Proof. Dimension is additive and positivity of g_{src} gives the inertia; induction over parents bounds every ending chain by the height and constructs an attaining chain; the edge inequality and transitivity of cone and ancestry give (F3); an enumeration along one axis with $\tau = n+1$, n the number of events, gives (F4); and the two geometric checks exclude every unsupported cone comparison in (F5). The finite statements are formalized [1, 9]. □

Read informally, the theorem combines two things observers have. Who read whose write is a partial order, and the longest chain of reads ending at an event gives it a height. The rank-three carrier gives three spatial directions and, with a real axis, a light cone. An event placement

that respects both turns the provenance order into a causal order in a four-dimensional Lorentzian carrier.

The first two clauses construct the carrier and signature from an independent real axis and the rank-three source carrier, without assuming a rank-four chart; source height enters only through placements, and the scaled height is no operational clock. A physical manifold requires a source-causal continuum certificate: one refinement family supplying physical event identification, source-selected order and cone embeddings, refinement maps, a calibrated count density, agreement across independent dimension and manifoldlikeness tests against Poisson-sprinkled controls, and convergence to a distinguishing time-oriented Lorentzian geometry [28, 29, 30].

Proposition 7.4 (Composition of the continuum certificate)**.** *For a source-causal continuum certificate, causal order fixes the conformal Lorentzian geometry of the admitted distinguishing limit and the calibrated count measure fixes its conformal scale [35, 36]; the local model is* $(1+3)$*, so a limit satisfying the dimension and topology clauses of the certificate is an effective four-dimensional time-oriented Lorentzian event manifold. If on the same family a causal-set d'Alembertian or scalar-curvature estimator converges as an independent curvature diagnostic [33, 34], an explicit tensor-curvature reconstruction converges to the Einstein tensor or the small-ball null-balance route identifies the limiting geometry tensor with* G_{ab}*, and stress and Ward–Bianchi data converge with calibrated coupling, then the smooth limit is* $G_{ab} + \Lambda g_{ab} = 8\pi G\, T_{ab}$*.*

A source-record population realizes the flat case.

Theorem 7.5 (A source-record causal and count limit)**.** *In the rank-three conservative repair carrier, let* $\varphi = (1+\sqrt{5})/2$*,* $L = 2/\sqrt{\varphi+2}$ *and* $q = F_n$ *the* n*-th Fibonacci number,* $n \geq 4$*. The population* $S_q = L\{b\varphi - \lfloor b\varphi \rfloor : 0 \leq b < q\}^3 \subset [0,L]^3$ *consists of* q^3 *distinct conservative source-record images. Retain it at each layer and let every next-layer event read all preceding-layer sites within* $a_q = L/\sqrt{q}$*, with model duration* $\Delta_q = a_q/c$ *for a declared speed* c*. As* $q \to \infty$ *the order converges away from null separation to the speed-*c *causal order, and with* $\rho_q = q^3/(L^3\Delta_q)$*, for fixed interior timelike diamonds,*

$$\frac{|I_q|}{\rho_q} \longrightarrow \mathrm{Vol}(I), \qquad \frac{2C_q}{|I_q|(|I_q|-1)} \longrightarrow \frac{1}{10},$$

where C_q *counts strict causally ordered pairs; for vertical duration* T*,* $\mathrm{Vol}(I) = \pi c^3 T^4/24$*.*

Corollary 7.6 (Proper-time ratios from record counts)**.** *If* I_q, J_q *approach interior timelike diamonds of proper durations* $\tau_I, \tau_J > 0$*, the observer readout* $R_q(I;J) = (|I_q|/|J_q|)^{1/4} \to \tau_I/\tau_J$*. Within positive volume-only readouts additive on collinear inertial intervals, the fourth root is unique up to units.*

At finite regulator the population is produced by the carriers themselves: each holds its record as port loads, reads its position through the rank-three response and decides its neighbors from that readback, and the precedence generated from their authenticated reads equals the layered order through $q = 89$, that is 704,969 carriers reading one another over ten rounds. On the largest interior diamond the inverted Myrheim–Meyer dimension reads 2.83, 4.38, 3.55, 3.84 and 3.92 at $q = 13, 21, 34, 55, 89$ [31, 32]; the outermost central diamonds, which the cube clips and which are therefore not interior readings, read 4.15, 3.61, 4.10, 4.09 and 4.07, and the readings at $q = 5$ and $q = 8$ lie below the regime of the theorem. The ordering fraction of the central diamond reads 0.088, 0.138, 0.092, 0.093 and 0.094 against the proved flat value 1/10, and the count clock reads 1.71, 2.09, 1.91 and 2.05 at $q = 13, 34, 55, 89$ against the model-time ratio 2. Covariance requirements

select the cone: a closed convex pointed displacement cone preserved by the irreducible icosahedral rotations and by boosts along one axis is the future or past cone of $t^2 - \|x\|^2$. The read law and model clock are declared, and the ordinal height of a history with self-waits is no proper-time instrument [9].

The executable carrier has its own finite causal bound. For a fixed seam schedule and shared local coins, dependence crosses at most one seam per ordered repair event, and retained signal witnesses attain the bound; this is a speed in seams per native event. Remote repairs can carry a difference to a receiver before its first local tick, so the bound is not one seam per receiver tick [87, 68]. Complete endpoint records give a forward-cone placement in the registered response geometry, and a four-event native execution has a Lean instance proving that implication and the failure of its converse [1].

8 Gauge structure and matter

The gauge construction treats gauge structure as the transport of overlap data. The matter hypothesis identifies stable massive matter with localized defects of that transport.

8.1 The forced gauge Lie type and one generation

Let A be the icosahedral adjacency operator and J the antipodal involution on the twelve ports. Exact incidence gives

$$10J = A^3 - 4A^2 - 5A + 10I, \tag{8}$$

and the real port module splits as $P_{12} \cong_{A_5} \mathbf{1} \oplus \mathbf{5} \oplus \mathbf{3} \oplus \mathbf{3}'$: the even coordinates carry $\mathbf{1} \oplus \mathbf{5}$ and the odd coordinates the rank-three frame channel and its Galois-conjugate companion.

The theorem below uses two premises distinguished in Section 2.2. The carrier layer of the specification supplies a complete reversible response: an injective response derivative $D : P_{12} \to \mathfrak{u}(H)$ whose image is commutator-closed with a positive trace pairing. The separately adopted response-endogeneity principle requires every proper carrier recharting to be implemented by the connected response group up to its centralizer; the library calls this a clause of A2.

Theorem 8.1 (Gauge Lie-type forcing)**.** *Let $D : P_{12} \to \mathfrak{u}(H)$ be the complete reversible response derivative of the carrier, and put $\mathfrak{g} = D(P_{12})$; let G_D^0 be the connected subgroup of the unitary group of H generated by* $\exp \mathfrak{g}$*. Suppose every proper carrier action $a \in A_5$ has a projective implementer satisfying*

$$\mathrm{Ad}_{[U_a]}\, D(v) = D(a \cdot v), \qquad [U_a] = [g_a c_a],$$

where $g_a \in G_D^0$ and c_a centralizes $\mathfrak{g}$ pointwise. Then

$$\mathfrak{g} \cong \mathfrak{u}(1) \oplus \mathfrak{su}(3) \oplus \mathfrak{su}(2).$$

The center is the unique trivial line. Up to exchanging $\mathbf{3}$ and $\mathbf{3}'$ by the outer automorphism of A_5, the $\mathfrak{su}(2)$ ideal carries $\mathbf{3}$ and the $\mathfrak{su}(3)$ ideal carries $\mathbf{3}' \oplus \mathbf{5}$.

Proof. Injectivity of D gives $\dim \mathfrak{g} = 12$, and the positive trace pairing makes the commutator-closed image a compact Lie algebra. For every $a \in A_5$ the centralizer factor acts trivially on $\mathfrak{g}$, so the action is inner, and covariance through the injective map D gives a homomorphism from A_5 to $\mathrm{Int}(\mathfrak{g})$. The transitive proper-carrier action fixes only the uniform port line, hence $\dim \mathfrak{g}^{A_5} = 1$. Compactness gives $\mathfrak{g} = \mathfrak{z} \oplus [\mathfrak{g}, \mathfrak{g}]$, and inner automorphisms fix the center pointwise, so $\dim \mathfrak{z} \le 1$. If $\mathfrak{z} = 0$, the classification of compact simple Lie algebras forces $\mathfrak{su}(2)^4$ [58]; inner automorphisms

preserve each ideal, and on each three-dimensional ideal A_5 acts trivially or faithfully, so its fixed space has dimension three or zero, since a finite rotation group fixing an axis is cyclic. The fixed-space dimension of $\mathfrak{g}$ would be a multiple of three, a contradiction. Hence $\dim \mathfrak{z} = 1$, the semisimple part has dimension eleven, and the classification leaves only $\mathfrak{su}(3) \oplus \mathfrak{su}(2)$. The three-dimensional ideal cannot carry the trivial action, so it carries $\mathbf{3}$ or $\mathbf{3}'$, and the eight-dimensional ideal carries the complementary triplet together with $\mathbf{5}$. □

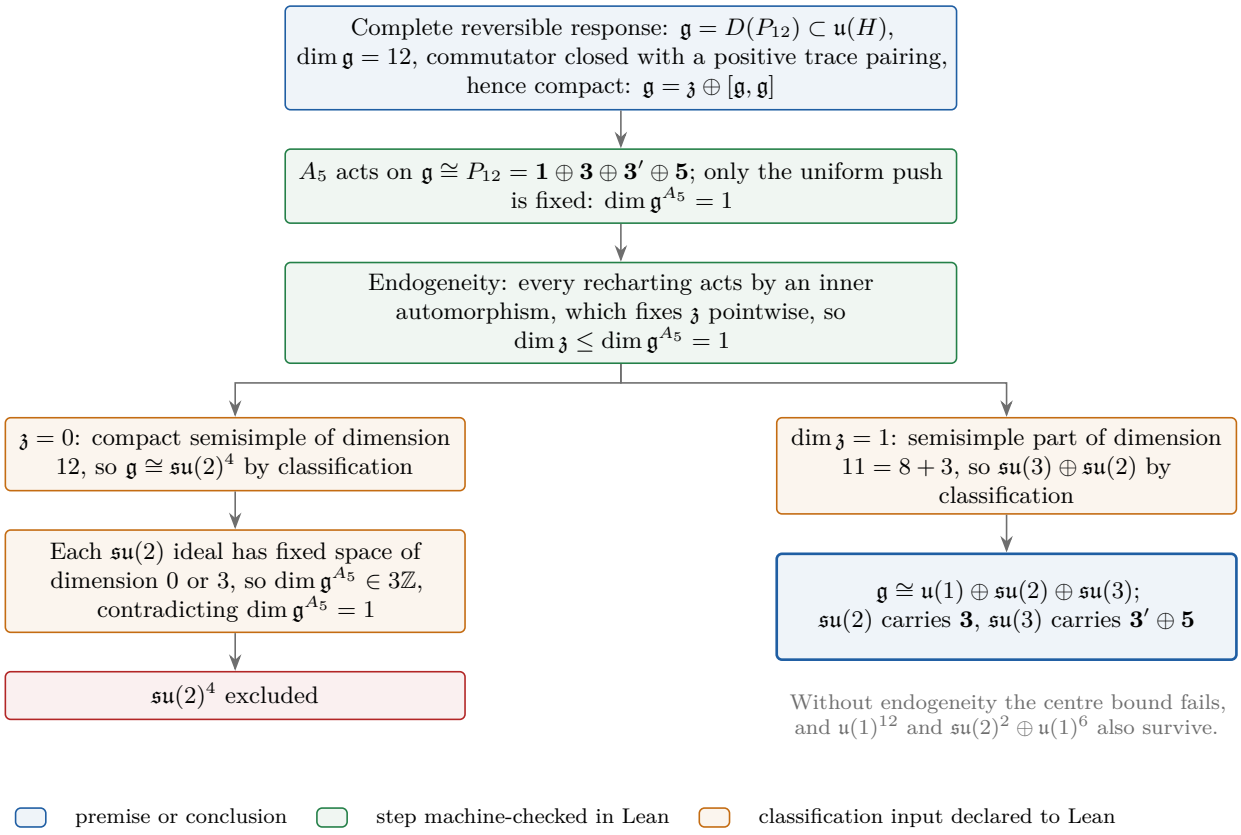


Figure 4: Steps of the gauge Lie-type theorem. Green steps are machine-checked in the Lean library; orange steps use the classification of compact simple Lie algebras and the fixed-space argument against $\mathfrak{su}(2)^4$, which enter Lean as declared hypotheses. The port count fixes the dimension twelve by declaration; the content of the theorem is the exclusion of $\mathfrak{su}(2)^4$ and a one-dimensional centre.

Figure 4 shows the steps and which of them Lean checks. The argument has a short intuitive form. Twelve ports give twelve independent ways to push the carrier, and complete response turns them into a twelve-dimensional compact Lie algebra. Internal implementation makes every recharting act by an inner automorphism, and inner automorphisms leave the center pointwise fixed, so the center lies in the directions the carrier symmetry fixes. Only the uniform push is fixed. The center is therefore at most one-dimensional, the remaining eleven dimensions must be semisimple, and $11 = 8 + 3$ is the only compact split.

No ambient compact group or candidate Lie algebra occurs among the premises. The port count twelve equals the dimension of the forced algebra, so the carrier fixes that dimension by declaration; the content of the theorem is the exclusion of the only other twelve-dimensional compact type

$\mathfrak{su}(2)^4$ through compactness and the one-dimensional fixed space, together with the forcing of the unique center. The native record laws cannot supply D: every native seam expectation annihilates a nonzero operator, so no composition of fixed native repairs is invertible, and D belongs to the reversible response layer of the carrier specification. The reconstruction of D from ordered reversible histories is consumed as a premise, and laboratory currents consume the realization map (M3) of Section 12.3.

The forced algebra has an explicit finite realization. Choose from each antipodal axis one of the certificate's exact coordinate vectors $u_i \in \mathbb{Q}(\sqrt{5})^3$, each with $\|u_i\|^2 = 2+\varphi$ for $\varphi = (1+\sqrt{5})/2$, and put $U = [u_1 \ \cdots \ u_6]$. For a real port field f with antipodal involution J, write

$$b_i = \frac{f_i + f_{Ji}}{2}, \qquad d_i = \frac{f_i - f_{Ji}}{2}, \qquad c = \frac{1}{6}\sum_i b_i, \qquad b^0 = b - c\mathbf{1},$$

so that the even coordinates b carry $\mathbf{1} \oplus \mathbf{5}$ and the odd coordinates d carry the rank-three frame channel detected by U and its Galois-conjugate companion detected by $\sigma(U)$, where σ is the field automorphism $\sqrt{5} \mapsto -\sqrt{5}$ applied entrywise. Let

$$\Phi_0(b^0) = \sum_{i=1}^{6} b_i^0 u_i u_i^{\mathsf{T}}, \qquad \widehat{x}\, y = x \times y.$$

The conditional matrix witness uses the space

$$H = \mathbb{C}^3_E \oplus \mathbb{C}^3_W,$$

two copies of $\mathbb{C}^3$ carrying the exact rotations R_g and their Galois conjugates $\sigma(R_g)$ defined below, and four nonzero rational coefficients $\lambda_1, \lambda_5, \lambda_3, \lambda_{3'}$, one on each source band. These are explicit branch premises. The two three-dimensional response blocks match the two rank-three odd channels; they are not asserted to be a derived physical Hilbert space. Write v_p for the signed coordinate vector at port p. For $g \in A_5$, let $R_g \in \mathrm{SO}(3)$ be the exact rotation satisfying $R_g v_p = v_{g(p)}$ on the oriented port frame, and define

$$\Pi(g) = \mathrm{diag}(R_g, \sigma(R_g)).$$

The target A_5 action on $\mathfrak{u}(H)$ is conjugation by $\Pi(g)$.

Theorem 8.2 (Conditional four-band finite port-response algebra)**.** *On the declared twelve-port carrier and supplied response representation, define*

$$\begin{aligned} K(f) = \mathrm{diag}\big(&\lambda_3 \widehat{Ud} + i\{\lambda_1 c I_3 + \lambda_5 \Phi_0(b^0)\}, \\ &\lambda_{3'} \widehat{\sigma(U) d}\big) \in \mathfrak{u}(H). \end{aligned} \tag{9}$$

Then K is injective, its image is commutator-closed, and

$$K(g\cdot f) = \Pi(g) K(f) \Pi(g)^*.$$

The induced A_5 action on $\operatorname{im} K$ is inner. Let $K^{-1} : \operatorname{im} K \to P_{12}$ denote the inverse of K onto its image. With the bracket pulled back from that image,

$$[x, y]_K = K^{-1}\big([K(x), K(y)]_{\mathrm{mat}}\big),$$

one has

$$(P_{12}, [\,\cdot\,,\cdot\,]_K) \cong \mathfrak{u}(3) \oplus \mathfrak{so}(3) \cong \mathfrak{u}(1) \oplus \mathfrak{su}(3) \oplus \mathfrak{su}(2). \tag{10}$$

Its center is the uniform one-dimensional port line, its derived algebra has dimension eleven, and the five-dimensional A_5 band is noncentral. The map has a positive-definite invariant Hilbert–Schmidt pullback.

Proof. Evaluating (8) on the four adjacency eigenspaces gives the antipode relation, and the source protocol solves the common farthest-shell filter to give the signed response $-J$. The constant coordinate maps to $i\mathbb{R}I_3$; the six rank-one axis matrices span the symmetric 3×3 matrices with $\sum_i u_i u_i^{\mathsf{T}} = (5+\sqrt{5})I_3$, so their sum-zero coefficients map isomorphically onto the traceless symmetric matrices under Φ_0; U is an isomorphism on the **3** band and annihilates $\mathbf{3}'$, with $\sigma(U)$ complementary. Hence K has rank 12 and image exactly $\mathfrak{u}(3) \oplus \mathfrak{so}(3)$. Equivariance, positive definiteness of the invariant pairing, and the statement that $\Pi(g)$ is the exponential of an element of $\operatorname{im} K$ are exact calculations in $\mathbb{Q}(\sqrt{5})$ over the full sixty-element domain, part of the proof by the exhaustive-computation rule of Table 2 [1]. □

Boundary 8.3. The A_5-module decomposition alone does not determine a Lie bracket: the same module also carries the zero bracket, whose trivial response-generated group fails the A2 endogeneity clause. The independent inputs to Theorem 8.2 are the carrier incidence and orientation, the supplied response representation, and four nonzero band coefficients. The block architecture and coefficients are declared. No measured particle, mass, or coupling value enters the exact verification. The theorem constructs the pulled-back bracket for this response map; it does not select K from source histories or prove a uniqueness claim over all A_5-equivariant brackets or finite carriers. The abstract Lie type is forced independently by Theorem 8.1.

Theorem 8.4 (Finite matter image and exhaustive exterior-menu selection)**.** *Let $V = \mathbb{C}^3_{-1/3} \oplus \mathbb{C}^2_{1/2}$ be an explicitly supplied trace-balanced matter representation of the forced Lie algebra, with $\widetilde{G} = \mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ acting by $(g_3, g_2, z) \cdot (v_3, v_2) = (z^{-2} g_3 v_3, z^3 g_2 v_2)$. Its exterior module branches as*

$$\Lambda^2 V \oplus \Lambda^4 V = Q \oplus u^c \oplus e^c \oplus d^c \oplus L \tag{11}$$

with $Q = (3,2)_{1/6}$, $u^c = (\bar{3},1)_{-2/3}$, $e^c = (1,1)_1$, $d^c = (\bar{3},1)_{1/3}$ and $L = (1,2)_{-1/2}$: fifteen states, all perturbative gauge and mixed gravitational anomalies vanishing, and an even number of weak doublets. The common kernel of the action is $\mathbb{Z}_6$, so the maximal faithful image is

$$S(U(3) \times U(2)) \cong \frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}. \tag{12}$$

Among all $2^{10} = 1{,}024$ subsets of the ten nontrivial irreducible summands of $\Lambda^\bullet V$, imposing nonemptiness, chirality and vanishing of the mixed gravitational–$\mathrm{U}(1)$, $\mathrm{SU}(3)^2\mathrm{U}(1)$, $\mathrm{SU}(2)^2\mathrm{U}(1)$ and $\mathrm{U}(1)^3$ traces leaves exactly two rank-fifteen masks, exchanged by charge conjugation.

Proof. The exterior-sum identity gives the five components and charges, and the anomaly sums are the standard ones [62]: for example $6(\frac{1}{6})^3 + 3(-\frac{2}{3})^3 + 1 + 3(\frac{1}{3})^3 + 2(-\frac{1}{2})^3 = 0$, and with three color copies of Q against one L the Witten parity is even [64]. The element $(e^{2\pi i/3} I_3, -I_2, e^{i\pi/3})$ acts trivially and generates a cyclic group of order six, and center-action enumeration excludes a larger kernel. The scan evaluates the four traces on all 1,024 masks; its only survivors are the two exterior-parity masks. □

The branching is the familiar $\mathbf{10} \oplus \overline{\mathbf{5}}$ of SU(5) [61], and the quotient is a possible global form of the Standard Model group [63]. One corollary concerns a channel the framework does not contain: the forced adjoint $(8,1,0) \oplus (1,3,0) \oplus (1,1,0)$ contains no mixed $(3,2,-5/6)$ generator, so the minimal simple grand-unified exchange channel is absent from the twelve-dimensional current algebra. Separately, Doplicher–Roberts reconstruction supplies a compact sector group from realized bosonic edge sectors [59, 60]; its equality with the finite response group is a separate commuting-square premise. Physical matter identification, family multiplicity, Yukawa couplings and masses consume (M3) and (M5) [11]; Section 8.2 states the matter mechanism and its status.

8.2 Matter as localized defects

The gauge theorems determine internal types and allowed currents on their stated module. A particle interpretation also needs a persistent excitation carrying those observables. The executable-model hypothesis proposes that a stable massive particle is a localized topological defect of native overlap transport: a bounded core whose surrounding records approach the reference regime, while its boundary transport carries a holonomy that admitted local repairs cannot erase. The obstruction is to trivializing the transport relative to the reference; it gives observers no license to disagree about a measured event. The repair process keeps maintaining the core, and the hypothesis identifies rest energy with that excess maintenance dynamics, read in the same physical clock and energy dictionary as every other energy and with one common signal speed. Moving the core transports its charge, and creation, annihilation, fusion and decay must follow from the same law. Unstable species would be metastable or internal defect excitations with derived decay channels.

Two constructions distinguish the topological invariant from its physical localization. Section 9 builds protected quantum transport sectors and their colored fermionic exchange. The older fixed-graph residual theorem supplies a complementary obstruction: on a fixed graph with positive mismatch weights and fixed edge offsets, nonzero cycle holonomy forces a strictly positive minimum mismatch, which no repair of the vertex assignments removes while the offsets are fixed [6]. The theorem supplies no update of the offsets, no localized moving core and no identification of the residual mismatch with excitation energy above a reference ground state; each is a further map.

The simplest local configuration candidate fails. Consider double occupation of one port in the native total-load-two C/E process. The twelve double-load classes feed one closed irreducible binary class of 66 states, so a scalar configuration label preserved by every admitted branch is constant on the 78-state domain. Under independent uniform choice of the thirty seams the double load survives n attempts with probability $(5/6)^n$, a mean lifetime of six seam attempts, and both complete C/E instruments contract the projector onto the collision subspace by exactly $5/6$ for arbitrary quantum inputs and untouched references, with no feed from the binary shell. Binary transport continues at zero drop of the quadratic potential, with nonzero current probability $5/33$ per attempt in the uniform stationary law, so repair activity is distinct from potential decrease. A proposed edge disagreement cost takes the values 8 and 10 with native transitions in both directions, so it is no monotone maintenance energy. Removing the five seams at one port creates closed binary classes of sizes 11 and 55, a sector made by changing the law [68]. These results reject this configuration defect and the tested automatic cost-to-energy identifications. They exclude no quantum, history or federated defect sector, no other admissible repair law, and not the defect hypothesis.

Several fixed mass prescriptions fail as well: all four earlier charged-fermion prescriptions miss the charged-lepton ratio windows, the declared candidate gives ratios $1 : 1.19 : 1.50$ against the measured $1 : 206.8 : 3477$, and the whole affine class of the native feature is bounded by 543.55 [68]. A specified coherent ordering memory does, however, give protected native exchange-history sectors, and its colored extension implements the CAR exchange of independent internal states. Source-executed field factors and a common calibrated clock provide a finite joint test of repair, propagation and particle creation. These results do not localize the protected charge in a physical defect or derive its measured mass from maintenance. That requires the same sector and admitted law to supply persistence, charge, spin, interactions and rest/kinetic energy in a common physical dictionary. An exhaustive invariant and localization no-go over every sector and admitted refinement of a frozen defect class stops that mechanism; incompatible required charge, spin, statistics or current structures exclude the realization they cover.

Matter operators and gravitational response

The finite response has an explicit conditional matter and interaction readout. A declared spin lift and trace redistribution act on $\Lambda^2\mathbb{C}^5 \oplus \Lambda^4\mathbb{C}^5$. Charge and Casimir projectors recover the five internal multiplets of one Standard Model generation. With an independently declared physical Weyl algebra, one scalar doublet admits the three charged-fermion interaction contractions. The optional neutrino interaction is dimension five. These exact current identities select no Yukawa coefficients, Higgs vacuum or observed masses. Identifying both family factors with the same unbroken irreducible source triplet instead forces scalar intertwiners and degenerate family masses.

The conditional Lorentzian scalar action also admits a common local tetrad dictionary for scalar, spinor and Maxwell principal symbols. A finite CAR construction shares its ideal spatial operator and incident-count clock. Completing a separate tetrahedral cochain control removes edge-cycle zero modes introduced by truncation and preserves its scalar action block. These statements keep their distinct global domains, boundary conditions and reference states; they establish no complete physical chiral field net.

For coefficient density $C = \mu\,\mathrm{diag}(-1, 1, a)$, the induced screen area is $dA = \mu\, d^2x$, while proper spatial volume is $dV = \mu N\, dr\, d^2x$, with $N^2 = \mu\sqrt{\det a}$. Counting stationarity consequently requires a further identification with fixed-volume geometric entropy stationarity. The static radial-product metric has vanishing radial null Einstein contractions. Its allowed scalar excitation can have positive relative null stress, so a fixed background, with the reference and other sources fixed, cannot satisfy both Einstein balances. A responsive geometry must supply the corresponding area focusing. This is a scoped restriction of that field/metric construction, independent of the validity of the conditional Einstein theorem.

The exact finite tensors and complete observation controls are given in `model/observer_particles/`. Their representation, field-operator and geometric witnesses instantiate finite hypotheses on their stated branches. They do not remove the separate source-selection, physical-quantity or controlled-refinement hypotheses.

9 One process for protected matter, fields and clocks

The preceding sections supply the carrier and record laws, complete quantum instruments, operational observers and the internal matter representation. This section combines them. Each change of carrier is explicit: first an ordering memory over the native two-load process, then independent internal states at occupied ports, then the full spatial CAR algebra required by propagation and particle creation. The final experiments use one implemented clock and retain the quantum reference and record outputs. This sequence is the constructive core of the finite model. The source-bound programs, complete outputs and independent checks are collected in the current construction archive [67]; these additions are distinct from the older release cited for the original observer model.

Domain	Realized connection	Scope of the connection
Native load and ordering memory	C/E branch maps preserve the memory charge; colored transport has the CAR exchange sign	A protected history sector; localization is a further physical construction
Internal matter module	Exact basis and current maps connect the fifteen-state module to the specified spin and family factors	A common representation; family number and Yukawas remain declared inputs
Twelve-site fermion field	The same pairing action, spatial source factors, full occupation space and calibrated clock	A one-dimensional finite field, retaining its additional lattice modes
Selected two-site factors	Complete repair, field, clock, energy and reference observations on the same sixteen-state field algebra	A direct source experiment with boundary hops omitted
Earlier screen and radial scalar	A screen limit and conditional scalar $(1+3)$ description on the stated radial product	A distinct domain; no fermionic refinement map to it is asserted

9.1 From repair topology to a field experiment.

A specified repair process admits protected quantum history sectors, fermionic exchange and an executable coupling to field and clock observables. Its ingredients are specified observer, scheduler, repair and memory laws; they need not be uniquely selected to define a mathematical model. What must be established is that the resulting operations share a carrier, retain complete quantum outputs and reproduce the intended observables. The constructions below establish several such joins, including a joint source experiment. They do not identify a complete physical theory.

9.2 Protected transport sectors.

On the native 66-state binary two-load configuration complex, an exact noncommutative reduction of the face and disjoint-move relations gives $\pi_1 = \langle g \mid g^2 \rangle = \mathbb{Z}/2$. The surviving loop has odd ordered-cover parity. This is topology of the declared transport and deformation domain, not topology of a localized field defect. The complete active branch $*$-algebra on either the unextended 78-state source or its binary shell is the full matrix algebra. A Hermitian source charge commuting with every branch is therefore scalar. This excludes that charge class, not every invariant subspace, expectation-conserved observable or controlled lifetime.

A coherent ordering bit changes the algebra. If a native branch maps x to $f_h(x)$, let $p_h(x)$ be the parity of the transported order and define

$$K_h = \frac{1}{\sqrt{2}} \sum_{x \in D_h} |f_h(x)\rangle\langle x| \otimes X^{p_h(x)}, \qquad Q = I_{66} \otimes X.$$

Every branch commutes with Q, so its two sharp sectors survive arbitrary finite record-adaptive schedules. An actual odd exchange loop returns the source and applies X to the bit: the memory participates in transport. Sector preservation does not assert preservation of every coherence between the sectors. For the all-seam C or E family, effect separation, a connected movement tree and the odd loop generate exactly $M_{66}(\mathbb{C}) \oplus M_{66}(\mathbb{C})$, with commutant $\mathrm{span}\{I, Q\}$, although the classical ordered cover is connected. The states $I_{66}/66 \otimes |\pm\rangle\langle\pm|$ have identical source and complete classical-record marginals; their distinction is quantum memory. Computational-basis reset or dephasing destroys it. Protection thus depends on the specified memory law preserving these sectors.

Literal source words implement this lift. Complete execution covers all 132 input columns on seam $(0,5)$, and exact word audits cover all 1024 addresses of the 90 seam/current feedback words. Native outcomes, padding and source-service failures remain in the accounting. The result is a protected sector of the enlarged process; a spatially localized source realization of its charge remains an additional construction.

This same carrier also separates operator action from scalar repair cost. Writing $P_- = (I-X)/2$ and $M_h = \sum_{x\in D_h} |f_h(x)\rangle\langle x|$,

$$A_h = \pi \sum_{x\in D_h} p_h(x)|f_h(x)\rangle\langle f_h(x)| \otimes P_-, \qquad K_h = e^{-iA_h}(M_h \otimes I)/\sqrt{2}.$$

Two closed histories with identical multisets of complete branch labels and identical compiled resource totals induce respectively I and X. An additive scalar cost of those labels cannot determine this response; the displayed operator action does. Its phase is not physical energy per unit time. Nor is the reduced memory autonomous: its one-step flip probability is $26/495$, while the two-step Z multiplier differs from the squared one-step multiplier by $398/245025$. The continuing source must be retained in any proposed Hamiltonian or memory description.

9.3 Internal states and fermionic exchange.

Give each occupied port an independent internal space V. The map

$$|\{x<y\}; a, b\rangle \longmapsto c^\dagger_{x,a} c^\dagger_{y,b}|0\rangle$$

fixes the CAR convention. Passing the occupied spectator reverses the internal slots and crosses one creation operator, hence

$$K_h^V = \frac{1}{\sqrt{2}} \sum_{x\in D_h} |f_h(x)\rangle\langle x| \otimes (-\mathrm{SWAP})^{p_h(x)}.$$

The complete effects remain the original domain projectors divided by two. Independent CAR calculations check every branch for $\dim V = 2, 3$; the crossing argument applies to every finite V. The odd loop acts as $-$SWAP, giving -1 on two equal internal states. Global Fock parity on this two-particle sector is $+1$ and cannot replace that exchange.

The ordering memory implements the same operation: for $J = -\mathrm{SWAP}$,

$$E_V\psi = (|0\rangle\psi + |1\rangle J\psi)/\sqrt{2}, \qquad (X\otimes I)E_V = E_V J.$$

Every native branch, and every common $U \otimes U$, intertwines through E_V. The negative exchange character acts on $\mathrm{Sym}^2 V$; the positive one acts on $\Lambda^2 V$. Both are compatible with identical fermions carrying independent internal states.

The finite algebra constrains but does not fix internal encodings. An untouched fiber of dimension m enlarges the repair commutant to $M_m \oplus M_m$. A separate collective-label encoding that puts all fifteen states of $\Lambda^2\mathbb{C}^5 \oplus \overline{\mathbb{C}^5}$ in the negative sector requires $m \geq 15$ and has an explicit inclusion at equality. This is not a universal memory bound for the two-particle construction. No element of the supplied $S(U(3)\times U(2))$ gauge group acts as $-I$ on the whole faithful fifteen-state module: its nonidentity involutions have signature $(7, 8)$. The smaller weak-parity encoding is consequently an internal-grading control, not an all-fermion exchange assignment.

An exact basis permutation joins that exterior module and all twelve currents to the retained chiral action. Specified two-component and three-family factors give $\dim V = 90$; neither family

count nor physical Lorentz spin is derived by this basis change. On twelve ports the hard-core two-particle dimension is 534,600, whereas the complete 1080-mode two-particle space has dimension 582,660. The remaining 48,060 states are same-port collisions and must be included when propagation or interactions can reach them.

9.4 One calibrated action and its observables.

The Weyl–Higgs action uses explicit charged Yukawa, Higgs-background and CKM inputs with recorded matching conventions. Recovering them from its response verifies implementation; it is not an independent mass prediction. The creation-pairing matrix $\Delta = \epsilon_{\text{spin}} \otimes \overline{K}$ is antisymmetric on 90 modes. At rest,

$$B_0 = \begin{pmatrix} 0 & \Delta \\ \Delta^\dagger & 0 \end{pmatrix}, \qquad H = E_{\text{vac}} + \sum_{a=1}^{90} E_a \gamma_a^\dagger \gamma_a.$$

The canonical chart retains 84 positive charged directions, six zero-energy neutrino annihilators and every particle–hole partner. Choosing the annihilator half of the zero block specifies the static reference; it does not supply a positive-energy continuation at nonzero momentum. This minimal action has massless neutrinos; a separate neutrino mass extension is not silently included.

Copying this internal system to twelve ports gives 1080 CAR modes. The static hard-core two-quasiparticle sector supports the colored construction, whereas a bare two-particle sector need not survive pairing. The compiled static convention subtracts $11E_{\text{vac}}$ from the replicated vacuum scalar $12E_{\text{vac}}$; the remaining common E_{vac} does not affect fixed-sector Ramsey gaps. Every extension across particle-number sectors must specify the same reference convention explicitly.

Complete source-controlled Ramsey observations recover an energy difference between equal-charge modes, with an opposite-charge partner. A separate neutral-ancilla phase probe reads the same single-particle charge, retaining all seven charge values and all pointer outcomes. Neither measures an absolute rest energy. The common electromagnetic phase cancels on the neutral pair and is not an electromagnetic-response signal. These executions retain all 128 padded internal addresses, native records and quantum output factors.

Currents are transformed with the Nambu chart, including their anomalous blocks. The neutral Higgs background has a nine-dimensional unbroken stabilizer, color and electromagnetism; three complementary gauge directions are broken. General weak operations cannot therefore be represented solely by $U \otimes U$ within fixed quasiparticle number. A time-dependent Higgs background requires the complete CAR domain.

9.5 Spatial dynamics, clock and particle creation.

A Hamiltonian cycle of the actual twelve-port seam graph supplies a finite spatial ring. With specified spacing and $c = 1$ units, its bare action is

$$h = -D \otimes \sigma_z \otimes I_{45}, \qquad D_{r,r+1} = -i/(2a), \quad D_{r+1,r} = i/(2a),$$

with the same local pairing matrix. The static symbol satisfies $E(k)^2 = \sin^2(k)/a^2 + E(0)^2$. All twelve momenta, all 2160 Nambu directions and both kinetic zeros remain. This is a one-dimensional witness, not a chiral $3+1$-dimensional limit. Its 1080 CAR modes retain every occupation sector, including collisions. The pairing graph splits into closed internal components of sizes one, two and six, so the largest spatial component has 72 modes. Complete four-state Fock source gates and fermionic routing determine their full Gaussian unitary maps without allocating

arbitrary 2^{1080}-component vectors. A source-prepared localized excess and its bond current have complete occupation readouts, with purifiers retained. The executed finite symmetric product is specified as such; its error relative to the summed Hamiltonian is separate from its source-gate error.

Two source-prepared clock hands, each with faithful eigenvalues $1/6, 5/6$ and a retained purifier, share the reference energy $\Omega_{\mathrm{ref}} = 1/4\,\mathrm{GeV}$. Both hands and every field factor use $\lambda_j = 1 + j/4$ for the actual native current. Complementary readouts infer $\tau_{\mathrm{eff}} = \sum_h \lambda_{j_h} \Delta\tau$ on a declared unaliased interval. The inverse receives complete probabilities and error bounds, not the programmed duration or a species mass. Rational enclosures and finite-shot bounds distinguish ensemble inference from a single clock reading. The energy-unit calibration and $\hbar = h/(2\pi)$ set seconds; repair counts and host runtime do not set this clock.

The particle-changing drive follows the actual common Higgs field $H = (0, ve^{i\phi})$ with fixed gauge background. Its creation phases have opposite signs in the up and down/charged-lepton channels, according to their conjugate-Higgs tensors. In a compatible static chart it comprises 42 neutral two-mode pairs and six massless spectators per site, with

$$H_{E,\phi} = E\cos\phi\,\mathrm{diag}(-1,0,0,1) + E\sin\phi\big(|00\rangle\langle 11| + |11\rangle\langle 00|\big).$$

Literal source gates retain each complete four-state space and all nonadjacent CAR signs. Across twelve sites this gives 504 pairs and 72 spectator modes. An externally driven quadratic field is not a quantized interacting gauge–Higgs theory.

Repair itself extends to every occupation sector. Binary endpoint occupancies follow the original C/E rule; nonbinary endpoints are held. A coherent extension includes that identity block in the existing zero-current Kraus operator. The binary and nonbinary output ranges are orthogonal, so the complete partial-injection effects sum to identity. On states with even occupancy at both endpoints the unread channel is exactly the identity, including coherences and references. The extension is unital before arbitrary record-conditioned controls; its faithful tracial reference has trivial modular flow. General completeness is algebraic. Literal source dilations verify the bounded collision and spectator-crossing cases, not a full 1080-mode dilation.

A separately flagged hold is retained as a control. It measures a vacuum/pair distinction that the coherent hold preserves. If A is the probability of no created pair on the repaired sites, a drive–repair–inverse echo returns $A^2 + (1-A)^2$ for the flagged policy and one for the coherent policy on this even-endpoint domain. For the stated drive at $\tau = 0.002\,\mathrm{GeV}^{-1}$, $j = 0$ and $\phi = \pi/2$, the flagged return is 0.63176674. Coherent collision holding does not remove backaction from the original current or E-parity records.

9.6 A joint source experiment.

Selected source factors test repair, propagation, particle creation and clock readout together. Four active CAR modes on two sites carry the same charged component. A spatial superposition is source-prepared with a path reference and an opposite-charge spectator at a third site. The quasiparticle-to-bare chart is fixed by both static and driven terms; diagonalizing the static mass alone does not suffice. Each of two events applies complete native repair, routed bare hopping in that chart, and the local Higgs factors. The same current advances the clock without reset. All 16 active states, spectator states, references and clock purifiers remain, including states with three active particles.

For both C/E laws and both collision policies, all $32^2 \times 4$ outcomes are retained: 16,384 in total. Complete effects are checked on the 64-dimensional field/clock input algebra, together with the untouched reference marginal. The coherent C process has collision population 0.23404814 before the second repair and final three-particle probability 0.53649765, versus 0.36223079 for the

flagged control. Their unread final-state trace distance is 0.20709787. E distinguishes the relevant odd binary occupancies, so the two policies agree on this witness within numerical admission. The composed all-outcome total-variation bound is 4.200002×10^{-7}, conditional on the full-column numerical source checks. It is not an interval-certified hardware bound. Boundary hops out of the selected sites are omitted: this experiment joins actual source factors but is not an invariant two-site restriction of the full spatial field.

9.7 Energy and modular observations on the same field.

The selected four-mode factors admit energy and reference-state calculations on their complete 16-state algebra. Let H be the sum of the static hopping and Higgs terms, with its vacuum scalar retained. The executed ordered product is not silently replaced by e^{-itH}. An explicit piecewise control schedule reproduces that product, separates switching work from energy transfer during complete repair, and uses the same clock duration. The latter transfer is not assigned to heat or controller work without an apparatus model. On the spatial action, onsite terms and half of each incident bond define

$$H = \sum_x h_x, \qquad J_{xy} = i[h_x, h_y] = -J_{yx}, \qquad \frac{dh_x}{dt} = -\sum_y J_{xy} + \partial_t h_x.$$

These identities hold on all occupation sectors. Equality of full unitaries after a scalar energy-reference shift requires the corresponding global source phase; density evolution, currents and gaps do not determine that scalar. Complete, charge-preserving source instruments on all 64 apparatus input columns reconstruct energy, drive-power and ring-current expectations from separate quadratic-group settings. This is expectation readout, not a projective measurement of the summed H.

For the declared $\beta = 1\,\mathrm{GeV}^{-1}$, the same static H gives a faithful $\rho = e^{-\beta H}/Z$. A complete eight-qubit source word prepares its field–purifier state. Signed CAR restrictions and compatible Petz observations compose on every matrix unit, and a native recovery isometry retains all four regional inputs, sixteen field states and four environment states. This is a new Gibbs-input experiment with the same field instrument, not a replacement of the earlier localized input. The modular convention is

$$\rho^{is} A \rho^{-is} = e^{-i\beta s H} A e^{i\beta s H} = \tau_{-\beta s}(A);$$

the dimensionless parameter s does not supply a second physical clock.

The observation is not a regional retraction. The cross-boundary hopping has vanishing partial traces and squared Hilbert–Schmidt norm exactly 4 in the specified units. Its nonadditivity makes the Gibbs density nonproduct for $\beta > 0$, obstructing a reference-preserving UCP retraction onto the full site factor. Compatible non-idempotent Petz observations remain valid. Nor is this Gibbs density stationary under the driven, repaired history. The complete quantum entropy balance is

$$S(\sigma) - S(\rho) = \beta \operatorname{Tr}[H(\sigma - \rho)] - D(\sigma \| \rho)$$

for normalized states; retaining classical records gives an identity involving conditional relative entropies. Numerical trace drift contributes $(\operatorname{Tr}\sigma - 1)\log Z$, and the quadratic Bregman remainder subtracts $\epsilon \operatorname{Tr}\delta$ from $D(\rho + \epsilon\delta\|\rho)$. These reported entropies concern the unnormalized continuing component. First-export abort weights and output families are retained separately; their entropy is not included. A geometric cap kernel, local stress dictionary and generalized-entropy stationarity do not follow from these finite identities.

9.8 The Weyl/contact branch and its finite premise closures.

The following constructions use the supplied Weyl/contact action with $c = 1/8$ and $U = 1$, separately from the calibrated Weyl–Higgs field above. Its dimer and plaquette restrictions retain their actual reference states. The charged-block service is first defined on decoupled dimers. The three-dimensional cube below instead uses bipartite eta symmetry to join a larger protected interacting sector to the complete field. The common local coefficients do not identify distinct global domains.

9.9 A complete charged internal sector.

The same lower charged dimer doublets admit a sharper finite native-record attachment than the fixed-column dictionary. Retain both internal columns at each of two occupied blocks among twelve identical dimers, with the same even half-filled ground at every other block. Their carrier has dimension $66 \cdot 4 = 264$. For the declared graded whole-block repair, a movement that crosses the spectator acts internally as minus the swap; a hold acts as the identity. In the singlet–triplet basis every complete native branch is therefore

$$K_{\text{spinful}} = K_+ \otimes I_1 \ \oplus \ K_- \otimes I_3.$$

An integer certificate checks all 720 raw C/E slots and 31,680 internal input columns, including zero slots. The triplet qutrit and its untouched reference correlations survive every admitted complete history without intermediate recovery. For initially correlated spatial/internal inputs, this assertion concerns the complete internal marginal; conditioning on a record can still steer it.

The raw effects separate all 66 positions. A connected transition tree and an odd exchange loop then classify the entire passive source commutant as $I_{66} \otimes (M_1 \oplus M_3)$, of complex dimension ten. Thus the three-dimensional factor is maximal under the full branch grammar; singlet–triplet coherences do not give a protected four-state factor. Positive all-seam coverage and the checked invariant maximally mixed state supply the corresponding fixed-algebra premises. This enlarged field carrier is distinct from the bare H66 source with scalar fixed algebra.

9.10 Complete extension and an exact spectral exclusion.

Let P be the sum of the orthogonal two-doublet/remaining-ground block projectors. The native outcomes have total effect P; adjoining the outside outcome $I - P$ gives a trace-preserving instrument on the full 48-mode Fock space. This is a formal operator construction, with bounded checks on all 4,096 states of three dimers, not a 48-mode source execution. Every definite occupied pair has the unchanged ground on its entire exterior. With the original supplied $K_j = H_j - N_j/2$, $c = 1/8$ and $U = 1$, the same defect has

$$\Delta = \frac{\sqrt{17}}{8} - \frac{1}{16}, \qquad (K_{\text{tot}} - 12E_g I)|_P = 2\Delta I_P.$$

Its excess CAR number is $q = -1$ for the hole and $q = +1$ for the particle; native currents transport q times the defect count. All three protected triplet states have the same gap and total charge $2q$. A common scale or chemical potential cannot split them into three different masses. The triplet is an internal spin-one representation, not three generations.

The same sector is not invariant under the joined uniform four-site Weyl/contact field. For its four charged-pair columns V, exact dyadic arithmetic gives

$$\|(I - VV^*)H_{\text{uniform}}V\|_F^2 = \frac{3}{256} > 0$$

for both particle and hole pairs, whereas the decoupled sum gives zero. Indeed, with $L = (I - VV^*)H_{\text{uniform}}V$, the stronger identity $L^*L = (3P_{\text{singlet}} + P_{\text{triplet}})/512$ excludes every nonzero nonleaking input, including every triplet state. This generator leakage preserves the distinction between the supplied protecting block service and the interacting field law. No binding, topological boundary charge, physical clock, Lorentz spin, observed species or absolute mass follows; the physical particle and common-field gates remain open.

9.11 A protected interacting sector on the original three-dimensional cube.

The source-bound `cube_eta_protection` packet retains the original open $2 \times 2 \times 2$ Weyl/contact action, $c = 1/8$, $U = 1$, on all $2^{16} = 65\,536$ occupation states. In the physical onsite basis define

$$\eta^+ = \sum_x (-1)^{x_1+x_2+x_3} c^\dagger_{x\uparrow} c^\dagger_{x\downarrow}, \qquad \eta^- = (\eta^+)^\dagger, \qquad \eta_z = (N-8)/2.$$

Onsite singlets are invariant under each supplied $i\sigma_i$ link; the bipartite signs cancel the hopping commutator. Thus $[H, \eta^+] = U\eta^+$, checked exactly on the complete field action. For $X = \eta^+|_{N=4} : \mathcal{H}_4 \to \mathcal{H}_6$,

$$X^\dagger X = (\eta^-|_{N=4})^\dagger (\eta^-|_{N=4}) + 4I, \qquad H_6 X = X(H_4 + UI).$$

Consequently X is injective and its polar isometry intertwines the full interacting four-particle parent with a protected candidate of dimension $\binom{16}{4} = 1820$. Its range is precisely the $\eta \geq 2$ subspace at $N = 6$.

The executed repair domain is a finite four-block service: the four x_1-directed dimers, native ports $(0, 1, 5, 7)$, and all five inherited seams $(0, 1), (0, 5), (0, 7), (1, 5), (1, 7)$. Two blocks carry the unchanged lower-hole doublet and two the unchanged dimer ground. This gives a 24-dimensional $N = 6$ support P. Its physical columns are expanded exactly as $A + \sqrt{17}B$, with integer Gaussian matrices A, B; $\eta^- A = \eta^- B = 0$, so the support has $\eta = 1$. The columns are also bound to the consumed normalized analytic dictionary: its exact normalization is $n = 136 - 32\sqrt{17}$, with $(A + \sqrt{17}B)^\dagger (A + \sqrt{17}B) = n^2 I$. Every native C/E coin and raw slot is checked against its actual graded whole-block permutation. With the complete outside outcome $Q = I - P$, all 130 slots are retained: 120 native and ten outside over the ten seam/law choices. Orthogonality gives

$$PX = 0, \qquad L_r X = 0 \quad (r \text{ native}), \qquad QX = X.$$

Every admitted complete repair is therefore exactly the identity on this sector, including arbitrary reference cross terms; interleaved field evolution remains unitary and interacting. This mechanism requires no alignment of the transverse dimer spin frames. It does not assert a twelve-block execution, arbitrary repair invariance, or maximality of the protected subspace.

9.12 Exact interacting spectral census and a local fixed-number probe.

Let $v = \eta^+|0\rangle$, $v^\dagger v = 8$, and $\Pi_3 = I - vv^\dagger/8$ on the 120-dimensional $N = 2$ space. For $T = (\eta^+)^2|_{N=2}$,

$$T^\dagger T = 60I + 24vv^\dagger, \qquad H_6 T = T(H_2 + 2UI).$$

Hence $T/\sqrt{60}$ is isometric on the rank-119 $\eta = 3$ parent. The hidden-spin census is 35 singlets and 84 triplet states; this is neither Lorentz-spin identification nor a family count. Exact integer characteristic polynomials in the actual onsite Hubbard frame give, with $z = 16(E_{N=6} - 2)$,

$$p_{119}(z) = z^{57}(z^2 - 12)^{21} q(z) c_{64}(z)^3 c_{128}(z)^3,$$
$$q(z) = z^2 - 16z - 12, \qquad c_a(z) = z^3 - 16z^2 - 12z + a.$$

These are eleven distinct energies: multiplicity 57 at zero, 21 at each $\pm\sqrt{12}$, one at each quadratic root and three at each of the six cubic roots. The last twenty states have contact-shifted energies. Squarefreeness, coprimality and rational root isolation are exact; approximate displayed response weights are separate diagnostics.

The $N = 6$ reference and one-site density probe are

$$|R\rangle = \frac{(\eta^+)^3|0\rangle}{\sqrt{2016}}, \qquad H|R\rangle = 3|R\rangle, \qquad |s\rangle = \frac{4n_0 - 3}{\sqrt{15}}|R\rangle.$$

Here $\langle n_0\rangle_R = 3/4$, $\mathrm{Var}_R(n_0) = 15/16$; $|s\rangle$ is an $\eta = 3$ singlet with mean energy 3 and variance 3/112. Its normalized two-particle parent is $(8|\uparrow\downarrow\rangle_0 - v)/\sqrt{56}$. The exact parent Krylov Gram determinant is $1488395072912184839188800532906554156908544 > 0$. Together with the annihilator $qc_{64}c_{128}$ this proves positive response weight at all eight interacting energies and zero weight at the three free-energy values.

The local number-preserving phase $F_0 = e^{i\pi n_0/2}$ satisfies

$$F_0|R\rangle = \frac{|R\rangle - \sqrt{15}|s\rangle}{4}, \qquad \mathrm{Tr}_0(F_0|R\rangle\langle R|F_0^\dagger) = \mathrm{Tr}_0\,|R\rangle\langle R|.$$

Thus the entire exterior reduced state is unchanged initially, while the global pure-state trace distance is $\sqrt{15}/4$ and the disturbed state has energy variance 45/1792. This is a finite local unitary disturbance of an entangled reference, not yet a boundary charge or an asymptotic localized particle. Both components remain protected. A fixed-$N = 6$ pointer Ramsey construction using controlled F_0, common field evolution and controlled $F_0^\dagger$ has

$$C(t) = \langle R|F_0^\dagger e^{-it(H-3)}F_0|R\rangle = \frac{1}{16} + \frac{15}{16}\sum_{j=1}^{8} w_j e^{-it(z_j/16-1)}, \qquad w_j > 0, \quad \sum_j w_j = 1.$$

No vacuum/charged-sector phase reference is needed in this protocol. The specified H and reference energy fix the frequency origin; these finite spectral gaps are not physical masses.

9.13 Native preparation and the remaining detector premise.

The actual source prepares the 56 signed three-doublon amplitudes using an eight-bit, 5379-pulse amplitude tree on the even modes, with the odd modes blank. Eight native CNOT words copy into the odd modes in 24 pulses, followed by the three-pulse one-site phase including its scalar phase. The compact tree evaluation is an exact spectator blank isometry; all 65 536 rows are retained for copying and the phase. The independent exact trigonometric and literal-port certificate bounds the complete source-state error by 8.48×10^{-13}, including every source export. All 130 following repair records are retained without success normalization. The native seventeen-bit time-dependent detector and native field continuation are not executed here: the fixed-number response above is analytic, while the reference preparation and local kick are executed. Hardware precision, growing-volume localization, deformed excitation transport, a continuum limit and empirical species identification remain open. The earlier small-carrier leakage and transverse hidden-spin obstruction remain valid; the larger eta sector supplies a different common protection mechanism.

The preceding preparation packet stops before the time-dependent detector. The following extension executes that detector and field continuation on the same cube, closing this finite readout premise.

9.14 An internal observer reads the protected response.

The `protected_particle_readout` construction attaches the preceding cube spectrum to complete native pointer records. Its identified object is the eight-dimensional cyclic response generated by the centered local density probe, inside the 119-state $\eta = 3$ sector and the larger 1820-dimensional protected space. It is a neutral, fixed-$N = 6$ response; these three dimensions are not interchangeable particle counts. The construction uses Ramsey interference to read a correlation function, a standard measurement principle, and supplies its actual source implementation for this field.[1]

The apparatus retains sixteen field bits and one pointer, hence all $2^{17} = 131\,072$ field/pointer occupation rows. The unchanged source prepares $|R\rangle$ and a pointer $|+\rangle$, then applies controlled $F_0 = \exp(i\pi n_0/2)$. There are two field segments of duration $1/2$: the first is followed by complete C repair on all five admitted seams, and the second by complete E repair on those seams. Controlled $F_0^\dagger$, a pointer phase $\phi \in \{0, \pi/2\}$, and the inverse preparation rotation precede the complete six-outcome pointer export. Every source first-failure record and all 130 repair records remain present. An outside repair outcome continues; a native repair outcome or source failure terminates in its own retained record. No probability is conditioned on survival.

With exact field evolution, each repair acts identically on both ideal arms. Immediately before the final pointer rotation their field vectors are

$$b_0 = e^{-itH}|R\rangle, \qquad b_1 = e^{i\phi}F_0^\dagger e^{-itH}F_0|R\rangle.$$

Consequently the unconditional ideal pointer-zero probability is

$$\begin{aligned} p_0(t,\phi) &= \tfrac{1}{2}\{1 + \mathrm{Re}[e^{i\phi}C(t)]\}, \\ C(t) &= \frac{1}{16} + \frac{15}{16}\sum_{j=1}^{8} w_j e^{-it\omega_j}, \qquad \omega_j = z_j/16 - 1. \end{aligned}$$

Thus the two settings read $\mathrm{Re}\, C = 2p_0(t,0) - 1$ and $\mathrm{Im}\, C = 1 - 2p_0(t,\pi/2)$. The phase $3t$ follows from the reference eigenenergy; it is not an adjustable frequency fit. A common scalar added to H, or a common number chemical potential on this fixed number domain, cancels between the arms. Absolute energy zero is therefore not determined by this experiment.

9.15 Complete source and field error, without a sector projection.

The executor composes success and every first-export failure map of each local word on at most three work bits, then applies these maps to the full field/pointer array. This is finite block contraction of the same native word, not replacement by an ideal gate. Independent tests compare the contracted maps with direct source execution, including order-one failure controls. The exact $N = 6$ restriction is used only by the independent original-H comparator, not by the native execution.

Each half-time field segment uses 32 symmetric product steps. The actual ordered support list has 24 hopping terms of norm $1/16$ and eight contact terms of norm one. Counting only possibly nonzero nested commutators gives the all-Fock coefficient $K = 629/256$. The two segments therefore contribute

$$\varepsilon_{\mathrm{split}} = 2\,\frac{629}{256}\,\frac{(1/2)^3}{32^2} = \frac{629}{1048576}.$$

Independent rational trigonometric and literal-port bounds add less than 8.56×10^{-11} for the preparation, controlled pointer operations, and both complete source field words. The complete

[1] For many-body Ramsey correlation measurements, see Knap et al., *Phys. Rev. Lett.* 111, 147205 (2013), `https://arxiv.org/abs/1307.0006`.

output-vector error is consequently $\varepsilon < 6.00 \times 10^{-4}$. Every halted record is part of this norm. Complete repair and export preserve the comparison by isometric extension or contraction; the all-input field bounds also hold with untouched references. The initial preparation bound consumes the specified initialized blank.

For normalized complete outputs, half trace distance is at most their vector distance. Each unconditional pointer probability therefore differs from its ideal value by at most ε, and each reconstructed quadrature by at most 2ε. Rational root, residue and trigonometric enclosures at $t = 1$ imply, for the actual specified source,

$$0.993649 < p_0(1, 0) < 0.994850, \qquad 0.501379 < p_0(1, \pi/2) < 0.502580.$$

Both intervals exclude the corresponding no-response values, one and one half. Numerical source residuals have their separate declared admission; they are not substituted for these proved source bounds. The executed construction uses 681 283 native pulses per setting. It gives pointer-zero probabilities approximately 0.9942497011 and 0.5019798894, versus the independent original-H values 0.9942494663 and 0.5019797169. The complete output-vector discrepancy is approximately 2.27×10^{-6}, below the rigorous bound.

The imaginary fringe also distinguishes the contact interaction from the same free cube. Set $U = 0$ while retaining the hopping, reference, local probe and ideal repair schedule. Bipartite occupation parity $\Gamma_{\mathrm{sub}} = (-1)^{\sum_{x\,\mathrm{odd}} n_x}$ satisfies

$$\Gamma_{\mathrm{sub}} H_0 \Gamma_{\mathrm{sub}} = -H_0, \qquad H_0|R\rangle = 0, \qquad \Gamma_{\mathrm{sub}} F_0|R\rangle = F_0|R\rangle.$$

The free characteristic function is consequently real at every time, so $p_{0,\mathrm{free}}(t, \pi/2) = 1/2$ exactly. Both free arms remain in the protected sector. The certified interacting interval therefore excludes this specified contact-free response. This is discrimination between two internal dynamics, not a fit of a physical particle mass.

For independent reset repetitions, Hoeffding's inequality and a union bound give simultaneous sampling error at most 1/2048 in these two probabilities using 2^{24} trials per setting, except with probability less than $4(3/8)^8$.[2] The sampling allowance still separates both fringes from no response. Independent resets are the sampling-service premise; a seeded numerical sample is an illustration of that model, not laboratory data. The recorded PCG64 illustration, with seed 20261101, gives respectively 16 680 454 and 8 423 380 pointer-zero outcomes out of 2^{24} trials; the complementary count includes every other complete outcome.

9.16 What eight-line resolution would additionally require.

The executed two settings attach the known protected spectral measure to an internally accessible record. They do not reconstruct eight unknown frequencies or individually resolve the eight lines. There is, however, a certified prospective experiment on the source-specified support. At times $t_n = 39n$, $n = 0, \ldots, 7$, use both pointer quadratures and set

$$y_n = \frac{16C(39n) - 1}{15}, \qquad V_{nj} = e^{-i39n\omega_j}, \qquad y = Vw.$$

Exact interval sine bounds in the Vandermonde product show

$$|\det V| = \prod_{j<k} 2\left|\sin\frac{39(\omega_k - \omega_j)}{2}\right| > 1400.$$

[2]The bounded-independent-sum inequality is imported from Hoeffding, *J. Amer. Statist. Assoc.* 58 (1963), 13–30, `https://doi.org/10.1080/01621459.1963.10500830`.

Hadamard's inequality bounds every 7×7 cofactor by $7^{7/2} < 343(8/3)$, so

$$\|V^{-1}\|_\infty < \frac{2744}{525}, \qquad \|\widehat{w} - w\|_\infty < \frac{43904}{2625}\, e,$$

where e bounds the total error of each pointer probability. The latter includes both quadratures and removal of the known elastic atom. Finite precision evaluation of an implemented inverse would need its own error allowance.

A sufficient prospective budget is systematic probability error $1/65536$ and sampling error $1/32768$ at each of the sixteen settings. With $9 \cdot 2^{29}$ independent trials per setting, the simultaneous failure probability is less than $32(3/8)^9 < 1/100$ and the weight error is less than $343/448000$. Twice that error is smaller than the least certified weight, so every line is separated from zero. The total is 77 309 411 328 trials and the largest model time is 273. This long-time accuracy and sampling schedule is not executed here. It identifies weights on the supplied support, not an arbitrary unknown spectrum from finitely many samples.

The finite result is an internally observable protected excitation response of the admitted simulator, with complete-history and finite-time errors. It does not require an external particle name or an SI calibration. It also does not establish that these neutral modes are single odd fermions, that the reference is a vacuum ground state, or that finite spectral lines are relativistic rest masses. Transported excitations, many-body continuum dynamics and comparison with nature remain separate constructions.

The one-time readout above leaves its long-time reconstruction prospective. The following certified accelerated calculation closes that known-support weight-resolution premise while retaining a bound to the full native word.

9.17 Full weight resolution by certified accelerated emulation.

The preceding sixteen-setting proposal can now be executed as a certified emulation of the same source word. The result resolves all eight weights on the source-specified support; it does not infer arbitrary unknown frequencies or identify physical species. The observable remains the eight-dimensional neutral cyclic response of the excited reference, with elastic weight $1/16$.

The actual chronological field word places the two disjoint spin hoppings of each spatial edge consecutively. They commute, and their sum is the complete Weyl edge commuting with physical η^+. The complete contact layer satisfies $D_6 T = T D_2 + 2T$, where $T = (\eta^+)^2$. Thus every ideal Strang step has the exact identity

$$S_6(h) T = e^{-2ih} T S_2(h), \qquad S_6(h)|R\rangle = e^{-3ih}|R\rangle.$$

The complete protected repairs act identically on these ideal states. This is a property of the grouped ideal word, not an exact projection identity for its rounded native implementation.

The actual Hubbard-frame singlet chart has dimension36, retaining all35 $\eta = 3$ singlets and the $\eta = 4$ parent. If U is the spatial forward half-hop matrix and $W = \mathrm{Sym}^2 U$ on normalized unordered pairs, the step is $W^T \exp(-ihD) W$. Transpose reverses the hopping order because every individual Hubbard hopping matrix is symmetric. Integer generator checks bind this chart to every original edge and the contact layer. Outward interval square roots supply the distinct-site normalization.

Use the common step $h = 39/2^{19}$. Time $39j$ takes $j2^{19}$ steps, with half before C repair and half before E repair. At time273 this is 3 670 016 steps. Their 38 755 368 960 field pulses are represented by the repeated source word; they are not individually executed. Instead, outward160-bit step entries are rounded to128-bit dyadics and powered by exact Gaussian-integer matrix multiplication.

For approximate unitaries with errors a, b, one rounded product has error at most $a+b+ab+36\,2^{-128}$. The longest power needs24 products and has operator error $249561087/2^{128} < 7.34 \times 10^{-31}$.

Independent literal-source templates bound the complete 10 560-pulse step error by 1.323294×10^{-12}. Telescoping includes all native first-export failures, and the full-Fock splitting error is $(629/256)t^3/n^2$. With preparation, pointer operations, complete repairs, certified arithmetic and exact binary64 probability-conversion error, every setting has systematic probability error below $8.568121 \times 10^{-6} < 1/65536$. Individual long-word failure weights are not materialized; their record addresses and complete-output norm bound are retained. No survival normalization is used.

The emulator draws counts from a binary64 proxy to the certified powered ideal-word probabilities, with PCG64 seed20261102. These are surrogate counts, not exact samples of the unmaterialized native failure instrument. There are $9\,2^{29}$ independent reset trials at each of sixteen settings, or77 309 411 328 overall. Zero is contrasted with all other complete outcomes. The stated independent binomial service gives simultaneous sampling error 1/32768 except with probability below $32(3/8)^9 < .01$. The same conditional recovery theorem applies to genuine native counts under that sampling event and the proved native systematic bounds.

Only the source frequency intervals enter the inverse. Rational Lagrange polynomials give a96-bit matrix B for $V_{nj} = e^{-i39n\omega_j}$; outward arithmetic certifies $\|B\|_\infty < 2$ and $\|BV - I\|_\infty < 10^{-18}$. Counts determine $\widehat{y}$ and $\widehat{w} = \mathrm{Re}(B\widehat{y})$. Since the true source weights form a probability vector, the conditional error is bounded by

$$\|\widehat{w} - w\|_\infty \le \|B\|_\infty \frac{16}{5}(\varepsilon_{\text{systematic}} + \varepsilon_{\text{sampling}}) + \|BV - I\|_\infty < 0.000258393828.$$

In ascending frequency order the recovered weights are

$$(.002457894673,\ .005641397395,\ .005890881572,\ .007131698528,\\ .004528321839,\ .421578515341,\ .415796456292,\ .136974834359).$$

Every resulting interval is strictly positive and contains the independent exact residue, checked after inference. This closes weight resolution on the given support in the certified emulator. It does not turn $E_H - 3$ into an absolute rest mass, identify an unknown support, or establish physical particle species or native hardware execution.

9.18 Comparing the actual interacting regional clocks.

The complete four-mode Weyl/contact Gibbs source supplies a useful finite alternative to assuming regional modular covariance. Let $W_S(s) = \exp(isH)$ be its state modular unitary at $\beta = 1$, and let $W_A(s) = \rho_A^{-is}$ be the embedded unitary of an actual CAR-reduced reference, with fixed scalar gauges understood. The relative operations

$$C_{AB}(s) = W_A(s)W_B(s)^*, \qquad C_{AB}(s)C_{BD}(s) = C_{AD}(s)$$

compare these clocks on every input and untouched reference. They obey the time-cocycle identity $C_{AS}(s+t) = C_{AS}(s)W_S(s)C_{AS}(t)W_S(s)^*$. The existing source words implement each global-to-regional correction at $s = 1/2$ with 2,826 pulses and ideal operator splitting bound 11/32768; the regional-to-regional comparison needs twenty pulses. All sixteen Fock columns and first export failures remain present. Such corrections are active: the first changes the common Gibbs reference by half trace distance about 0.00444. They do not assert that both regional flows preserve the same global state.

9.19 The stronger identification fails on this reference.

For a linear modular intertwiner, an eigenoperator coefficient satisfies $(\nu_j - \omega_i)L_{ji} = 0$. The actual dimer at speed $1/8$, contact $U = 1$, $\beta = 1$ and $\mu = 0$ has three positive regional frequencies, approximately 0.0001258906, 0.9988218345 and 0.9989477251. An exact source characteristic polynomial and nine partial-trace moment identities determine its reduced spectral weights. Rational bounds on the square root, exponential and logarithm then separate all three frequencies from every global energy difference by more than 10^{-4}, in all 243 comparisons. Consequently every same-parameter linear intertwiner has range in the six-dimensional regional modular centralizer and cannot transmit a moving local clock hand. The group and coefficient implications are Lean checked; the concrete spectral and interval attachment is checked separately by exact arithmetic. This excludes a strong identification on one finite reference, while the explicit relative operations provide a functioning comparison. The supplied action, coherent services, reference preparation and dimensionless parameter remain visible inputs. No physical duration, common energy calibration, geometric modular flow or Einstein premise is discharged.

9.20 Controlled regional modular energy.

The same complete-Fock Weyl/contact action now supplies a quantitative regional modular attachment on its four-site open plaquette. All 256 field states and all sixteen states of the two-site region are retained. For the normalized signed partial trace E, put $H_c = H - I$ and $X(\beta) = E(e^{-\beta H_c})$. Exact Gaussian-integer and rational arithmetic constructs the first eight coefficients of $-\log X$. In particular, with $M_j = E(H_c^j)$,

$$C_1 = M_1, \qquad C_2 = \tfrac{1}{2}(M_1^2 - M_2), \qquad C_3 = \tfrac{1}{6}M_3 - \tfrac{1}{4}(M_1M_2 + M_2M_1) + \tfrac{1}{3}M_1^3.$$

For $0 \le x = \beta B < \log 2$, where $B \ge \|H_c\|$, the positive coefficient majorant $f(x) = -\log(2 - e^x) = \sum a_j x^j$ gives

$$\left\| -\log X - \sum_{j=1}^{p} \beta^j C_j \right\| \le f(x) - \sum_{j=1}^{p} a_j x^j.$$

This follows from the norm-convergent noncommutative logarithm series, without replacing the reduced reference by a local Gibbs ansatz. At the declared $\beta = 1/32$ and exact bound $B = 3$, an outward rational evaluation gives eighth-order error below 1.917×10^{-9}. Hence every Hermitian trace-zero regional variation obeys the entropy-tangent error bound $1.917 \times 10^{-9}\|\delta\|_1$. The normalization scalar in $K_A = -\log X + \log \operatorname{Tr}_A X$ cancels only on that tangent space.

The first nonscalar boundary correction is cubic and does not commute with the bare regional energy. Removing the contact interaction removes its interaction-dependent terms; removing the crossing bonds removes all nonscalar corrections. A full-versus-empty regional matrix-element witness and the same rational tail prove distance greater than 2.7×10^{-8} from $\beta C_1 + aI$ for every scalar a on this instance. This supplies both a corrected finite modular generator and a certified boundary to its bare-energy identification. The source preparation below supplies this plaquette reference to a separate controlled error. The action, temperature and lattice remain declared; the norm bound can grow with volume. Geometric cap weights, null stress, fixed-volume entropy–area stationarity, a physical scaling limit and positive Newton normalization remain additional obligations.

9.21 A separately prepared interacting Gibbs reference.

On the complete 256-state plaquette at $\beta = 1/32$, the degree-eight Gibbs ensemble retains all 256 preparation records and every export abort. Independent outward arithmetic certifies its literal-source preparation error below 1.700×10^{-12} on the initialized input. The actual regional success logarithm, including the exact normalization scalar, is within 2×10^{-9} of the preceding P_8 comparison. The all-input continuation has a separate source and splitting bound. The proof in `model/certified_reference/README.md` supplies finite Gibbs preparation; it does not assert invariance under the complete repair. The following stationary reference has a different modular generator.

9.22 The actual stationary source weight.

One identical literal native $Y(-\pi/3)$ request and complete dephasing export per occupation mode prepare $\sigma_M = \operatorname{diag}(a,b)^{\otimes M} = a^M e^{\gamma N}$, where $a = |U_{00}|^2$, $b = |U_{10}|^2$ for the actual represented source unitary and $\gamma = \log(b/a)$. The exact one-mode error is below 1.894×10^{-16}, proving $a, b > 0$ and $b > a$; abort blocks remain separate. Number conservation and complete unital repair preserve this weight exactly. Tracing complete outside outputs before regional compression gives a k-mode logarithm error at most $k\delta/(1/4 - \delta)$, with $\delta = (2+\epsilon)\epsilon$. This charge clock is trivial within fixed number. On the earlier six-site $N = 4, S = 1$ active sector, every common stationary modular generator stays at distance at least $|\beta|\sqrt{13/48}$ from βK, even allowing scalar and protected-spin terms. Thus the stationary reference and interacting Gibbs energy remain distinct. The exact source and reference proofs are in `model/continuing_reference/README.md`; the following construction closes its previously restricted scheduler premise.

9.23 A stationary finite controller for native feedback.

The reference can now continue under an explicit record-adaptive scheduler. Use two classical working bits (c, d); $c = 0$ selects the complete native C repair on seam $(0, 1)$, and $c = 1$ the complete E repair on $(0, 5)$. Append every raw record r, including zero slots and the outside outcome $Q = I - P$, and update

$$(c, d) \longmapsto (d \operatorname{XOR} f(r), c), \qquad f(r) = \text{coin XOR (raw mod 2)}, \quad f(Q) = 0.$$

At output (u, v) every incoming term uses the same instrument v. Completeness and unitality of the unchanged repair therefore give

$$\sum_r L_r^{(v)} \frac{\sigma_M}{4} L_r^{(v)*} = \frac{\sigma_M}{4}, \qquad \Omega = \sigma_M \otimes I_4/4, \quad \sigma_M = \operatorname{diag}(a,b)^{\otimes M}.$$

Here a, b are the actual literal-source weights, so the successful block still has trace $(a+b)^M$. All initial export-failure sectors are retained and frozen, emitting a bypass record. Separate unitary layers selected by the current working memory and generated by the original number-conserving field Hamiltonian also preserve this marginal. The controller is a declared reversible classical law on $\mathbb{C}^4$, not a claimed quantum pulse circuit.

Two unchanged native events on an explicitly initialized binary spectator, occupied at ports $2, 3$ and repaired on $(0, 1)$, leave it fixed and supply two exact fair coins. Their retained records seed $I_4/4$ without an angle-error assumption. This uses the supplied simulator basis initialization; no hidden interacting-field preparation of that spectator is asserted. The working memory remains two bits, while the archive grows by at most five bits per tick, besides the two seed records and all

6^M source readouts. Stationarity traces these records; it never normalizes the successful field block or declares the growing archive stationary. The modular generator remains $-\gamma N - M \log a + \log 4$, $\gamma = \log(b/a)$, and every complete step intertwines the charge flow $e^{it\gamma N}$. No physical energy-temperature identification follows. Earlier local literal-source bounds remain uniform in volume and ideal adaptive history length; implemented per-step errors must still be added.

9.24 A complete finite mixing certificate and its boundary.

On the invariant three-position highest-spin repair corner, with no joined Hamiltonian step inserted, the field/controller algebra is $\bigoplus_4 M_3$. Each E outcome has rank at most one. Exact integer enumeration finds at least one eight-tick history encountering E for every one of the 12^2 input/output population pairs, with common probability denominator 2^8. The corresponding rank-one Kraus histories prove

$$\Phi^8 = \frac{3}{64}\mathcal{D} + \frac{61}{64}\mathcal{R}, \qquad \mathcal{D}(X) = \mathrm{Tr}(X) I_{12}/12,$$

where $\mathcal{R}$ is completely positive, trace preserving and unital. Thus $I_{12}/12$ is the unique stationary density on this CQ algebra, and the archive-traced error contracts by $(61/64)^{\lfloor n/8 \rfloor}$, including arbitrary untouched quantum references and all field coherences. This is neither whole-Fock mixing nor mixing with interleaved Hamiltonian evolution. Indeed the full retained history recovers the initial corner basis state and memory. As a negative control, the recordwise reversible one-bit policy $c \mapsto c \,\mathrm{XOR}\, f(r)$ sends the uniform six-state input to $(3, 1, 2, 1, 3, 2)/12$, a trace distance $1/6$ from uniform. The two-register incoming-row balance is a substantive premise, not a consequence of memory reversibility alone.

9.25 Exact energy Ward cells under coarse-graining.

The same complete-Fock Weyl/contact action supplies a conserved energy column on its standard three-dimensional lattice. With the declared half-bond convention $H = \sum_x h_x$, its range-two currents are $J_{xy} = i[h_x, h_y]$. Distributing each current equally over all shortest nearest-neighbour paths preserves its incidence $\delta_x - \delta_y$: intermediate vertices cancel on every path. The resulting oriented face current $F_{x,i}$ therefore satisfies $\dot{h}_x + \sum_i (F_{x,i} - F_{x-e_i,i}) = 0$ exactly. The rational Clifford certificate checks the actual $3 \times 3 \times 3$ action at $c = 1/8, U = 1$: all 351 site pairs, 81 nonzero currents, 54 faces and 27 local equations, on the complete 54-mode algebra. Its coverage of all 2^{54} Fock states is a symbolic operator identity, not execution of that many amplitudes. Independent complete small-Fock matrices attach all three hopping axes and the interaction to the existing source action.

For a finite time cell, the exact flux is $I_{x,i}(n) = \int_{t_n}^{t_{n+1}} F_{x,i}(t)\, dt$. Hence

$$h_x(t_{n+1}) - h_x(t_n) + \sum_{i=1}^{3} \big(I_{x,i}(n) - I_{x-e_i,i}(n)\big) = 0.$$

For any normalized input and untouched reference, setting $T_{00}(n, x) = \langle h_x(t_n) \rangle$ and $T_{i0}(n, x) = \langle I_{x-e_i,i}(n) \rangle$ supplies the $j = 0$ forward-difference Ward column of the finite tensor interface. These are cell energy and integrated flux, not pointwise continuum stress. Summing spatial blocks cancels internal faces, so $D_{\mathrm{coarse}} P_{\mathrm{faces}} = P_{\mathrm{sites}} D_{\mathrm{fine}}$. Successive block maps compose exactly; successive time cells telescope. All 27 rectangular width choices, including ragged blocks, pass the exact source certificate. Generic finite-sum divergence, continuity, block composition and time telescoping are also Lean-checked; the concrete CAR source binding remains the exact rational certificate.

The existing local source bounds control the approximation uniformly in ambient volume. With $A = U+3c$, one has $\|h_x\| \le A$ and $\|F_{x,i}\| \le B = 14A^2$: a face receives absolute shortest-path weight at most seven. Its support fits a radius-three cube. If L_h, L_F are the existing normalized locality tails and ϵ is the patch unitary error, an endpoint energy error is at most $A\min(2, L_h + 2\epsilon)$ and an integrated-face error at most $tB\min(2, L_F + 2\epsilon)$. Replacing the integral by m midpoint samples adds the explicit bound $t^2B\,343(U + 6c)/(2m)$; endpoint sampling is not exact. A cell residual is bounded by two endpoint errors plus six face errors, with supplied readout errors added separately. The receipt evaluates these resource bounds without claiming to execute the large sample count. Input preparation, hardware errors and controller/failure-space energy remain separate. This closes a finite same-action energy-column attachment, not the three momentum columns, a symmetric Lorentz tensor, geometric modular weights, area stationarity, a continuum scaling theorem or the full Einstein premise register.

9.26 Local momentum balance retains its force.

On the open Weyl/contact box, the three local central-difference momenta obey exact balances with kinetic face stress and quartic contact force. All spatial blocks and time cells retain that force. On the 3^3 box at $U = 1$ its total-force Gram matrix is $(9/4)I_3$, so an internal closed divergence cannot absorb it. Conserved number-preserving quadratic charges of the connected action reduce to number and hidden spin. These exact results in `model/lattice_momentum_balance/README.md` concern local quadratic candidates. The following periodic construction supplies conserved global momenta with a separate locality boundary.

9.27 Global periodic momentum and the local question.

On a declared periodic box, graded translations T_j commute with the full interacting action. Their principal Hermitian logarithms P_j are exactly conserved. They differ from additive one-body momentum by integer many-body winding. On odd-prime rings every number-preserving translation logarithm has nonzero one-particle spin-trace kernels at all nonzero separations. For the principal branch, every range-R density sum obeys $\|P_j - A_R\| \ge 1/(R + 1)$ when $L > 2(R + 1)$. These full-space statements, proved in `model/periodic_momentum/README.md`, leave controlled restricted state domains available. The following bounds quantify one such domain and retain its actual interaction leakage.

9.28 A collision-suppressed local Ward bound.

The same periodic Weyl/contact action supplies a controlled local approximation beyond its nonlocal conserved momentum. Let B be a spin-complete set of spatial Fourier modes, $f = |B|/V$, and let Ω restrict the initial state to that band and $N \le N_{\max}$. Write $D = \sum_x n_{x\uparrow}n_{x\downarrow}$ and $d = \lfloor N_{\max}/2 \rfloor$. Band annihilators have norm $\sqrt{f}$, so $\Omega D\Omega \le f\min(N_\uparrow, N_\downarrow) \le fd$. Since $D^2 \le dD$ on this number sector,

$$\|UD\Omega\| \le Ud\sqrt{f}, \qquad \|(I - \Omega)e^{-itH}\Omega\| \le \min\{1, |t|Ud\sqrt{f}\}.$$

The second bound uses free invariance of Ω and Duhamel's formula; the full interacting action is never projected. For two particles, the exact contact principal angles are $\lambda_K = |B \cap (K - B)|/V$, giving escape norm squared $\max_K \lambda_K(1 - \lambda_K)$ before multiplication by U^2. Thus projecting the outgoing contact back into B would miss its leading square-root leakage.

Let $s_j(x)$ be the local sine-momentum density. Write $\Pi_{xy,j}$ for the antisymmetric kinetic stress across a nearest-neighbor face. The exact identity is $i[H, s_j(x)] + \sum_y \Pi_{xy,j} = F_j(x)$, with $F_j(x) =$

$i[UD, s_j(x)]$. Normal ordering its quartic contact terms gives $\|F_j(x)\| \le U$ and $\|F_j(x)\Omega\| \le 4Uf$. For preparation at zero and $0 \le t_0 \le t_1$, the actual integrated Ward residual therefore obeys

$$\left\| \left[s_j(x,t_1) - s_j(x,t_0) + \sum_y \int_{t_0}^{t_1} \Pi_{xy,j}(t)\, dt \right] \Omega \right\|$$
$$\le \min\left\{ U(t_1 - t_0),\, 4Uf(t_1 - t_0) + \frac{U^2 d\sqrt{f}}{2}(t_1^2 - t_0^2) \right\}.$$

The preparation clock is not restarted in later cells. Spatial blocks cancel their internal faces and sum the site bounds; time cells telescope. The complete 54-mode 3^3 source certificate checks all 81 local equations, 81 kinetic faces per column and 27 rectangular coarse partitions per column. Finite-sum transport of the errors is also Lean-checked. This is symbolic full-Fock coverage, not execution of all its amplitudes.

The two-particle total sine momentum has multiplier $2\sin(K_j/2)\cos(k_j - K_j/2)$. The uniform contact vector gives the sharp complete-fiber force norm $\|i[UD, S_j]\| = \sqrt{2}U|\sin(K_j/2)|$. If initially $|k_j| \le \kappa$ and $2\kappa < \pi$, total momentum stays within $|K_j| \le 2\kappa$ even after relative-momentum scattering. Combining this conserved domain with the sine remainder gives

$$\|(P_j - S_j)e^{-itH}\Omega\| \le \frac{\kappa^3}{3} + 4\kappa \min\{1, |t|U\sqrt{f}\}.$$

For a 64^3 counting example with $B = \{-1, 0, 1\}^3$, $N_{\max} = 2$, $U = 1$ and $t = 1/8$, the certified leakage is below 0.00127, the local Ward defect below 0.000132, and the momentum error after division by κ below 0.0083. The separate eight-site exact-Hamiltonian diagnostic retains a nonzero unforced local defect, so the contact force has not been discarded. These are fixed-number dilute bounds. The critical band-edge scaling $f \sim a^3$, $U \sim a^{-2}$ instead makes $U\sqrt{f} \sim a^{-1/2}$, for which this estimate supplies no vanishing leakage. A finite-density or critically scaled local stress limit, Lorentz symmetry, physical units and the remaining Einstein premises remain separate. Proofs, finite domains and replay are in `model/dilute_momentum_ward/README.md`.

9.29 Interacting lattice scattering and its zero threshold.

The same Weyl/contact action now admits an on-shell two-fermion scattering theorem. On the physical exchange subspace of $L^2(\mathbb{T}^3, \mathbb{C}^4)$ at total momentum K, its free multiplication operator $H_{0,K}$ is bounded self-adjoint and the contact perturbation is $U|d\rangle\langle d|$, with $\|d\| = 1$. The difference is trace class with norm U. The Kato–Rosenblum theorem therefore gives complete wave operators on the absolutely continuous subspaces and a unitary scattering operator.[3] With $g_+(E) = a(E) - i\pi\rho_0(E)$, its contact-channel eigenvalue is

$$S_c(E) = \frac{1 - Ua(E) - i\pi U\rho_0(E)}{1 - Ua(E) + i\pi U\rho_0(E)}.$$

The orthogonal reducing complement of the contact-generated subspace has identity scattering. Point spectrum, including the isolated pair and the free dark zero modes, remains outside the incoming scattering domain. Completeness here concerns two-body absolutely continuous scattering; it is not many-body or chiral-continuum completeness.

[3] The imported trace-class and finite-rank scattering results are reviewed in `https://arxiv.org/abs/1711.00528` and `https://arxiv.org/abs/0902.3568`; the attachment here verifies their premises for the stated Weyl/contact exchange fiber.

At $K = 0$, the singlet resolvent is $z/(z^2 - 4c^2|\sin k|^2)$. Coarea with all eight sine-map preimages gives, for $0 < |E| < 2c$,

$$\rho_0(E) = \frac{E^2}{4\pi^2 c^3} \sum_{n\geq 0} A_n \left(\frac{E}{2c}\right)^{2n}, \qquad A_0 = 1, \quad A_1 = \tfrac{1}{2}, \quad A_2 = \tfrac{11}{40}, \quad 0 < A_n \leq 1.$$

The coefficients are exact sphere moments of the binomial Jacobian. After degree M, the dimensionless remainder is at most $r^{2M+2}/(1 - r^2)$, $r = |E|/(2c)$; rational arithmetic and a Machin bound for π give certified density intervals. Local analyticity and evenness give $g_+(E) \to 0$ at zero. Hence $\rho_U(E) = \rho_0(E)/|1 - Ug_+(E)|^2 \sim E^2/(4\pi^2 c^3)$. This supplies nontrivial scattering and a gapless contact spectral support on one interacting lattice action while retaining its isolated pair. Zero model energy is not an identified physical vacuum threshold, so no massless physical particle is assigned. The source-bound proof and imported scattering theorems are recorded in `model/contact_scattering/ONSHELL.md`.

9.30 Interacting edge scattering and a moving bound pair.

The same two-body source family now has a controlled moving bound band. Set $c_a = \sqrt{3}/a^2$ and retain the repulsive coupling

$$U_a^{-1} = \frac{a^2 W_0}{2} - \frac{a^3 \gamma}{\pi}, \qquad W_0 = \left\langle \frac{1}{\sum_j \cos^2 k_j} \right\rangle, \qquad \gamma > 0.$$

The normalized full-torus integral satisfies $2/3 \leq W_0 < 5$; $a\gamma \leq 1$ makes the coupling positive. The single global comparison is $J_a = 6/a^2 - H_a$, explicitly shifting energy and reversing time. Neither γ nor this clock/energy choice is inferred from physical data.

At zero total momentum the threshold estimate is

$$\left| \frac{g_0(6/a^2 + z) - a^2 W_0/2}{a^3} + \frac{\sqrt{z}}{\pi} \right| \leq 19a|z|$$

for real $|a^2 z| \leq 1/8$, using the upper boundary square root; the exterior constant improves to 11. Coarea for all eight valleys, an inner ball of radius $1/2$ and its complementary measure give the respective Green remainders 18η and $36|\eta|$ before rescaling. Thus, for $0 \leq s \leq S$, $a^2 S \leq 1/8$ and $836aS \leq 7\gamma$,

$$S_J(s) \longrightarrow \frac{\gamma - i\sqrt{s}}{\gamma + i\sqrt{s}}, \qquad \sup_{[0,S]} |S_J - S_{J,\lim}| \leq \frac{1672aS}{7\gamma}.$$

This is the earlier nonidentity scalar contact-channel limit; its orthogonal free channels remain present. Lattice contact renormalization is an established mechanism, not a new particle identification; compare Koerber–Berkowitz–Luu (2019) and Winkler et al. (2006).

9.31 Nonzero total momentum has a controlled pole.

Write $K = aP$, $q = k - K/2$, $d_j = \sin q_j \cos(K_j/2)$, $e_j = \cos q_j \sin(K_j/2)$ and $R^2 = \sum_j \cos^2(K_j/2)$. For $|K_j| \leq \pi/2$, concavity gives the exact complete free edge $B_a(K) = 2c_a R$, attained at all eight $q_j = \pm\pi/2$. The four-spin singlet resolvent is

$$G_K(q, E) = \frac{E(E^2 - 4c_a^2 e^2)}{E^4 - 4c_a^2(d^2 + e^2)E^2 + 16c_a^4 (d \cdot e)^2}.$$

At $E = B_a(K) + z$, put $t = E/(2c_a)$, $\eta = t^2 - R^2$ and $V = \sum_j \cos K_j \cos^2 q_j$. Equivalently,

$$g_K(E) = \frac{t}{2c_a} \left\langle \frac{1 - e^2/t^2}{V + \eta + (d \cdot e)^2/t^2} \right\rangle.$$

Let $\kappa = \max |K_j| \le 1/2$ and $h = a^2 z \le 1/8$, with $z \ge 0$. Writing $Q = \sum_j \cos^2 q_j$, the denominator lies between $(1 - \kappa^2/2)(Q + h + h^2/12)$ and $Q + h + h^2/12$; $e^2 \le \kappa^2 Q/4$ and $(d \cdot e)^2/t^2 \le e^2$. The complete-torus integral and prefactor therefore give

$$|g_K(B_a(K) + z) - g_0(6/a^2 + z)| \le 3a^2\kappa^2.$$

Combining this with the exterior threshold estimate proves

$$\left| \frac{g_K(B_a(K) + z) - a^2 W_0/2}{a^3} + \frac{\sqrt{z}}{\pi} \right| \le a(11z + 3P_{\max}^2), \qquad |P_j| \le P_{\max}.$$

Under $256a(\gamma^2 + P_{\max}^2) \le \gamma$, the pole denominator has opposite signs at $z = \gamma^2/4$ and $z = 4\gamma^2$. Strict monotonicity of the exterior Stieltjes transform supplies exactly one upper pole $E_a(K) = B_a(K) + \delta_a(K)$, uniformly on that momentum cube:

$$\frac{\gamma^2}{4} \le \delta_a(K) \le 4\gamma^2, \qquad |\delta_a(K) - \gamma^2| \le \frac{22}{7} a\gamma(132\gamma^2 + 9P_{\max}^2).$$

This does not invoke the older, stronger sufficient condition $U > B_a$.

9.32 An explicit bound-band wavepacket comparison.

The same global J_a has bound energy $\epsilon_a(P) = 6/a^2 - B_a(aP) - \delta_a(aP)$. Rationalizing B_a and bounding the sine remainder gives

$$\sup_{|P_j| \le P_{\max}} \left| \epsilon_a(P) - \left(\frac{|P|^2}{4} - \gamma^2 \right) \right| \le \mathcal{E}_a := \frac{a^2 P_{\max}^4}{4} + \frac{22}{7} a\gamma(132\gamma^2 + 9P_{\max}^2).$$

In particular, subtracting $B_a(K)$ separately at each momentum would incorrectly remove the center-of-mass kinetic term.

Let $R_K = (E_a(K) - H_{0,K})^{-1}$ and $\psi_{a,K} = R_K d/\langle d, R_K^2 d\rangle^{1/2}$. Its positive contact overlap fixes its phase, and the pole equation makes it a normalized, measurable, exchange-antisymmetric eigenvector. With consistent measures $dK/(2\pi)^3$ and $dP/(2\pi)^3$, the formula

$$(W_a f)(K, k) = a^{-3/2} f(K/a) \psi_{a,K}(k)$$

defines an isometry on packets supported in the stated cube. Its range reduces the two-body source action; the remaining source states are retained. If h_{pair} multiplies by $|P|^2/4 - \gamma^2$, then for arbitrary untouched references,

$$\|e^{-itJ_a} W_a - W_a e^{-ith_{\text{pair}}}\| \le \min\{2, |t|\mathcal{E}_a\}.$$

This proves coherent moving-pair propagation with an explicit rational spacing schedule. The limiting inertial mass is twice that of a single free edge in these model coordinates; neither an SI mass nor a relativistic rest energy is identified. The exact bounds, independent four-spin and torus checks, and wrong-subtraction/time-orientation controls are retained in `model/moving_bound_pair/README.md`.

The result concerns one composite sector. Nonzero-K on-shell scattering, a physical vacuum/detector map, many-pair interactions, relativistic massive/massless sectors and Einstein dynamics remain separate. The earlier fixed-zero $c = v/a$ result also remains valid: $\int_{-M}^{M} |S - I|^2 dE \leq 1024M^2/(63c)$ for arbitrary real finite $U(a)$ and $M \leq c/4$, so every fixed energy-normalized packet becomes transparent while tuned shrinking resonances can persist. Its scaling and energy comparison differ explicitly from the bound-band construction.

9.33 Identified model energies and the physical particle boundary.

The model energies and inertial mass are identifiable. The cube has eight exactly characterized interacting energies, each reached by the local probe with positive weight. The moving pair has $\epsilon(P) = |P|^2/4 - \gamma^2$ for the reflected generator $J_a = 6/a^2 - H_a$, hence binding energy γ^2 and inertial mass 2 in those model units. If physical energy and length units $E_{\mathrm{u}}, L_{\mathrm{u}}$ are independently attached and $p_{\mathrm{phys}} = \hbar P/L_{\mathrm{u}}$, then $M = 2\hbar^2/(E_{\mathrm{u}}L_{\mathrm{u}}^2)$ and $E_{\mathrm{bind}} = E_{\mathrm{u}}\gamma^2$, with $E_{\mathrm{u}}T_{\mathrm{u}} = \hbar$ for the common clock. These conversion formulas do not select the units or fix an absolute rest-energy offset. Binding energy is not total rest mass. For the cube, the following rounded displays have exact rational energy and weight intervals in the receipt, with the original contact unit $U = 1$:

H energy	$E - 3$	Probe weight
1.816498920	−1.183501080	0.002465366
1.860537501	−1.139462499	0.005644624
1.955137632	−1.044862368	0.005881219
2.108692846	−0.891307154	0.007137664
2.167639734	−0.832360266	0.004528123
3.015861346	+0.015861346	0.421577939
3.030769653	+0.030769653	0.415789141
3.044862368	+0.044862368	0.136975924

Weights refer to the normalized centered density probe; the phase-kicked target has 15/16 times these weights plus an elastic weight 1/16 at energy 3. The lowest displayed line is not the $N = 6$ ground: $-3\sqrt{3}/8 \leq E_{0,N=6} \leq -\sqrt{3}/8$, by the free one-particle bound and a polarized zero-contact variational state. These are finite energy levels and signed transitions, not eight rest masses. The physical species assignment count remains zero. On the cube the density probe, its phase-kicked reference and all spectral components remain in $N = 6$: the transition operator is number-neutral and even under $\Gamma = (-1)^N$. An isometric dictionary intertwining this grading cannot map that response to a single odd fermion, since an odd creation operator changes parity. This does not exclude other odd model excitations or a fermion–antifermion response; even parity alone does not establish Bose exchange or Lorentz spin. Hidden spin and model number have not been identified with physical spin and electric charge. The physical target conventions are the PDG electron and quark quantum numbers and relativistic kinematics.[4]

The supplied cube reference also fails the vacuum ground-state spectral condition for its own H. Its positive density-probe spectral measure has five negative and three positive frequencies $\omega = z/16 - 1$; all eight weights are strictly positive by the exact Krylov certificate. In particular $z_- = 8 - 2\sqrt{19}$ has weight $(1 - 4/\sqrt{19})/14 > 0$ and gives an $N = 6$ eigenenergy $5/2 - \sqrt{19}/8 < 2$, below the reference energy 3. A number-preserving unitary swapping these two states lowers the energy, so the reference is not passive for the full field unitary algebra. This is a mathematical witness, not an executed extraction protocol: the actual local phase preserves the mean energy.

[4] `https://pdg.lbl.gov/2025/listings/rpp2025-list-electron.pdf`; `https://pdg.lbl.gov/2025/reviews/rpp2025-rev-quark-model.pdf`; `https://pdg.lbl.gov/2025/reviews/rpp2025-rev-kinematics.pdf`.

These are frequencies of a positive greater correlation, not the two-sided frequencies of a retarded commutator. A scalar or common number chemical potential leaves the gaps unchanged, and time reversal still leaves mixed signs.

The moving-pair limit $\epsilon(P) = |P|^2/4 - \gamma^2$ is a genuine controlled nonrelativistic band. For any fixed affine energy map $E = \alpha\epsilon + \beta$, $\alpha \neq 0$, and fixed invertible linear momentum map $p = LP$, the residual $E^2 - c^2|p|^2 - m^2c^4$ has nonzero quartic coefficient $\alpha^2/16$ along a coordinate line. It cannot vanish on an open momentum set. Thus these maps do not turn this limit into an exact massive or massless Lorentz shell. A low-momentum approximation remains possible: after matching constant and quadratic terms, the difference between $E_0 + Ar^2$ and $\sqrt{E_0^2 + 2E_0Ar^2}$ is $A^2r^4/(E_0 + Ar^2 + \sqrt{E_0^2 + 2E_0Ar^2})$, for rest energy $E_0 > 0$ and $A > 0$.

`model/particle_identification` binds these scoped tests and the remaining vacuum, gauge-charge, spin, clock, detector and mass-shell requirements. The earlier dimer electron mass was a calibration input, not an independent mass prediction; its retrospective equality comparison does not identify the new cube or pair modes. No mass fitting, new eligible physical comparison, or whole-program exclusion is added.

The preceding identity audit leaves a relativistic construction open. The following extension constructs a massless finite-particle limit on the growing lattice. Its carrier and reference remain distinct from the fixed protected cube, and its limiting interaction vanishes.

9.34 A controlled massless relativistic sector of the unchanged action.

The same Weyl/contact family admits a simultaneous eight-valley continuum comparison on every fixed finite-particle momentum domain. Set

$$H_a = d\Gamma(h_a) + U(a)D, \qquad h_a(k) = -\frac{v}{a}\sum_{j=1}^{3}\sigma_j \sin k_j, \qquad D = \sum_x n_{x\uparrow}n_{x\downarrow},$$

with fixed $v > 0$ and finite $U(a) \geq 0$. This declares the scaling $c = v/a$; it adds no mass term, energy subtraction or time reversal. For $\nu = \pi b$, $b \in \{0,1\}^3$, let $C = [-\Lambda, \Lambda]^3$ and $a\Lambda < \pi/2$. With both Fourier measures normalized by $(2\pi)^{-3}$,

$$(W_a f)(\nu + ap) = a^{-3/2} f_b(p), \qquad W_a : \bigoplus_b L^2(C, \mathbb{C}^2) \longrightarrow L^2(\mathbb{T}^3, \mathbb{C}^2)$$

is an isometry, with zero output outside the eight disjoint patches. There is no additional factor $1/\sqrt{8}$. Its exterior powers give an isometry $W_a^{\leq N}$ on all particle numbers through a fixed N, with the vacuum and every valley retained.

Writing $s_j = (-1)^{b_j}$ and $\chi_b = -s_1s_2s_3$, the limiting blocks are

$$h_b(p) = -v\sum_j s_j\sigma_j p_j, \qquad E_\pm(p) = \pm v|p|, \qquad E^2 - v^2|p|^2 = 0.$$

All eight continuum masses are zero. Four blocks have each chirality. The proper frame $R_b = -\chi_b \operatorname{diag}(s)$ has a constant $SU(2)$ lift and spin $S_{b,j} = R_{b,jj}\sigma_j/2$, so $h_b = 2\chi_b v S_b \cdot p$. The positive-energy helicity is $\chi_b/2$; the negative-energy helicity is $-\chi_b/2$. At $p = 0$ the two spin states are degenerate and helicity is undefined. In these frames an opposite pair has $\alpha_j = \operatorname{diag}(-\sigma_j, \sigma_j)$. With off-diagonal identity β, the matrices $\gamma = (\beta, \beta\alpha_1, \beta\alpha_2, \beta\alpha_3)$ satisfy the Clifford relations for signature $(1,-1,-1,-1)$, checked exactly. Thus the limiting differential equation is massless

Weyl/Dirac in coordinate $x^0 = vt$, not merely a fitted linear dispersion.[5] The fixed cutoff and the lattice regulator are not Lorentz invariant; neither the valley pairing nor this spin frame selects physical species.

9.35 The full interacting evolution has an explicit comparison bound.

The sine remainder gives, uniformly on C,

$$\|h_a(\nu + ap) - h_b(p)\| \le \frac{va^2}{6} \sum_j |p_j|^3 \le \frac{va^2\Lambda^3}{2}.$$

For $a\Lambda \le 1$ and $p \neq 0$, both spectral-projector errors are at most $a^2\Lambda^2/6$. The squared-energy defect lies between $-v^2a^2 \sum_j p_j^4/3$ and zero. No projector at the degenerate node is assigned.

Let Ω_a project onto the initial Fock band and $d = \lfloor N/2 \rfloor$. On the infinite lattice the site-diagonal band weight is exactly $f_a = 8(a\Lambda/\pi)^3$. The positive countable partial sums defining D are bounded by d, hence converge strongly and satisfy $D^2 \le dD$. On band states each annihilation has norm at most $\sqrt{f_a}$ and preserves the band for the remaining particles. Summing its squared norms gives $\Omega_a D \Omega_a \le f_a d$, separately on each up/down number sector, and therefore $\|D\Omega_a\| \le d\sqrt{f_a}$. This proves the infinite-volume attachment of the finite dilute-band estimate without substituting a continuum fraction for a finite count.

Duhamel's formula now yields

$$\|e^{-itH_a} W_a^{\le N} - W_a^{\le N} e^{-itd\Gamma(h_{\mathrm{W}})}\| \le \min\{2, |t|\epsilon_a\},$$
$$\epsilon_a = \frac{Nva^2\Lambda^3}{2} + |U(a)|d\sqrt{8(a\Lambda/\pi)^3}.$$

The left evolution is the full source Hamiltonian: no band projection is inserted and every outgoing contact state remains present. The estimate also holds with arbitrary untouched references; it bounds half trace distance and individual probability errors, capped by one. For fixed N, Λ, v and bounded times it tends to zero whenever $U(a) = o(a^{-3/2})$. At $N \le 1$ the contact is exactly zero for every $U(a)$. Thus the proved limit is free; the critical pair scaling $U \sim a^{-2}$ and finite-density limits are outside this conclusion. For $N = 2$, $\Lambda = v = U = t = 1$ and $a = 1/64$, the complete-vector bound is less than 0.001237 and the projector bound is $1/24576$.

The finite source check retains all 128 one-particle Fourier columns on a periodic 4^3 cube. An eight-valley two-particle state there has exact $\|D\psi\|^2 = 1/8$ and $\|(1 - \Omega)D\psi\|^2 = 7/64$, so the contact escape is real. Propagation on all 8,128 two-particle states at $t = 1/8$ gives a numerical vector difference about 0.0441641, below the rigorous $\sqrt{1/8}/8$. These matrix diagnostics and exact source identities do not constitute an additional native preparation or detector execution.

9.36 Resulting energies and the physical boundary.

At $v = 1$, $a = 1/64$, every valley has the following energies; both signs are retained, and only their common positive magnitude is displayed. The receipt gives rational enclosures for the rounded lattice values.

[5] For the standard Weyl equation and helicity convention, see D. Tong, *Quantum Field Theory*, Sections 4.4 and 4.7, `https://www.damtp.cam.ac.uk/user/tong/qft/qfthtml/S4.html`.

p	lattice $\lvert E_a\rvert$	continuum $\lvert E\rvert$	continuum mass
$(0,0,0)$	0	0	0
$(1,0,0)$	0.999959310393	1	0
$(1,1,0)$	1.414156018578	$\sqrt{2}$	0
$(1,1,1)$	1.731980331101	$\sqrt{3}$	0

These are original-generator model energies, without an SI calibration. The finite-spacing mass-shell defect is a regulator error, not a tachyonic mass. This growing-lattice branch is distinct from the protected eight-site cube: neither its repair invariance nor the identity of its eight response lines is transferred to the valleys. The empty Fock state is not a ground state: already at finite spacing, a one-particle mode at $(\pi/2, 0, 0)$ has energy $-c$, with zero contact for every U. This compares different number sectors; it is not a number-preserving extraction protocol from the empty sector. A filled vacuum, its interacting stability and a physical antiparticle dictionary remain unconstructed. A single 2×2 constant Hermitian mass anticommuting with all three kinetic Pauli matrices is exactly zero; this limited obstruction does not exclude valley coupling or many-body mass generation. The package `model/relativistic_sector_audit` therefore adds a controlled massless relativistic dynamical sector, not massive physical species, a photon assignment, an interacting relativistic vacuum theory or Einstein dynamics.

The preceding continuum construction does not itself execute a detector. The following finite periodic construction attaches model flavor and energy labels to a source-compiled occupation readout, and compares the resulting massless sector with observed particle data.

9.37 A detector identifies the model's propagating Weyl flavors.

The massless sector now has a concrete model-particle readout. On the periodic 8^3 lattice at $c = 1/8$, retain all 1,024 physical CAR modes of the original Weyl/contact action. For each $b \in \{0,1\}^3$, choose momentum labels $\ell_b = (4b_x, 4b_y, 4b_z + 1)$, with $k_b = \pi\ell_b/4$. For component eigenvalue $r = \pm1$ of σ_z, the normalized mode is

$$\psi_{b,r}(x,s) = \frac{e^{ik_b\cdot x}}{\sqrt{512}}\delta_{s,(1-r)/2}, \qquad H\psi_{b,r} = E_{b,r}\psi_{b,r}, \qquad E_{b,r} = -r(-1)^{b_z}\frac{\sqrt{2}}{16}.$$

The contact term is exactly zero on the entire one-particle sector for every supplied $U \geq 0$. These sixteen modes are orthonormal, checked in Gaussian integers with amplitudes $(A+\sqrt{2}B)/32$. Their rank-one projectors satisfy $H\Pi_{b,r} = E_{b,r}\Pi_{b,r}$; the remaining one-particle projector commutes with H. Thus a label gives an actual source-generator energy, not a fitted interpretation of a peak. All modes have $N = 1$, odd CAR parity and fermionic exchange statistics from the supplied CAR algebra.

The operational species are eight Weyl flavors indexed by measured valley momentum. Their common energy magnitude is $\sqrt{2}/16 \simeq 0.0883883476483$ in original model units; it alone does not distinguish the flavors. With $\chi_b = -(-1)^{b_x+b_y+b_z}$, the positive-energy branch has cone helicity $\chi_b/2$ and the negative branch has the opposite helicity.

flavor b	momentum labels ℓ_b	positive-branch component	positive cone helicity
000	$(0,0,1)$	1	$-1/2$
001	$(0,0,5)$	0	$+1/2$
010	$(0,4,1)$	1	$+1/2$
011	$(0,4,5)$	0	$-1/2$
100	$(4,0,1)$	1	$+1/2$
101	$(4,0,5)$	0	$-1/2$
110	$(4,4,1)$	1	$-1/2$
111	$(4,4,5)$	0	$+1/2$

Here helicity uses the explicit proper valley frame $R_b = -\chi_b \operatorname{diag}((-1)^{b_j})$; it is $(-1)^{b_x+b_y} r/2$ on this family. The raw detector measures momentum and CAR component, so bare $\sigma_z/2$ is not silently identified with this cone spin or with physical particle spin. The total directional current $J_z(k) = -c \cos k_z \, \sigma_z$ has eigenvalue $E_{b,r}$ here, with nonzero current $\pm\sqrt{2}/8192$ on every forward z bond. Current is stationary on these selected modes; $[H, J_z] \neq 0$ at generic momenta. The witness is propagating flux, even for a uniform-density plane wave.

9.38 The Fourier readout is compiled with complete native errors.

The detector applies the fermionic extension of $F[k,x] = e^{-2\pi i k\cdot x/8}/\sqrt{512}$ separately to each component, then copies all 1,024 occupations to fresh pointers. Its output includes every binary occupation string, the vacuum, all other one-particle addresses, multiple occupations and every export abort. On the chosen sixteen effects it measures the energies above; the other 1,008 one-particle records remain present without that energy assignment. On higher-number inputs these same records measure mode occupations, not the total interacting energy.

An explicit three-axis FFT uses 4,608 two-mode butterflies and 768 bit-reversal transpositions. Each butterfly is a mode phase followed by the native hopping word with coefficient $-i$ and duration $\pi/4$. All nonadjacent operations use graded FSWAP routes, including their double-occupation sign. Exact arithmetic in $\mathbb{Z}[\zeta_8]/(\zeta_8^4 + 1)$ proves the 64 entries and 64 Gram sums of the one-axis transform; tensoring gives the full mode map. One transform requests 1,260,800 adjacent FSWAPs and 32,909,824 native pulses. Native excitation from the vacuum, inverse-transform preparation, forward-transform readout and all occupation pointers total 65,822,721 formal pulses. Original-H propagation is a separate specified model-law service, not an uncounted synthesized field word.

The complete literal-source error for preparation and detection is

$$\epsilon_{\text{source}} < 8.31 \times 10^{-9}.$$

The estimate retains all first-export failures and is valid with untouched references. It does not assume that literal native maps are exactly Gaussian or number preserving. Unitary original-H propagation on the full Fock space does not amplify preparation error. Therefore, at every model time, a prepared catalog member produces its correct native label with unconditional probability at least $1 - \epsilon_{\text{source}}$; all wrong, outside, vacuum, multiple-occupation and abort outputs together have probability at most ϵ_{source}.

The replay executes the ideal Gaussian comparison on all 1,024 one-particle rows and all sixteen inputs. At time eight its complete output-vector discrepancy from the exact spectral prediction is about 1.94×10^{-15}, separate from the rigorous native bound. A seeded 65,536-record illustration retains all sixteen label counts and every zero outside count for the exact uniform input mixture. These are model-simulated counts, not a literal 2^{1024}-row native execution or laboratory data. No survival normalization is used.

9.39 Mass readout and comparison with measured particles.

For the periodic family $L = 8 \cdot 2^m$, $a = 2^{-m}$ and $c = (1/8)/a$, the model period stays eight and the momentum offset stays $p_z = \pi/4$. The exact branch energies satisfy

$$E_\pm(a) = \pm\frac{\sin(a\pi/4)}{8a} \longrightarrow \pm\frac{\pi}{32}, \qquad m_{\text{continuum}} = 0.$$

This is a direct finite-torus consequence of the original spectrum. The finite-spacing mass-shell defect is a regulator correction, not a tachyonic mass. No SI scale is assigned.

A separate retrospective comparison uses fixed PDG 2025 targets, which never enter the source or detector calculation:

physical target	measured mass in MeV/c^2	present continuum mass
electron	$0.51099895000 \pm 0.00000000015$	0
muon	$105.6583755 \pm 0.0000023$	0
tau	1776.93 ± 0.09	0

No common positive unit conversion maps zero to these positive masses; equal finite-momentum energies are not a rest-mass hierarchy.[6] The photon has spin one, so direct spin-preserving identification with these cone-helicity-1/2 modes also fails. Three such massless flavors give zero mass-squared differences, outside the nonzero ranges of the specified three-neutrino fit in PDG Table 14.7.[7] A single massless neutrino or a composite photon in another sector is not excluded. The model's Weyl flavors are now operationally identified; direct identifications with measured physical species remain zero. The empty reference is not a ground state, and the negative branch is retained rather than renamed as a physical antiparticle. Interacting vacuum stability, mass generation, physical charges and the empirical dictionary remain additional work.

The missing ground-reference and mass-generation steps admit further constructions with explicit domains. The original cube has a global interacting ground, and its one-dimensional restriction has an interaction-generated canonical charge gap. A specified refinement of that restriction also permits a controlled relativistic mass calculation. The following results retain the reference and coupling choices rather than combining these distinct domains into an assumed physical sector.

9.40 A global interacting ground on the original cube.

The unchanged open $2 \times 2 \times 2$ Weyl/contact source, with $c = 1/8$ and $U = 1$, now admits a certified comparison across its entire 65 536-dimensional Fock space:

$$E_G = -\frac{\sqrt{3}}{4}, \qquad N_G = 4, \qquad \dim\ker(H - E_G) = 5.$$

The five states form one hidden-spin $S = 2$ multiplet. No chemical potential, energy recentering, mass term, or replacement Hamiltonian is introduced. This is a finite interacting ground reference, not a claim of a unique or Lorentz-invariant infinite-volume vacuum.

The existing onsite unitary frame sends the original hopping to the spin-independent signed cube adjacency $A/16$, where

$$A = \begin{pmatrix} 0 & B \\ B^T & 0 \end{pmatrix}, \quad BB^T = 3I_4, \quad K = \begin{pmatrix} \sqrt{3}I_4 \\ -B^T \end{pmatrix}, \quad K^*K = 6I_4.$$

[6] PDG 2025 listings: `https://pdg.lbl.gov/2025/listings/rpp2025-list-electron.pdf`; `https://pdg.lbl.gov/2025/listings/rpp2025-list-muon.pdf`; `https://pdg.lbl.gov/2025/listings/rpp2025-list-tau.pdf`.

[7] The photon listing is `https://pdg.lbl.gov/2025/listings/rpp2025-list-photon.pdf`. The selected normal-ordering fit gives $\Delta m^2_{21} \in [6.92, 8.05] \times 10^{-5}\,\text{eV}^2/c^4$ and $\Delta m^2_{32} \in [2.388, 2.531] \times 10^{-3}\,\text{eV}^2/c^4$; these are reported 3σ fit ranges, not exact bounds: `https://pdg.lbl.gov/2025/reviews/rpp2025-rev-neutrino-mixing.pdf`.

Filling the four negative orbitals $K/\sqrt{6}$ with hidden spin up gives an explicit normalized state G. Its occupation coefficients belong to $\mathbb{Q}(\sqrt{3})$, it has no doublons, and the original physical-frame eigenvalue equation is checked exactly. With $\lambda = \sqrt{3}/16$, the $N = 4$ bound $H \geq -4\lambda$ is saturated precisely by negative-band states annihilated by every local pair annihilator. On the full 70-dimensional four-particle negative-band space, the stacked contact map has spin-weight block dimensions $1, 16, 36, 16, 1$ and ranks $0, 15, 35, 15, 0$. Nonzero integer minors modulo101 certify the rank lower bounds; five explicit spin-lowered vectors certify the matching upper bounds. This proves the complete five-dimensional ground kernel in that number sector.

For $N \leq 3$, positivity gives $H \geq -N\lambda > -4\lambda$. For $N \geq 9$, the occupancy bound $D \geq N - 8$ gives $H \geq -N\lambda + N - 8 > 0$. For $N = 5, 6, 7, 8$, hidden $SU(2)$ symmetry ensures that every eigenvalue occurs in the balanced spin weight. Those block dimensions are $1568, 3136, 3920, 4900$. A numerical factorization proposes an integer matrix Q; only the following subsequently checked integer identity enters the proof:

$$H = 2^{-40} QQ^T - \frac{27}{64} I + R, \qquad \|R\|_2 \leq \|R\|_\infty < \frac{1}{512}.$$

All integer product and row-sum overflow bounds are checked before evaluation. Since QQ^T is positive semidefinite for every proposal, the argument requires no trusted floating eigenvalue, factorization error estimate, or nonzero-pivot premise. The exact comparison

$$\frac{\sqrt{3}}{4} > \frac{441}{1024} > \frac{217}{512}$$

then proves that every $N \neq 4$ lies more than $7/1024$ above E_G. This last bound concerns charged sectors; no numerical lower bound for neutral excitations inside $N = 4$ is asserted. Any density matrix supported on this finite ground space is passive for the stated H.

9.41 Positive charged spectral data above this ground.

At the origin the onsite frame is the identity, so the probes below are the original physical CAR modes. Their exact local spectral data are

probe or component	N	weight	$\omega = E - E_G$
$c_\uparrow G$	3	1/2	λ
$c_\uparrow^\dagger G$	5	1/2	λ
$c_\downarrow G$	3	0	absent
$S = 5/2$ component of $c_\downarrow^\dagger G$	5	1/10	λ
$\eta = 5/2$ component of $c_\downarrow^\dagger G$	5	1/10	$1 + \lambda$

Thus the resolved model gaps are approximately 0.108253175473 and 1.108253175473 in source energy units. They are finite-volume gaps, not relativistic rest masses. The remaining down-addition weight is $4/5$, with strictly positive spectral support and nonzero variance.

More explicitly, set $v = c_\downarrow^\dagger G$, with $\|v\| = 1$. The exact vectors

$$v_s = \frac{1}{5} S^- S^+ v, \qquad v_\eta = \frac{1}{5} \eta^+ \eta^- v, \qquad v_r = v - v_s - v_\eta$$

are mutually orthogonal and have squared norms $1/10, 1/10, 4/5$. The first two obey the original Hamiltonian eigenvalue equations at -3λ and $1 - 3\lambda$. Consequently the complete greater addition measure satisfies

$$\mu_\downarrow = \frac{1}{10}\delta_\lambda + \frac{1}{10}\delta_{1+\lambda} + \frac{4}{5}\mu_r, \qquad \operatorname{supp} \mu_r \subset (7/1024, \infty).$$

The displayed 1/10 values are exact symmetry-resolved contributions; without further resolution of μ_r, coincident residual atoms are not excluded. All moments below are checked over $\mathbb{Q}(\sqrt{3})$:

$$\mathbb{E}\omega = \frac{1}{2}, \qquad \mathbb{E}\omega^2 = \frac{131}{256}, \qquad \mathrm{Var}(\omega) = \frac{67}{256},$$

$$\mathbb{E}_r\omega = \frac{1}{2} - \frac{\sqrt{3}}{64}, \qquad \mathbb{E}_r\omega^2 = \frac{131}{256} - \frac{\sqrt{3}}{64}, \qquad \mathrm{Var}_r(\omega) = \frac{1069}{4096} > 0.$$

The contact expectation in v is 1/2; deleting that interaction changes its mean gap from 1/2 to zero. This channel therefore probes the actual contact interaction. Addition and removal carry model number changes $+1$ and -1 and odd CAR grading relative to G; electric charge and physical Lorentz spin do not follow from these labels.

9.42 Attachment to the declared repair process.

The existing four-dimer repair completion has24 active columns, all entirely in $N = 6$. Its complete130-outcome dictionary therefore acts on $N = 3 \oplus N = 4 \oplus N = 5$ as zero for every active Kraus outcome and as identity for the outside projector. This6748-dimensional statement retains cross-number coherences and arbitrary reference systems. Since the original H preserves number, it also survives every finite interleaving with the ordinary field evolution. This is passive outside-service protection, not active correction or autonomous preparation of the ground.

The new finite ground removes the negative-frequency objection that applied to the earlier excited reference. It does not retroactively turn that earlier preparation into ground-state preparation. A native source and detector for G, the unresolved charged spectrum, a controlled growing-volume interacting vacuum, relativistic massive poles, physical dictionaries and units, and the geometric Einstein premises remain separate obligations. The exact certificate and negative controls are retained in `model/interacting_vacuum`.

9.43 An interaction-generated charge scale in the original one-dimensional law.

There is a positive many-body charge threshold without inserting a Dirac mass. On the unchanged periodic Weyl/contact geometry $(L, 1, 1)$, $L = 4m$, the onsite frame $G_x = (-1)^x (i\sigma_x)^x$ satisfies $G_x(i\sigma_x)G_{x+1}^\dagger = -I$ and has period four. Its CAR lift fixes local doublons, so the original action is exactly the repulsive Hubbard chain with $t = c/2 = 1/16$ and $U = 1$. Four exact local frame identities, source-matrix checks through $L = 16$, and an independent 256-state full-Fock witness bind this join. Unlike the earlier logarithmic-reference alternative Hamiltonian, no coupling is reassigned.

For the *half-filled canonical ground sector*, the imported Lieb–Wu charge-gap theorem gives

$$\Delta_c = \lim_{L\to\infty} [E_L(L+1) + E_L(L-1) - 2E_L(L)] = U - 4t + 8t \sum_{k\geq 1} \binom{1/2}{k} (2t/U)^{2k-1} \eta(2k-1).$$

Here the series is used only for $2t/U < 1$. Twelve binomial terms and 96-term rational Euler bounds for each Dirichlet eta value enclose $\Delta_c = 0.7715517157380375\ldots$ with width below $1.67\,10^{-25}$. The binomial tail is bounded by $8t|\binom{1/2}{K+1}|(2t/U)^{2K+1}/[1-(2t/U)^2]$; the nonnegative Euler coefficients give tail 2^{-96}. Floating quadrature is an independent check only.

Particle–hole symmetry fixes the charge costs for the original H:

$$E_+ = \frac{U + \Delta_c}{2} = 0.885775857869\ldots, \qquad E_- = \frac{\Delta_c - U}{2} = -0.114224142131\ldots.$$

Thus half filling is a specified canonical reference, not the original $\mu = 0$ grand-canonical vacuum. The symmetric threshold $\Delta_c/2 = 0.3857758578690188\ldots$ uses the explicitly centered $K = H - (U/2)N$; Δ_c itself is invariant under any common $-\mu N$. The spin sector remains gapless, contact still vanishes in the one-particle sector, and no preparation or complete-repair preservation of this many-body reference is established by the gap theorem.

9.44 The fixed-interaction continuum boundary.

The equivalent positive integral for the charge gap and the elementary K_1 integral bounds imply, for every $t, U > 0$,

$$\Delta_0 \le \Delta_c \le \Delta_0 \frac{1 + 3/(8A)}{1 - e^{-2A}}, \qquad A = \frac{2\pi t}{U}, \qquad \Delta_0 = \frac{8}{\pi}\sqrt{Ut}\, e^{-A}.$$

Consequently fixed U with $t = v/(2a)$ has $\Delta_c \to 0$ as $a \to 0$: this lattice gap does not itself supply a nonzero continuum mass. A sufficient additional running family is

$$a(A) = \frac{4ve^{-A}}{M\sqrt{2\pi A}}, \qquad t(A) = \frac{v}{2a(A)}, \qquad U(A) = \frac{\pi v}{a(A)A},$$

for a *chosen* $M, v > 0$. It gives $\Delta_0/2 = M$ exactly and $1 \le \Delta_c/(2M) \le [1 + 3/(8A)]/[1 - e^{-2A}] \to 1$. This is a quantitative charge-threshold limit, not a derivation of M from the fixed source, a complete relativistic scaling theory, or a $3+1$-dimensional massive species. The receipt `model/mass_generation_audit/results.json` keeps these reference, coupling-family, and dimensional premises explicit.

The preceding charge-threshold result alone does not prove a mass shell. The next construction supplies energy and momentum together, with a controlled relativistic limit along an explicitly chosen coupling family.

9.45 A controlled relativistic charge mass on an explicit running family.

The preceding charge gap can be promoted to a relativistic dispersion *in one spatial dimension on a specified coupling trajectory*. The source-bound half-filled Hubbard reference and centered generator $K = H - (U/2)N$ remain explicit. The staggered particle–hole map $d_{x,\downarrow} \mapsto (-1)^x d^\dagger_{x,\downarrow}$, with $d_{x,\uparrow}$ unchanged, preserves hopping and reverses the centered interaction. A complete 256-state CAR witness checks this duality and the translation identity $PTP^\dagger = -(-1)^{N_\downarrow}T$. Consequently the attractive spin dispersion becomes the repulsive charge dispersion, with the appropriate momentum-carrier shift. Here p is relative to that low-energy branch; the onsite source frame also carries its specified uniform internal rotation. The even-ring SO(4) selection rules are retained: an elementary thermodynamic quasiparticle is not automatically an isolated finite-ring state or a local-CAR pole.

Writing $A = 2\pi t/U$, $\theta = A\sin k$, $|k| < \pi/2$, and $v = 2ta$, the imported exact dressed dispersion has the source-normalized form

$$E = \frac{8t}{\pi} \sum_{\substack{m \ge 1 \\ m \ \mathit{mod} \ d}} \frac{K_1(mA)}{m} \cosh(m\theta), \qquad vp = \frac{8t}{\pi} \sum_{\substack{m \ge 1 \\ m \ \mathit{mod} \ d}} \frac{K_0(mA)}{m} \sinh(m\theta).$$

On $a(A) = 4ve^{-A}/[M\sqrt{2\pi A}]$, $t = v/(2a)$, $U = \pi v/(aA)$, the leading Bessel coefficient is exactly M. For $A \ge 1$ and $|\theta| \le \Theta < A$, put

$$q = e^{-2(A-\Theta)}, \qquad r_E = \frac{1 + 3/(8A)}{1 - q} - 1, \qquad r_p = \max\{1/(8A), q/(1-q)\}.$$

The Gamma-integral tangent bounds $1-1/(8z) \le K_0(z)/b(z) \le 1$ and $1 \le K_1(z)/b(z) \le 1+3/(8z)$, with $b(z) = \sqrt{\pi/(2z)}e^{-z}$, and the positive odd-mode tail prove

$$|E/M - \cosh\theta| \le r_E \cosh\Theta, \qquad |vp/M - \sinh\theta| \le r_p \sinh\Theta,$$

$$\left|(E^2 - v^2p^2)/M^2 - 1\right| \le \cosh^2\Theta(2r_E + r_E^2) + \sinh^2\Theta(2r_p + r_p^2) \;\longrightarrow\; 0.$$

At $\Theta = 1$, the last bound is below 0.034 for $A = 64$ and 0.017 for $A = 128$. Thus the limiting charge-doublet dispersion has $E = M\cosh\theta$, $p = (M/v)\sinh\theta$, and the calculated rest mass is $m = M/v^2$. Exact rational finite-parameter enclosures and independent Bessel/integral controls are retained in `model/hubbard_relativistic_sector/results.json`.

For a concrete trajectory through the original point, take $A_0 = \pi/8$, $v = 1/8$ and $M_0 = e^{-\pi/8}/\pi$. Then $a(A_0) = 1$, $t(A_0) = 1/16$, $U(A_0) = 1$, and

$$M_0 = 0.2149329913552644\ldots, \qquad m_0 = \frac{64e^{-\pi/8}}{\pi} = 13.75571144673692\ldots \quad \text{in model mass units.}$$

This is the mass of the *chosen running trajectory*, not a unique mass selected by the original bare point. Indeed, for any $M > 0$, multiplying $a_M(A)$ by $1 + (M/M_0 - 1)e^{-(A-A_0)}$ gives the same starting point and limiting rest energy M. The initial lattice half-gap is different, and the fixed-U continuum family has zero limiting charge mass. No native ground preparation, repair-invariant charged pole, $3+1$-dimensional massive species, or empirical particle-mass identification follows from this one-dimensional result.

The original three-dimensional action also permits a growing-volume ground construction at fixed spacing. This closes ground-state existence and positive excitation energy without selecting the one-dimensional canonical reference or its coupling continuation. Its local response is then connected to the existing geometry theorem interfaces.

9.46 An original-action infinite-volume ground representation at fixed spacing.

The full three-dimensional Weyl/contact action now admits a translation-invariant, gauge-invariant ground state at its original zero chemical potential, without choosing a particle-number sector. This is a compactness existence theorem, not a computed or natively prepared state. Keep fixed finite $c > 0$, $U > 0$ and the complete quasi-local CAR algebra with two modes per site. The previously proved full-Fock locality estimate supplies the strongly continuous dynamics τ_t. On each even-sided cube Λ_L of side $2L$, take the normalized projector onto the absolute minimum of H_{Λ_L}, extend this even state by the exterior vacuum, and average its translates over all sites of Λ_L. A diagonal subsequence of the bounded local matrix entries defines a state ω; positivity, normalization and CAR compatibility pass to the limit. The vanishing boundary-to-volume ratio makes ω translation invariant.

For a local A, including an odd charged probe, the local derivation $[H, A]$ contains only interaction terms touching its support. Whenever the support and its nearest neighbors lie inside the box,

$$\omega_L(A^*[H, A]) = \operatorname{Tr}\rho_L A^*(H_{\Lambda_L} - E_L)A \ge 0.$$

Bad translates occupy a vanishing fraction. Their real-part lower defect and imaginary magnitude are each bounded by $2\|A\|^2|\operatorname{supp} A|(U + 6c)$, so $\omega(A^*[H, A]) \ge 0$ in the limit. This uses the full ground minimum and would fail for a merely canonical ground with an unrestricted charged probe.

For a radius-R buffer, let $B_R(t)$ denote the existing volume-uniform local-dynamics boundary bound. Comparing both finite and infinite dynamics to this intermediate cube gives, for $\|A\| \le 1$,

$$|\nu_L(\tau_t(A)) - \nu_L(A)| \le \min\Big\{2,\, 2B_R(t) + 2\left[1 - (1 - R/L)^3\right]\Big\}.$$

First $L \to \infty$, then $R \to \infty$, establishes stationarity. The same comparison applies to two-point correlations. Every finite full-ground correlation has a positive spectral measure supported on $[0, \infty)$; compact-time convergence and bounded correlations pass all Schwartz Fourier tests supported on negative energies to the limit. The GNS implementation therefore satisfies

$$U_t \pi(A)\Omega = \pi(\tau_t(A))\Omega, \qquad U_t = e^{itL_\omega}, \qquad L_\omega \geq 0, \quad L_\omega \Omega = 0.$$

The generator measures excitation energy in this representation; no chemical potential has been inserted into the source action.

The state can be chosen at nonzero density. Tiling each even box by the already certified four-particle cube vector gives contact zero, energy $-2\sqrt{3}c$ per eight-site tile, and zero expectation for every cross-tile hopping. Hence $E_L/|\Lambda_L| \leq -\sqrt{3}c/4$. Cartesian difference operators commute and the Pauli mixed terms cancel, giving $h^2 \leq 3c^2 I$ and $H \geq -\sqrt{3}cN$. Thus every such full ground has density at least $1/4$. The exact real-bipartite particle–hole identity $E(2V-N) = E(N) + U(V-N)$ excludes $N > V$ from an absolute minimum when $U > 0$. Spatial averaging before the subsequence passes both bounds:

$$\frac{1}{4} \leq \omega(n_{0\uparrow} + n_{0\downarrow}) \leq 1.$$

Half-filling and a particular density are not assumed or concluded. The original ground energy per site also has a limit e_∞. For every even cube side ℓ, tiling larger boxes gives the computable bracket, conditional on its finite full-ground energy,

$$\frac{E_\ell}{\ell^3} - \frac{3c}{\ell} \leq e_\infty \leq \frac{E_\ell}{\ell^3}, \qquad -\sqrt{3}c \leq e_\infty \leq -\frac{\sqrt{3}c}{4}.$$

The lower surface term counts intertile hopping norms; a product of gauge-invariant tile grounds gives the upper bound. Untiled boundary terms vanish per volume, and the bracket width tends to zero. For the constructed averaged ground, $\omega(h_0) = e_\infty$, where h_0 assigns half of each incident bond to the origin. This supplies an energy-density observable, not a state-convergence rate or rest mass.

The contact interaction remains represented:

$$[H_U - H_0, a_0^* a_2] = U a_0^* a_2 (n_1 - n_3) \neq 0.$$

Every unital GNS representation is injective on each full local matrix algebra, so this quartic operator cannot disappear in representation. This faithfulness of the representation does not assert faithfulness of the ground-state functional. The claim is a nonzero represented interaction derivation, not a scattering or connected-vacuum-correlator theorem.

An exact source countercontrol prevents simply substituting the free filled sea. For a pure Slater projector with $P_{xx} = I_2/2$,

$$\mathrm{Var}(D) = \frac{V}{16} + \sum_{x \neq y} |\det P_{xy}|^2 \geq \frac{V}{16}.$$

If it is also a kinetic eigenstate, its interacting energy variance is $U^2 \mathrm{Var}(D) > 0$. On the original cube, the complete 3844-term negative-band filling has $N = 8$, $\langle D \rangle = 2$ and $\mathrm{Var}(H) = 2/3$ at $U = 1$. The exact checks and proof are retained in `model/thermodynamic_ground/results.json`. For local CAR probes, the same subsequence also supplies the analytic retarded response

$$G_{AB}(z) = -i \int_0^\infty e^{izt} \omega(\{\tau_t(A), B^*\})\, dt, \qquad \mathrm{Im}\, z > 0.$$

Each time tail after T is at most $2\|A\|\|B\|e^{-\eta T}/\eta$ for $\operatorname{Im} z \geq \eta > 0$. Compact-time locality and weak-star convergence, followed by this tail bound, give uniform convergence on compact subsets of the upper half-plane. The unknown finite-state convergence error is retained; no real-axis pole, local inverse or metric is inferred. The result closes fixed-spacing ground-state existence and the positive-energy representation for the actual interacting lattice. It supplies no selected ground functional, convergence rate, native preparation, repair stationarity, uniform gap, Lorentz continuum, physical particle assignment or Einstein dynamics.

9.47 Exact matter response before a geometric identification.

The unchanged open $2\times2\times2$ Weyl/contact cube at $c = 1/8$, $U = 1$ now supplies a state-dependent retarded response, including its exact eliminated-state memory. In the existing onsite hidden-spin frame, write $h = A/16$, where A is the real signed cube adjacency and $A^2 = 3I$. The retained full-Fock ground theorem gives the normalized highest-weight reference G, with four occupied negative-band up orbitals, $E_G = -\sqrt{3}/4$, and a strict gap $\delta = 7/1024$ to every different particle-number sector. The second exact stationary reference is $G_+ = \sqrt{2}\, c_{0\uparrow}^\dagger G$, of energy $-3\sqrt{3}/16$. No chemical potential, fitted mass, or new geometry law is inserted. Physical CAR responses are obtained by the existing onsite unitary frame; hidden spin is not assigned a physical Lorentz or electric-charge meaning.

For either reference ψ of energy E, the complete eight-site fermionic response, for $\operatorname{Im} z > 0$, is

$$\begin{aligned}\mathcal{G}_{xy}(z) &= -i\int_0^\infty e^{izt}\langle\{c_x(t), c_y^\dagger\}\rangle_\psi\, dt \\ &= \langle c_x[z-(H-E)]^{-1}c_y^\dagger\rangle_\psi + \langle c_y^\dagger[z+(H-E)]^{-1}c_x\rangle_\psi.\end{aligned}$$

There is an exact isometric dilation

$$\begin{aligned} V_y &= c_y^\dagger\psi \oplus \overline{c_y\psi}, & V^\dagger V &= I, \\ L &= (H_{N+1}-E) \oplus -\overline{(H_{N-1}-E)}, & \mathcal{G}(z) &= V^\dagger(z-L)^{-1}V. \end{aligned}$$

The conjugate removal space fixes the matrix indices; the present source arrays happen to be real. With $P = VV^\dagger$, $Q = I - P$, define $M_1 = V^\dagger LV$, $W = QLV$ and $B = QLQ|_{\operatorname{ran} Q}$. Exact block elimination gives

$$\mathcal{G}(z) = \left[zI - M_1 - \Sigma(z)\right]^{-1}, \qquad \Sigma(z) = W^\dagger(z-B)^{-1}W, \qquad W^\dagger W = M_2 - M_1^2.$$

Both L and B are self-adjoint; consequently $-\operatorname{Im}\mathcal{G}(z) > 0$ and $-\operatorname{Im}\Sigma(z) \geq 0$ in the upper half-plane. The eight observed columns do not replace the source Hilbert space: their exact complement remains in the memory kernel.

9.48 A computed change of response at fixed microscopic geometry.

For the up channel of both references, the full cyclic identity $LV = Vh$ gives $\mathcal{G}_\uparrow(z) = (z-h)^{-1}$ and $W = 0$. For G, the up dilation retains all 56 addition and 56 removal states. The ground down channel has no removal component and retains all 560 states with $N_\uparrow = 4$, $N_\downarrow = 1$. Exact arithmetic over $\mathbb{Q}(\sqrt{3})$ gives

$$M_{1,\downarrow} = h + \tfrac{1}{2}I, \qquad W_\downarrow^\dagger W_\downarrow = \tfrac{1}{4}I.$$

Thus a static Hartree term alone does not reproduce this interacting response. Indeed $W_x = (n_{x\uparrow} - 1/2)c^\dagger_{x\downarrow}G$; orthogonal down locations and up occupation variance $1/4$ give the displayed Gram matrix directly.

For G_+, the down channel retains its entire 448-state invariant block. In lexicographic order of the eight cube sites, its exact matrices are

$$M^+_{1,\downarrow} - h = \operatorname{diag}(1, \tfrac{2}{3}, \tfrac{2}{3}, \tfrac{1}{2}, \tfrac{2}{3}, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}),$$
$$(W^+_\downarrow)^\dagger W^+_\downarrow = \operatorname{diag}(0, \tfrac{2}{9}, \tfrac{2}{9}, \tfrac{1}{4}, \tfrac{2}{9}, \tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{4}).$$

The first-moment change is exactly the added orbital density, $(1/2, 1/6, 1/6, 0, 1/6, 0, 0, 0)$. Removing the contact term in the explicit $U = 0$ control removes both this Hartree term and the ground down memory, giving $(z - h)^{-1}$. That comparison does not declare G a full free-Fock ground. The excited reference receives no positive-gap assertion.

9.49 The complete local charged spectral measure.

For the original physical origin probe $v = c^\dagger_{0\downarrow}G$, the retained calculation now resolves all 78 distinct positive spectral lines, including the previously unresolved weight $4/5$. A monic degree-78 polynomial p over $\mathbb{Q}(\sqrt{3})$ is admitted by the exact source equation $p(16H)v = 0$. A nonzero 78×78 Hankel determinant modulo 97, with $\sqrt{3} \mapsto 10$, proves the matching cyclic rank lower bound. Exact moments determine a numerator q such that

$$\langle v, (z - 16H)^{-1} v\rangle = \frac{q(z)}{1296\, p(z)}.$$

The denominator 1296 is the squared norm of the inherited integer state numerator. Modular reconstruction and a numerical eigensolver only propose coefficients and brackets. Integer annihilation and rank checks, 78 disjoint strict sign changes, and outward 1024-bit interval arithmetic certify every root and its strictly positive residue $q(x)/(1296p'(x))$. Each charged gap is $(x + 4\sqrt{3})/16$.

The smallest gap is approximately 0.029950938342053226, with weight 0.3130113772155167; the largest is approximately 1.8360744654423855, with weight $2.711810625844545 \times 10^{-10}$. The receipt retains every gap and weight in rational intervals rounded outward to 24 decimal places. Polynomial substitution proves that the two lines $\sqrt{3}/16$ and $1 + \sqrt{3}/16$ each have total weight exactly $1/10$. The remaining 76 lines have total weight $4/5$, leaving zero unresolved weight for this local probe. This is the complete local cyclic measure, not the entire 560-state spectrum, a native detector execution, an infinite-volume pole theorem, or a relativistic mass assignment.

9.50 Controlled low-frequency coefficients.

For the interacting ground down channel, put $K = H_{4,1} - E_G$. The full-ground certificate gives $K > \delta I$. Moreover, $D \le 1$ in this block, the kinetic norm is at most 5λ, and $-E_G = 4\lambda$, with $\lambda = \sqrt{3}/16$; hence $K \le 1 + 9\lambda < 2$. The same bounds hold for its compression B. For $|z| \le r < \delta$ the exact memory has the convergent expansion

$$\Sigma(z) = -\sum_{n=0}^{p} z^n C_n + R_p(z), \qquad C_n = W^\dagger B^{-n-1} W \ge 0,$$
$$\|R_p(z)\| \le \frac{r^{p+1}}{4\delta^{p+1}(\delta - r)}.$$

These are finite gapped frequency coefficients, not a spatial derivative expansion or a relativistic cone. All retarded moments $M_0, \dots, M_4$ are also computed exactly; the corresponding large-frequency remainder is $2^{p+1}/[R^{p+1}(R-2)]$ for $|z| \geq R > 2$.

The static response is computed with an exact residual certificate. For $C = V^\dagger K^{-1} V = -\mathcal{G}_\downarrow(0)$,

$$\|C - (12.9058381883I - 4.9026412320A)\| < 10^{-7},$$
$$\|\Sigma(0) - (-0.3633623064I - 0.0105943755A)\| < 10^{-6}.$$

A numerical solve only proposes a matrix with denominator 2^{40}. Integer arithmetic encloses its full $\mathbb{Q}(\sqrt{3})$ residual; division by the proved δ bounds the solution error. The exact isometry then bounds C, and $C > I/2$ bounds its inverse. An independent rational inverse residual certifies $\Sigma(0) = C^{-1} - M_1$. No numerical tolerance admits the theorem, and exact equality to an affine adjacency polynomial is not inferred.

9.51 The existing geometry interface and its remaining join.

The same-action thermodynamic-ground construction separately supplies analytic local responses along its ground subsequence on compact subsets of $\operatorname{Im} z > 0$. The cube calculation does not supply a rate of convergence to that state, and its finite gap is not transferred to infinite volume. The existing `wave_density_metric` constructor takes a complete local action density $\mathcal{C} = \mu \operatorname{diag}(-w, a)$, with $\mu, w > 0$ and $a > 0$, and recovers $g = \sqrt{-\det \mathcal{C}}\, \mathcal{C}^{-1}$. The existing `source_tetrad` interface in the common matter join then constructs the common coframe and the scalar, Weyl, Dirac and Maxwell principal-symbol identities for its supplied density. The present result supplies an actual matter-dependent inverse response from which such coefficients must be derived. It does not substitute a Hartree diagonal for a spatial principal symbol, identify the cube with the distinct screen source, or complete the common action-normalization join. Fixed microscopic geometry is compatible with this emergent-response route; a new fundamental metric register is not required. Common local limits, modular and area response, full stress, and the same-family Einstein balance premises remain to be proved. No new physical species, native response detector, or completed G1–G6 premise is claimed.

The finite ground and its charged probe can now be prepared and read by the declared observer service. The full matrix spectrum extends the local probe calculation. Separately, the infinite ground admits a spatially local inverse response with nonzero interaction memory at complex frequencies separated from the real axis. These constructions supply additional inputs to the existing field and geometry interfaces.

9.52 Native preparation of the same ground and its charged response.

The original cube ground and its local charged spectral measure now have an explicit native source preparation and readout. The exact highest-weight ground G, with $N = 4$ and $E_G = -\sqrt{3}/4$, has 62 nonzero hidden-up occupation amplitudes $(a + b\sqrt{3})/36$, where a, b are integers and $ab = 0$. Its squared amplitudes are therefore exact integers divided by 1296. These source-derived weights and signs determine an eight-bit binary amplitude tree; no fitted spectrum or unknown-state oracle is used. The existing compiler emits 5,379 literal native pulses. An independent rational interval check consumes every event, inverts the Walsh layers, and verifies the sine and cosine of each conditional rotation against the exact signed square-root ratios. Disjoint control blocks give a maximum error per level, and telescoping the levels and complete native load/export maps bounds the full initialized preparation, including aborts.

Eight blank down modes are included by an exact isometric embedding into all 65,536 Fock rows. A further 184 native pulses apply the inverse onsite frame. Each local frame word is checked on its full four-state Fock space, including vacuum and doublon phases, using $e^{is\pi P/4} = (I + isP)/\sqrt{2}$ and Gaussian-integer products. Its literal-angle and native-instrument errors are separately enclosed. The resulting complete preparation-state error is below 10^{-10}; the retained exact bound is approximately 8.67×10^{-13}.

At the origin the hidden and original frames coincide, and the down mode is empty in G. Thus

$$\gamma_1 = Z_0 X_1 = c_1 + c_1^\dagger, \qquad \gamma_1 G = c_1^\dagger G, \qquad \|\gamma_1 G\| = 1.$$

The Jordan–Wigner sign is essential: replacing $Z_0 X_1$ by X_1 produces an orthogonal probe because the exact up occupation is $1/2$. This unitary creation preserves the entire initial exterior CAR state: disjoint even observables commute, and exterior odd expectations vanish in both definite-parity states. The origin-down occupation changes from zero to one and total number from four to five. Its mean added energy is $1/2$ with variance $67/256$; this is a localized initial defect with a conserved number flag, not a persistent localization or mass-eigenstate claim. A 27-pulse pointer word prepares $(|0\rangle G + |1\rangle \gamma_1 G)/\sqrt{2}$. Its controlled Majorana is the product of commuting Pauli rotations in $\exp[i\pi(I - Z_{\text{pointer}})(I - \gamma_1)/4]$. This explicitly supplied native qubit service maintains a phase relation between the $N = 4$ and $N = 5$ arms; it does not derive a physical fermion-parity reference from passive repair.

Both arms evolve under the same original Hamiltonian. With a second controlled Majorana and pointer analyzer of phase ϕ,

$$C(t) = \langle \gamma_1 G, e^{-it(H - E_G)} \gamma_1 G \rangle,$$
$$P_0(\phi, t) = \tfrac{1}{2} \left[1 + \mathrm{Re}(e^{i\phi} C(t)) \right].$$

The ground-energy phase is supplied by the evolving reference arm; no separate scalar offset is inserted into the source law. The two executed settings have $t = 1$ and $\phi = 0, \pi/2$. Each setting uses two half-time segments, each with 32 symmetric splitting steps and 337,920 native pulses, followed by a 29-pulse analyzer. Including its ground preparation and pointer, a complete setting uses 681,459 pulses on the full 17-bit field/pointer register.

9.53 Protection and complete output accounting.

The five C repair seams follow the first time segment, and the five E seams follow the second. The complete repair theorem acts as identity on $N4 \oplus N5$, including cross-number coherences: all native branches vanish there, and the outside branch is identity. This is passive protection during the waits. Preparation and control words are not claimed to remain in this subspace at every intermediate pulse. The final inverse Majorana can create an $N = 6$ component after the charged particle has moved; no repair is applied after that analyzer.

Every source first-export failure is retained as its own terminal record. The numerical execution also retains all 130 repair outcome slots, including off-subspace outputs from control error, and all six final pointer outcomes. A native repair outcome terminates with that record; continuation uses the outside branch without renormalization. The lossless archive retains the prepared full ground, both full final states and every first-failure weight table. Individual failure vectors remain addressable from their actual source prefixes.

9.54 Connection to the resolved spectrum.

All field and pointer words are consumed against the exact source templates. Their complete operator errors, the preparation error and the full-Fock Strang errors of both time segments give a complete output error below 3/5000 for either setting. Contractivity through the complete repair and readout channels preserves the telescoping bound. The same bound controls each unconditioned pointer probability. Independently, the 78 certified spectral intervals evaluate

$$C(t) = \sum_{j=1}^{78} w_j e^{-i\Delta E_j t}$$

with rational sine and cosine enclosures. Adding the source error gives certified pointer intervals, checked against the literal execution and an independent exponential of the original Hamiltonian. The $\pi/2$ outcome is separated from $1/2$; the $U = 0$ comparator has the purely real response $\cos(\sqrt{3}t/16)$. Zero-time native runs check the analyzers as well. These two settings measure the charged characteristic function. They do not independently resolve all 78 lines, select an autonomous ground, prepare a thermodynamic vacuum, establish laboratory units, or turn finite energy gaps into relativistic particle masses.

9.55 The complete finite matrix spectrum and its exact spatial algebra.

The 78-line local down-addition result extends to all eight source columns $V_x = c^\dagger_{x\downarrow} G$. On the entire 560-state invariant sector, exact arithmetic proves $q(16H)V = 0$ for the same monic degree-78 polynomial q and verifies all 78 matrix moments in $\operatorname{span}\{I, A\}$, where $A = A^{\mathsf{T}}$, $A^2 = 3I$ and $\operatorname{tr} A = 0$. Polynomial division of the resolvent then gives

$$\mathcal{G}(z) = \frac{16\big[p_I(x)I + p_A(x)A\big]}{1296q(x)}, \qquad x = 16z - 4\sqrt{3}.$$

The numerator coefficients are exact convolutions of q and those source moments. This is a rational resolvent identity, rather than a truncated large-frequency expansion.

A second exact polynomial calculation gives $q \mid (p_I^2 - 3p_A^2)$. At each simple root x_i, define $\alpha_i = p_I(x_i)/(1296q'(x_i))$ and $\beta_i = p_A(x_i)/(1296q'(x_i))$. The earlier interval certificates give $\alpha_i > 0$; the new identity gives $3\beta_i^2 = \alpha_i^2$, and strict interval signs determine $s_i = \operatorname{sign}\beta_i$. Therefore

$$R_i = \alpha_i I + \beta_i A = 2\alpha_i P_{s_i}, \qquad P_\pm = \tfrac{1}{2}(I \pm A/\sqrt{3}).$$

Both $P_\pm$ have rank four. There are 40 plus-band lines and 38 minus-band lines; each band's $2\alpha_i$ weights sum to one. The rank of $E_i V$ equals that of its Gram matrix R_i, so the reducing cyclic space generated by all eight probes has dimension $78 \cdot 4 = 312$. The remaining 248 source dimensions are orthogonal to these probes and their time translates; their spectrum is not claimed.

Writing $\mathcal{G} = FI + JA$, both band functions have strictly negative imaginary part for $\operatorname{Im} z > 0$. Hence

$$\mathcal{G}^{-1} = \frac{FI - JA}{F^2 - 3J^2}, \qquad \Sigma = zI - (A/16 + I/2) - \mathcal{G}^{-1}.$$

The full finite inverse response and memory are exactly onsite plus the original signed nearest-neighbor adjacency. This special cube algebra does not supply an infinite-volume spatial limit or a Lorentz metric.

The certified common spectral support lies strictly between $g = 299/10000$ and $b = 37/20$. On the reducing cyclic space, both $K = H - E_G$ and its compression perpendicular to the observed

columns obey these bounds. The exact coupling $W = KV - VM_1$ lies in that same space, so the orthogonal unobserved sector contributes zero to the Schur memory. Using $W^\dagger W = I/4$, its order-p Taylor remainder satisfies

$$\|R_p\| \le \frac{r^{p+1}}{4g^{p+1}(g-r)}, \qquad |z| \le r < g.$$

This improves the earlier conservative full-sector disk without assigning the stronger gap to the unresolved sector or to infinite volume. Evaluation at $x = -4\sqrt{3}$ now computes $-\mathcal{G}(0)$ and $\Sigma(0)$ exactly over $\mathbb{Q}(\sqrt{3})$, including the previously bounded coefficients 12.9058381883 and -4.9026412320. No floating linear solve, native detector execution, physical species assignment or relativistic rest-mass inference is used in this extension.

9.56 Excluding a complete finite-gap mass dictionary.

The full support gives an exact test of assigning these visible single-charge gaps to charged-lepton rest energies under one positive unit factor. Its largest-to-smallest ratio is below 62, including the positive up-channel line. The retained retrospective PDG 2025 benchmark has muon/electron ratio above 206 and tau/electron ratio above 3477, even in the declared central-value plus/minus reported-uncertainty boxes. If a source ratio is at most R and the target ratio at least $Q > R$, simultaneous relative errors at most $\epsilon < 1$ require

$$\frac{1-\epsilon}{1+\epsilon} \le \frac{R}{Q}, \qquad \epsilon \ge \frac{Q-R}{Q+R}.$$

The exact rational calculation therefore forces an error above 54 percent for at least one member of an electron/muon assignment, and above 96 percent for an electron/tau assignment, for every common positive scale and every choice of visible lines. The target values do not enter the source law or spectrum. This rejects the stated finite-gap mass dictionary, not a continuum, an unobserved or composite sector, or OPH as a whole. The finite gaps are not thereby proved to be relativistic rest masses, and no eligible prospective comparison is added.

9.57 A local inverse response on the interacting infinite ground.

The original zero-chemical-potential Weyl/contact action on $\mathbb{Z}^3$ now admits a controlled spatial inverse response on the previously constructed thermodynamic ground, at fixed spacing and away from real frequency. This does not extrapolate the special two-eigenvalue cube identity. Let ω be the translation-averaged full-ground limit, L_ω its positive GNS generator, and

$$L = L_\omega \oplus -\overline{L_\omega}, \qquad V_{x\sigma} = c^\dagger_{x\sigma}\Omega \oplus \overline{c_{x\sigma}\Omega}.$$

The CAR give an isometry V from $\ell^2(\mathbb{Z}^3;\mathbb{C}^2)$. Define J in the same way from $b_{x\sigma} = n_{x\bar\sigma}c_{x\sigma}$ and write $\rho_{x\sigma\tau} = \omega(c^\dagger_{x\tau}c_{x\sigma})$. Exact onsite CAR identities, with vanishing different-site anticommutators, give

$$R = \bigoplus_x (\mathrm{tr}(\rho_x)I - \rho_x), \qquad V^\dagger J = J^\dagger V = J^\dagger J = R,$$
$$LV = Vh + UJ, \qquad 0 \le R \le I, \quad \|LV\| \le \sqrt{3}c + U.$$

Finite-support approximants and closedness of L extend the last identity to the entire one-particle input space. With $P = VV^\dagger$, $Q = I - P$, the bounded off-diagonal coupling is $W = QLV =$

$U(J-VR)$. Subtracting $WV^\dagger+VW^\dagger$ from L proves self-adjointness of its compressed complement $T=QLQ|_{Q\,\mathrm{Dom}\,L}$. The exact infinite-space Schur identity is therefore

$$\begin{aligned}\mathcal{G}(z)&=V^\dagger(z-L)^{-1}V=[zI-h-UR-\Sigma(z)]^{-1},\\ \Sigma(z)&=W^\dagger(z-T)^{-1}W,\qquad\qquad W^\dagger W=U^2(R-R^2).\end{aligned}$$

In particular $\|\Sigma(z)\|\le U^2/(4\,\mathrm{Im}\,z)$.

This ground has strictly nonzero memory for $U>0$. The source hidden-spin frame obeys $R_{x+t}=\pm R_tR_x$, so translations normalize its global SU(2) symmetry. Every full-box ground projector and the exterior vacuum are invariant; translated averages and weak-star limits retain the symmetry. Hence $\rho_x=rI$, with the earlier density theorem giving $1/8\le r\le 1/2$, and

$$\frac{7U^2}{64}I\le W^\dagger W=U^2r(1-r)I\le\frac{U^2}{4}I.$$

The spectral theorem gives $iy\Sigma(iy)\to W^\dagger W$ strongly as $y\to\infty$. Thus the same-mode response cannot be that of a number-conserving static quadratic Hamiltonian with zero memory. Neither a scattering theorem nor faithfulness of the local ground-state density matrices follows.

9.58 A quantitative connected four-point witness.

The same ground has a non-Gaussian multi-time field, a stronger statement than nonzero two-point memory. Fix $a=c_{x\sigma}$ and $n=n_{x\bar\sigma}$, and set

$$A(t)=\{\tau_t(a),a^\dagger\},\qquad D(t)=A(t)-\omega(A(t))I,\qquad F(t)=\omega(D(t)^\dagger D(t)).$$

The local commutator gives $D(0)=0$ and $D'(0)=-iU(n-rI)$. Since $n^2=n$, $F(t)/t^2\to U^2r(1-r)\ge 7U^2/64$. An explicit remainder uses $M=9c^2+12cU+U^2$ and $\delta=[H,\cdot]$:

$$\|\delta^2(a)\|\le 3c(3c+U)+U(6c+3c+U)=M,\qquad \|D(t)+itU(n-rI)\|\le 2Mt^2.$$

Thus, for $U>0$,

$$F(t)\ge\frac{U^2t^2}{64}>0,\qquad 0<|t|\le\frac{U}{16M}.$$

For odd zero-mean operators define the connected fermionic function

$$\begin{aligned}\kappa_4(B_1,B_2,B_3,B_4)=\omega(B_1B_2B_3B_4)&-\omega(B_1B_2)\omega(B_3B_4)\\&+\omega(B_1B_3)\omega(B_2B_4)-\omega(B_1B_4)\omega(B_2B_3).\end{aligned}$$

With $(B_1,B_2,B_3,B_4)=(a,\tau_t(a)^\dagger,\tau_t(a),a^\dagger)$, $F(t)$ is the sum of κ_4 on words $1234,1243,2134,2143$. All crossed contractions, including anomalous ones, cancel pairwise. At least one connected function therefore has magnitude $U^2t^2/256$ or greater on the displayed interval. This excludes a quasifree representation of the complete multi-time field by linear fermionic fields, including pairing and auxiliary Gaussian modes. Indeed any quadratic evolution on an enlarged CAR algebra preserves linearity of an observed mode, making its anticommutator scalar and $D(t)=0$ in every reference state. Thus even a non-Gaussian reference cannot realize this witness through such a linear quadratic embedding. It does not exclude nonlinear encodings or establish continuum scattering.

9.59 The full ordered Euclidean hierarchy.

The same positive ground generator $L_\omega \geq 0$ defines, for ordered real times $s_1 \leq \cdots \leq s_n$ and bounded local CAR operators,

$$S(A_1, s_1; \ldots; A_n, s_n) = \langle \Omega, A_1 e^{-(s_2-s_1)L_\omega} A_2 \cdots e^{-(s_n-s_{n-1})L_\omega} A_n \Omega \rangle, \qquad |S| \leq \prod_j \|A_j\|.$$

Every finite approximant uses one common translated full-ground state and its positive generator $H_\Lambda - E_\Lambda$ for the whole word, with zero exterior Hamiltonian, before averaging over translations. For each positive gap the spectral theorem gives

$$e^{-sL} = \int_{\mathbb{R}} \frac{s}{\pi(s^2+t^2)} e^{itL}\, dt, \qquad L \geq 0.$$

The product integral is dominated by $\prod_j \|A_j\|$. Its real-time integrand is the expectation of $A_1 \tau_{t_1}(A_2) \tau_{t_1+t_2}(A_3) \cdots$. Local-dynamics bounds, telescoping the product, and the same weak-star ground subsequence give uniform convergence on compact time boxes. The omitted product-measure tail is at most $\sum_j 2s_j/(\pi T_j)$, capped by one; a finite-versus-limit comparison costs twice this tail times the product of operator norms. Dominated convergence therefore supplies every ordered Euclidean n-point limit on this one ground, without another subsequence or a state-convergence rate. Zero gaps are handled by the identity operator.

For a positive-time word $F = (A_1, t_1; \ldots; A_n, t_n)$ define

$$\Psi_F = e^{-t_1 L_\omega} A_1 e^{-(t_2-t_1)L_\omega} A_2 \cdots e^{-(t_n-t_{n-1})L_\omega} A_n \Omega.$$

The reflected list is $(A_n^*, -t_n; \ldots; A_1^*, -t_1)$, in that order. Concatenation with a second positive-time word gives $S(\theta F, G) = \langle \Psi_F, \Psi_G \rangle$; hence every reflected correlation matrix is positive semidefinite. No CAR factors are reordered. Exact rational two-mode controls check the Gram identity, all 31 principal minors of a five-word example, and a negative odd-word cross entry. This proves ordered time-reflection positivity at fixed spacing. It does not establish $O(4)$ covariance, bosonic permutation symmetry, a spatial continuum, or the full relativistic Euclidean axioms.

9.60 Uniform inverse and spatial decay bounds.

Fix $\operatorname{Im} z \geq \eta > 0$, $|z| \leq Z$, with $Z \geq \eta$. For a unit vector $v = Vf$, Cauchy–Schwarz applied to $\langle (\bar{z} - L)v, (z-L)^{-1} v \rangle = 1$ yields

$$-\operatorname{Im} \mathcal{G}(z) \geq B_0^{-1} I, \qquad \|\mathcal{G}(z)^{-1}\| \leq B_0, \qquad B_0 = \frac{(Z + 2c + U)^2}{\eta}.$$

Both $\mathcal{G}$ and its adjoint are bounded below; its range is therefore closed and dense, which proves surjectivity in infinite volume. No determinant or finite-volume gap is used.

Let $d = \|x - y\|_\infty$. The original graded locality bound gives

$$\|\mathcal{G}_{xy}(z)\| \leq A(1+d)^2 e^{-2\mu d}, \qquad \mu = \min\{1/8, \eta/(384c)\}, \qquad A = 192c/\eta^2 + 8/\eta.$$

For $d \geq 2$, approximate the evolved odd operator in the radius $d-1$ cube, whose anticommutator with the remote mode is zero. Split the Laplace integral at $(d-1)/(192c)$ and use the previous full-CAR boundary estimate on the first part and the uniform norm bound on its tail. This proves the displayed constants without assuming spatial clustering of ω.

Put $q = (1+\mu)^{-1}$ and $T_n(q) = \sum_{d\geq 1} d^n q^d$. These are rational functions with positive Eulerian coefficients. Define

$$\widehat{S} = 24T_5(q) + 48T_4(q) + 26T_3(q) + 4T_2(q) + 2T_1(q),$$
$$\alpha = \min\{\mu, (2B_0 A\widehat{S})^{-1}\} > 0.$$

For any bounded real one-Lipschitz function f, the block Schur test gives

$$\|e^{\alpha f}\mathcal{G}e^{-\alpha f} - \mathcal{G}\| \leq \alpha A\widehat{S} \leq (2B_0)^{-1}.$$

A Neumann series bounds the conjugated inverse by $2B_0$. Choosing $f(u) = \min\{\|u-y\|_\infty, R\}$ with $R \geq d$ proves

$$\|(\mathcal{G}^{-1})_{xy}(z)\| \leq 2B_0 e^{-\alpha d},$$
$$\|\Sigma_{xy}(z)\| \leq (Z + U + 2B_0 + 2c)e^{-\alpha d}.$$

Only bounded weights were used. The constants are conservative bounds, not estimates of a physical correlation length.

9.61 The controlled spatial symbol and its geometric boundary.

Translation invariance and exponential summability give an analytic two-by-two inverse-response symbol $K(k,z)$ for $\|\operatorname{Im} k\|_1 < \alpha$. For real momentum increment p, its total-degree-n Taylor remainder is bounded by

$$\frac{2B_0\|p\|_1^{n+1}}{(n+1)!}\sum_{x\in\mathbb{Z}^3}\|x\|_\infty^{n+1}e^{-\alpha\|x\|_\infty}.$$

The shell count $24d^2+2$ and the same rational T_n functions give explicit upper bounds on this convergent sum. A frequency circle of radius $\eta/2$ gives, for $|w| < \eta/2$, the additional remainder $B_*(2|w|/\eta)^{n+1}/(1-2|w|/\eta)$, where B_* is the above inverse bound evaluated at $(\eta/2, Z+\eta/2)$.

These estimates supply actual local spatial coefficients on the interacting infinite ground, with a controlled complex-frequency domain. They do not compute the ground functional or those coefficients, and deteriorate as $\eta \to 0$. A real hyperbolic principal symbol, common matter coefficients and full action normalization are still required before applying the existing `wave_density_metric` and `source_tetrad` interfaces. The joint modular, area, stress and Einstein balance requirements remain separate.

For a refinement $c = v/a$ at fixed $v, U, Z, \eta > 0$, the retained estimates have $\alpha = O(a^9)$ and therefore physical momentum width $\alpha/a = O(a^8)$; the connected-witness time window is $O(a^2)$. These shrinking bounds do not prove uniform continuum control or a no-go theorem for the action.

9.62 The full ground-family spectrum and its internal spin.

The complete five-dimensional $N = 4$ ground space closes a remaining finite spectral ambiguity. Work in the exact hidden-spin frame, with G_0 its spin-two highest-weight ground, $\lambda = \sqrt{3}/16$, and $V_{\uparrow,x} = c^\dagger_{x\uparrow}G_0$, $V_{\downarrow,x} = c^\dagger_{x\downarrow}G_0$. The actual 56×560 raising matrix $T = S_+$ obeys

$$TV_\downarrow = V_\uparrow, \qquad TT^\dagger = 5I, \qquad H_{5,0}T = TH_{4,1}.$$

The full spatial Gram is $V_\uparrow^\dagger V_\uparrow = P_+$, and $(H - E_G)V_\uparrow = \lambda V_\uparrow$. At magnetic weight $3/2$,

$$V_{\text{high}} = T^\dagger V_\uparrow/5, \qquad V_{\text{low}} = V_\downarrow - V_{\text{high}}, \qquad TV_{\text{low}} = 0, \qquad V_{\text{high}}^\dagger V_{\text{high}} = P_+/5.$$

The retained residue at λ is exactly $P_+/5$. Positivity therefore leaves no low-spin contribution at that line: it has total hidden spin 5/2, while the other 77 visible lines have total hidden spin 3/2.

This conclusion covers every ground vector. Put $G_k = S_-^k G_0$, with $\|G_k\|^2 = k!4!/(4-k)!$, $0 \le k \le 4$. Exact occupation-space CAR give, for every site,

$$c_\downarrow^\dagger G_k = S_-^{k+1} V_\uparrow/5 + S_-^k V_{\mathrm{low}},$$
$$c_\uparrow^\dagger G_k = (5-k) S_-^k V_\uparrow/5 - k S_-^{k-1} V_{\mathrm{low}}.$$

The second term vanishes at $k = 0$. All eighty creator columns are checked on the original source; the Hamiltonian commutes with every ladder. No coherent ground superposition or linear CAR probe can introduce another addition frequency.

The full $N = 5$ space has spin multiplicities $56, 504, 1008$ at $S = 5/2, 3/2, 1/2$, respectively. Four reachable copies occur at λ, and four at each of the other 77 lines. Thus the complete reachable dimension is

$$4 \cdot 6 + 77 \cdot 4 \cdot 4 = 1256 \quad \text{inside} \quad \binom{16}{5} = 4368.$$

The dark complement has dimensions $312, 784, 2016$ in those three spin sectors. These are dimensions of source subspaces, not full energy degeneracies; their dark eigenvalues remain uncomputed. All annihilator probes similarly generate sixteen $N = 3$, spin-3/2 states with positive removal cost λ, leaving 544 dark dimensions.

9.63 The invariant mixed-reference energy readout.

For $\rho_G = P_G/5$, the exact addition and retarded matrices are

$$\overline{\mathcal{G}}_{\mathrm{add}}(z) = \frac{1}{2}\left[\mathcal{G}_\downarrow(z) + \frac{P_+}{z-\lambda}\right] \otimes I_2,$$
$$\overline{\mathcal{G}}_{\mathrm{ret}}(z) = \frac{1}{2}\left[\mathcal{G}_\downarrow(z) + \frac{P_+}{z-\lambda} + \frac{P_-}{z+\lambda}\right] \otimes I_2.$$

At each site and fixed spin, the addition line λ has weight 3/10, and each other line has half its previous weight. The total addition weight is 3/4; conditional weights are therefore 2/5 at λ and $2w_i/3$ elsewhere. Removal has weight 1/4 at the signed retarded pole $-\lambda$. All 78 positive lines survive. The 79 signed poles have rank-eight residues, giving a minimal mixed-state response realization of dimension 632. This realization keeps sixteen observed columns and differs from the physical reachable range that keeps all five ground inputs distinct.

Direct full-source Gram matrices verify every spin and spatial entry of the zeroth and first moments, including

$$\overline{M}_{0,\mathrm{ret}} = I_{16}, \qquad \overline{M}_{1,\mathrm{ret}} = (A \otimes I_2)/16 + I_{16}/4.$$

The actual physical-frame block is $g_{xy} R_x^\dagger R_y$; intersite spin coherence is retained, while local blocks are scalar. The same positive-energy envelope and finite charged-lepton ratio exclusion now hold for every ground choice and linear single-CAR probe. These total internal spins include the spin-two background. They do not identify intrinsic particle or Lorentz spin, a measured species, or a relativistic mass. The result does not exclude composite probes or a continuum spectrum, and does not execute native preparation or complete finite-shot spectroscopy of the mixed reference.

9.64 Complete finite-resolution energy readout.

For a unit one-particle probe f, the same ground defines the probability measure on signed addition and removal energies

$$\mu_f(B) = \langle Vf, 1_B(L)Vf\rangle.$$

The complete Kraus pair $c(f)^\dagger, c(f)$ labels addition and removal; each branch has a nonnegative physical excitation energy. Only the response-frequency variable assigns a minus sign to removal. The exact identities $LV = Vh + UJ$ and $V^*J = J^*J = rI$ give the operator moments

$$M_0 = I, \qquad M_1 = h + UrI, \qquad M_2 = h^2 + 2Urh + U^2rI.$$

The second-moment identity is a quadratic-form statement and does not require $Vf \in \operatorname{Dom} L^2$. For a local mode at $c = 1/8, U = 1$, the mean lies in $[1/8, 1/2]$, the second moment in $[19/128, 67/128]$, and the variance in $[17/128, 35/128]$. Thus $\mu_f(|E| \geq R) \leq 67/(128R^2)$. More generally, $\|hf\| \leq K$ gives a second moment at most $K^2 + UK + U^2/2$, useful for uniformly controlled physical smearing.

Choose $M \geq 2$ and $\tau > 0$. The positive effects

$$K_k(E) = \left| \frac{1}{M} \sum_{j=0}^{M-1} e^{-ij(\tau E - 2\pi k/M)} \right|^2, \qquad q_k = \int K_k(E)\, d\mu_f(E), \quad 0 \leq k < M,$$

sum exactly to one by finite Fourier orthogonality. Writing $\phi_f(t) = \int e^{-itE}\, d\mu_f(E)$, their complete probabilities are

$$q_k = \frac{1}{M} + \frac{2}{M} \sum_{n=1}^{M-1} \left(1 - \frac{n}{M}\right) \operatorname{Re}\big(e^{2\pi ikn/M} \phi_f(n\tau)\big).$$

The principal energy centers are modulo $2\pi/\tau$ and are readout bins, not asserted eigenvalues. For $0 < \tau\Delta \leq 3$, circular error has probability at most $\min\{1, 10/(M\tau^2\Delta^2)\}$, improved to $22/(7M\tau\Delta - 44)$ when its denominator is positive. This follows by summing the two inverse-square grid tails. A second-moment bound C adds an alias term at most $C/(3/\tau - \Delta)^2$. Consequently the probability of a real energy error at least Δ is controlled, without a pointwise spectral density or an isolated-pole assumption.

The complete frozen 78-line cube measure now gives certified distributions for $M = 8, 16, 64$ at $\tau = 1$. At $\Delta = 1/2$, direct positive-kernel and Fourier interval calculations bound the respective error probabilities by 0.100284517428, 0.055193158232, and 0.020209031434. All outcomes are retained. The previously executed native time-one quadratures check the first harmonic; higher harmonics here are source calculations, not new native long-time or finite-shot measurements.

Along the existing full-ground subsequence, the finite positive spectral measures converge weakly. The uniform fixed-spacing spatial kernel bound also promotes entrywise convergence to strong convergence of the compressed resolvents on the common one-particle space. No identification of the varying GNS Hilbert spaces is assumed. An explicit finite readout error adds the local-dynamics, native-control, and statistical errors to the still-unspecified finite-ground state discrepancy; the last term is never set to zero. Sharp interval probabilities require zero limiting mass at their endpoints. These results do not supply physical particle identifications, relativistic rest masses, or an interacting continuum.

9.65 An all-valley positive-energy continuum field net.

There is now a spacing-limit construction beyond fixed-lattice response. Take $c(a) = v/a$, fixed $v > 0$, and $0 \leq U(a) \leq U_{\max} < \infty$. At each spacing retain the full original Weyl/contact dynamics

and a translation-invariant full ground. On centered cells $C_{a,n} = a(n + [-1/2, 1/2)^3)$ define

$$(J_a f)(n) = a^{-3/2} \int_{C_{a,n}} f(x)\, dx, \qquad P_a = J_a^* J_a.$$

Cellwise Cauchy–Schwarz and coordinate averaging give $\|J_a\| \le 1$ and $\|P_a f - f\| \le a \sum_j \|\partial_j f\|$ for $f \in H^1$. Thus $P_a \to I$ strongly. For every $b \in \{0,1\}^3$ set $M_b u(n) = (-1)^{b\cdot n} u(n)$ and $C_{a,b}(f) = c(M_b J_a f)$, with the conventional antilinear annihilator. All eight valleys evolve under the original full Hamiltonian. Writing $D_{a,j} f = [f(\cdot + ae_j) - f(\cdot - ae_j)]/(2a)$ gives

$$h_a M_b J_a f = M_b J_a \left[iv \sum_j (-1)^{b_j} \sigma_j D_{a,j} f \right], \qquad \|h_a M_b J_a f\| \le v \sum_j \|\partial_j f\|.$$

For $b \ne d$, shifting one lattice step in a coordinate where they differ reverses the cross-valley pairing. Subtracting the shifted pairing proves

$$|\langle M_b J_a f, M_d J_a g\rangle| \le \frac{a}{2} \left(\|\partial_j f\| \|g\| + \|f\| \|\partial_j g\| \right).$$

The limiting equal-time fields therefore have the full continuum CAR over $L^2(\mathbb{R}^3; \mathbb{C}^2) \otimes \mathbb{C}^8$.

The necessary time estimate holds in the full CAR operator norm. For $B(u) = \sum_{x\sigma} \overline{u_{x\sigma}} n_{x\bar\sigma} c_{x\sigma}$, the onsite anticommutator is positive and obeys $\{B(u_x), B(u_x)^*\} \le \|u_x\|^2 I$. Different-site odd anticommutators vanish, hence $\|B(u)\| \le \|u\|$. The exact contact commutator consequently gives

$$\|\tau_{a,t}(C_{a,b}(f)) - \tau_{a,s}(C_{a,b}(f))\| \le |t - s| \left(v \sum_j \|\partial_j f\| + U_{\max} \|f\| \right).$$

This avoids an extensive interaction-norm estimate.

A diagonal subsequence of spacings makes every word in a countable dense set of smooth smears, rational times and all eight valleys converge simultaneously. The displayed time and norm estimates extend the limits to all L^2 smears and real times. Polynomial positivity, norm bounds and the CAR pass to the limit and construct bounded fields in one GNS representation. Common time shifts are strongly continuous. Nearest *even* lattice translations approximate every physical translation while keeping all valley phases equal to one, so continuous spatial translations also act on the limiting net. For every multi-time polynomial X, the correlations $\omega_a(X_a^* \tau_{a,t}(X_a))$ have nonnegative energy support and converge uniformly on compact time intervals. Testing against smooth negative-energy spectral functions and using dominated convergence therefore proves a nonnegative generator on the entire limiting GNS space, not only on linear-field vectors.

Graded locality also survives with an explicit bound. Suppose f is supported in $[-r, r]^3$, g outside $[-R, R]^3$, and put $\delta = (R - r)/2 - 48vT > 0$. For $0 < a \le \min\{1, (R - r)/8\}$ and $|t| \le T$, the original full-CAR bound, with lattice radii $\lceil r/a \rceil + 1$ and $\lfloor R/a \rfloor - 1$, yields

$$\begin{aligned} \|\{\tau_{a,t}(C_{a,b}(f)), C_{a,d}(g)^{\#}\}\| &\le 192 v T (2r + 5)^3 (2R + 1)^2 \|f\| \|g\| a^{-6} e^{-\delta/a} \\ &\le 192 \cdot 7! v T (2r + 5)^3 (2R + 1)^2 \|f\| \|g\| \frac{a}{\delta^7}. \end{aligned}$$

Here # allows an adjoint or no adjoint. Inserting this norm estimate between arbitrary polynomial vectors makes the limiting anticommutator the zero operator. A finite partition of unity extends the result to separated compact supports. The continuum net thus has exact graded locality at sup-norm distance greater than $96v|t - s|$. This is a conservative propagation cone, not a Lorentz metric.

There is a further uniform response bound on every physical momentum window $|p| \le P$ around any retained valley. These windows reduce the translation-invariant response and have kinetic norm at most vP. The exact Schur formula gives

$$\|\mathcal{G}_a(z)^{-1}\|_P \le |z| + vP + \frac{U_{\max}}{2} + \frac{U_{\max}^2}{4\,\mathrm{Im}\,z}, \qquad -\mathrm{Im}\,\mathcal{G}_a(z) \ge \frac{\mathrm{Im}\,z}{(|z| + vP + U_{\max})^2} I_P.$$

These bounds are uniform in spacing away from real frequency. The cell-average map differs from the exact band isometry by at most $a^2P^2/24$, so the two conventions have the same limiting correlations. Weak convergence alone does not allow interchanging the inverse and limit.

9.66 A norm-controlled free limit and the remaining interacting step.

A Duhamel comparison with the free lattice evolution uses only $\|B(u)\| \le \|u\|$ and gives an error at most $|U(a)t|\|u\|$. The central-difference remainder then gives, for $f \in H^3$ and $h_b = iv\sum_j (-1)^{b_j}\sigma_j\partial_j$,

$$\|\tau_{a,t}(C_{a,b}(f)) - C_{a,b}(e^{ith_b}f)\| \le |t|\left(U(a)\|f\| + \frac{va^2}{6}\sum_j \|\partial_j^3 f\|\right).$$

If $U(a) \to 0$, the limiting dynamics is exactly the free eight-valley Weyl dynamics. Positive energy forces positive-band annihilators and negative-band creators to annihilate the ground. The one-particle zero-energy kernel is zero in L^2; spectral-band approximation and the CAR recurrence therefore uniquely determine the filled-sea quasifree state and all its moments. This is a controlled free continuum, not the required interacting one.

For bounded positive limiting U, the construction still gives the full positive-energy continuum field net, but it has not proved that a connected interaction survives, selected a unique limiting state, or derived Lorentz covariance. Its propagation cone and positive energy supply no common physical stress, geometric modular flow, renormalized area stationarity or completed G1–G6 Einstein package.

9.67 Ground occupation and energy constraints under refinement.

The original zero-chemical-potential ground satisfies a useful uniform constraint before any continuum particle identification. With normalized Brillouin measure, let $\rho(k)$ be its one-body covariance and $P_\pm(k)$ the spectral projectors of the original Weyl hopping. The free filled sea has onsite covariance $I/2$ and double occupation $1/4$. Its finite-box restrictions give the variational bound

$$0 \le e_{\mathrm{kin}} - e_{\mathrm{free}} + Ud = e_U - e_{\mathrm{free}} \le U/4, \qquad d = \omega(n_\uparrow n_\downarrow).$$

Here $e_{\mathrm{free}} = -c\int \sqrt{\sum_j \sin^2 k_j}$; no free reference is substituted for the interacting state. Set

$$w(k) = \mathrm{tr}(P_+\rho) + \mathrm{tr}(P_-(I-\rho)), \qquad D = \int w(k).$$

Then $0 \le w \le 2$ and the kinetic excess is $c\int \sqrt{\sum_j \sin^2 k_j}\, w(k)$. The region with $\sqrt{\sum_j \sin^2 k_j} \le \delta$ has normalized volume at most δ^3 for $0 < \delta \le 1$, since $2\arcsin\delta/\pi \le \delta$. Splitting this integral proves

$$D \le \min\{2, 2\delta^3 + U/(4c\delta)\}, \qquad |n-1| \le D.$$

For $U \le 24c$, choosing $\delta = (U/(24c))^{1/4}$ gives $D \le 8(U/(24c))^{3/4}$. Thus the density tends to one per site, and the per-spin density r tends to $1/2$, when $U/c \to 0$. At $v = 1/8$, $U = 1$, $a = 1/65536$, $c = v/a$ and $\delta = 1/16$, the exact bound is $1023/1024 \le n \le 1$.

The same defect controls every fixed microscopic local word. The free-sea annihilation parts $c(P_+f)$ and $c^\dagger(P_-f)$ of a unit site-basis field have interacting GNS norm at most $\sqrt{D}$. Moving the first annihilation part to the right by CAR gives the free Wick recursion, with two boundary errors at most $\sqrt{D}$. Hence for an even word of length m,

$$|\omega(A_1 \cdots A_m) - \omega_{\text{free}}(A_1 \cdots A_m)| \le C_m\sqrt{D}, \qquad C_0 = 0, \quad C_m = 2 + (m-1)C_{m-2}.$$

In particular $C_4 = 8$ and $d \to 1/4$. This controls fixed microscopic local observables; shrinking momentum pockets can still carry a full normalized continuum packet, so no Gaussian continuum conclusion follows.

For cell-averaged physical H^1 packets and $c = v/a$, bounded $U(a) \to U_0$ fixes the first spectral moment to $\langle f, h_{\text{W}} f\rangle + (U_0/2)\|f\|^2$ in every weak spectral limit. The uniform second-moment bound supplies the required first-moment tail control. The scalar term is retained without a chemical-potential shift. Second moments need not pass: the positive measures $(1 - n^{-2})\delta_0 + n^{-2}\delta_n$ converge weakly to δ_0 while their variances tend to one. Thus even a nonzero memory Gram need not survive the limiting response. These constraints supply ground and energy inputs, not relativistic rest masses or an Einstein balance. The proof and exact controls are in `model/continuum_ground_constraints`.

9.68 A complete native finite-clock energy measurement.

The original zero-chemical-potential cube now has a finite native preparation of its invariant ground reference $P_G/5$. Five azimuths $2\pi j/5$ and three polar cosines $-\sqrt{3/5}, 0, \sqrt{3/5}$, with polar weights $5/18, 4/9, 5/18$, twirl the spin-two highest-weight ground exactly. The azimuth sum cancels every nonzero magnetic difference through four, and the three-node quadrature integrates the remaining degree-four binomial weights to $1/5$. A native four-bit selector has weights $[5, 8, 5]$ repeated five times, followed by zero, divided by 90. Its fifteen conditional full-Fock spin rotations and the original inverse frame are executed; all selector and export outcomes are kept. The complete prepared-state error is below 10^{-9}.

At the origin, a native occupation-copy followed by $\gamma_1 = Z_0X_1 = c_1 + c_1^\dagger$ gives the complete charge instrument with Kraus maps $c_1^\dagger, c_1$. Both branches are retained. Two clock bits control original-action evolutions for model times one and two, with the independently known phase from $E_G = -\sqrt{3}/4$. The positive Fourier transform produces

$$K_k(E) = \left|\frac{1}{4}\sum_{n=0}^{3} e^{-in(E-2\pi k/4)}\right|^2, \qquad \sum_{k=0}^{3} K_k(E) = 1.$$

This is a complete positive energy measurement. Both physical charge branches have nonnegative excitation cost; principal bin centers are defined modulo 2π, and the removal tag retains the sign needed for the retarded frequency. For the invariant ground, the local addition measure is $\mu_\downarrow/2 + \delta_\lambda/4$ and the removal measure is $\delta_\lambda/4$, where $\lambda = \sqrt{3}/16$. The complete frozen 78-line measure therefore determines every joint charge/bin probability without fitting.

All eighteen-bit clock rows and thirty classical selector/charge columns are propagated through the actual native local instruments. The exact column decomposition used for computation discards no quantum coherence or outcome. Controlled hopping and contact words use at most three local

work bits; no abstract controlled Hamiltonian is supplied as a primitive. With sixteen and thirty-two Strang steps, the full-Fock split errors sum to 1887/65536. Including every preparation, control, scalar-phase and Fourier error gives complete state and total-variation error below 3/100. A longest successful history contains 612666 native pulses. All 3240 joint selector/tag/clock slots, the selector stops and every first-export failure are retained. The independent original-Hamiltonian exponential is a comparator only.

This construction executes coherent histories through model time three and closes the stated finite native preparation-and-energy-POVM task. Four broad bins do not separately resolve seventy-eight lines. It neither identifies physical species nor converts a finite excitation energy into a relativistic rest mass. It also does not prepare the separate centered-action reference, establish infinite-ground state convergence, or supply the remaining geometric premises.

9.69 A CPT constraint on the unchanged charged fields and clock.

For a real, even, unit scalar Schwartz packet ϕ, average the signed addition/removal measure over all sixteen continuum field components. Writing this probability measure as ν_ϕ, spin trace cancels the Weyl first moment and its kinetic cross term. The exact finite-spacing sum rules, $r_a \to 1/2$, and weak lower semicontinuity give

$$\int E\, d\nu_\phi(E) = \mu = U_0/2, \qquad \int E^2\, d\nu_\phi(E) \le C_\phi = v^2\|\nabla\phi\|^2 + U_0^2/2.$$

Only the second-moment inequality passes to the limit. Negative response frequency denotes removal; the two charge-tagged excitation energies are nonnegative with respect to the original positive GNS generator L.

Suppose a proposed physical identification supplies an antiunitary Θ preserving the same ground and L, and mapping the sixteen smeared annihilators A_s into their adjoints by a unitary matrix Q:

$$\Theta\Omega = \Omega, \qquad \Theta L\Theta^{-1} = L, \qquad \Theta A_s\Theta^{-1} = \sum_t Q_{st}A_t^*.$$

The real even packet is preserved by the spatial inversion and conjugation in the usual CPT action. For every spectral set B, antiunitarity and unitarity of Q give

$$\sum_s \|P_L(B)A_s^*\Omega\|^2 = \sum_s \|P_L(B)A_s\Omega\|^2.$$

Thus the averaged addition and removal measures agree, ν_ϕ is reflection symmetric, and its first moment is zero. This precise same-field, same-ground, same-clock CPT identification is therefore impossible for $U_0 > 0$. No CPT theorem is applied without its hypotheses: this does not exclude Lorentz covariance alone or other field dictionaries.

With ν_ϕ^- the reflected measure and total variation defined as $\sup_B |\nu_\phi(B) - \nu_\phi^-(B)|$, Cauchy–Schwarz proves

$$\mathrm{TV}(\nu_\phi, \nu_\phi^-) \ge \frac{\mu^2}{C_\phi}, \qquad W_1(\nu_\phi, \nu_\phi^-) \ge 2\mu.$$

There is a bounded odd witness: for $R = C_\phi/(2\mu)$, clipping E to $[-R, R]$ gives f_R with $|E - f_R(E)| \le E^2/(4R)$ and $\int f_R\, d\nu_\phi \ge \mu/2$. Gaussian packets of width ℓ have $C_\phi = 3v^2/(2\ell^2) + U_0^2/2$. At $v = 1/8$, $U_0 = 1$ and $\ell = 1$, the total-variation lower bound is 32/67; broad packets make it tend to 1/2. These are rigorous bounds, not computed infinite-ground spectra or executed native measurements.

A scalar charged-field rephasing changes the generator to $L - \kappa Q_{\mathrm{charge}}$ and the signed first moment to $\mu - \kappa$. Consequently $\kappa = U_0/2$ is the unique necessary centering for this CPT condition.

It is a change of charged dynamics, not an additive energy-origin change or a rest mass. Positivity on the old ground is not automatic: the empty two-mode ground of $H = n_0 + 3n_1$ has a centered response of zero mean at $\kappa = 2$, while $H - 2N$ has eigenvalues $0, -1, 1, 0$. The new reference and its ground property require proof.

The free $U(a) \to 0$ control has physical addition and removal energies $v|p|$, zero rest masses, and four Weyl chiralities of each sign. A massive Dirac control, $H_D = Z \otimes (\sigma \cdot p) + mX \otimes I$, has $H_D^2 = (|p|^2 + m^2)I$ and zero trace: its mass couples chiralities and does not create the scalar signed-energy shift. Exact full-CAR counterexamples and the conditional proof are retained in `model/continuum_cpt_constraint`; no physical species, interacting Lorentz completion or Einstein premise is thereby supplied.

9.70 A resolved canonical contact-force channel.

The bounded-contact refinement also resolves a nonlinear operator in the original action. At each site set

$$b_\sigma = n_{\bar\sigma} c_\sigma, \qquad d_\sigma = (2n_{\bar\sigma} - 1)c_\sigma = 2b_\sigma - c_\sigma.$$

The onsite unitary $Z = I - 2n_\uparrow n_\downarrow$ obeys $Zc_\sigma Z = -d_\sigma$, so d is itself an exact CAR family, with $\|d(u)\| = \|u\|$. It is a composite of the original fields, not an added microscopic mode. The unchanged Hamiltonian satisfies

$$[H, c(u)] = -c((h + U/2)u) - (U/2)d(u).$$

Thus retaining the scalar term does not justify dropping the composite. For $B(u) = \sum_{x,\sigma} \overline{u_{x\sigma}} b_{x\sigma}$, exact local algebra gives

$$\{B(u), c(v)^*\} = \{B(u), B(v)^*\} = Q(u, v) = \sum_x \mathrm{d}\Gamma(q_x), \qquad q_x = (u_x^* v_x)I - v_x u_x^*.$$

Each q_x has rank at most one and $\|q_x\| = \|q_x\|_{\mathrm{HS}} = \|u_x\|\|v_x\|$. The anomalous mixed anticommutator is the pair sum

$$P(u, v) = \{B(u), c(v)\} = \sum_x \overline{u_{x\downarrow} v_{x\uparrow} - u_{x\uparrow} v_{x\downarrow}} c_{x\uparrow} c_{x\downarrow}.$$

Let D_a be the occupation defect above, $L = \|u\|\|v\|$, and $I_4 = \sum_x \|u_x\|^2 \|v_x\|^2$. The uniform four-field Wick comparison, $\omega_a(Q) = r_a \langle u, v\rangle$, and the free covariance identity $\mathrm{Var}_{\mathrm{free}}(\mathrm{d}\Gamma(q)) = \mathrm{Tr}[\Pi q^*(I - \Pi)q] \le I_4$ imply

$$\omega_a(Y_a^* Y_a) \le 4\big[I_4 + (8\sqrt{D_a} + D_a/2)L^2\big]$$

for either $Y_a = 2Q - \langle u, v\rangle I$ or $Y_a = 2P$. For $u = M_b J_a f$, $v = M_e J_a g$ with bounded smooth compactly supported smears, $I_4 \le a^3 \|f\|_\infty^2 \|g\|^2$ and the coefficient supremum is at most $a^3 \|f\|_\infty \|g\|_\infty$. These ground fluctuations therefore vanish.

The passage is an operator statement on the full field hierarchy. A degree-m CAR polynomial obeys $\|[\mathrm{d}\Gamma(q), A]\| \le m\|q\|\|A\|$ for $q = q^*$: its one-body unitary orbit has frequencies in $[-m\|q\|, m\|q\|]$, and the Bernstein interpolation bound applies. The same argument gives $\|[\sum_x n_{x\uparrow} n_{x\downarrow}, A]\| \le m\|A\|$, with degree at most $m + 2$ after the commutator. In the free interaction picture the order-N Dyson remainder is consequently bounded by

$$\|A\|(2U_{\max}|s|)^N \frac{(m/2)_N}{N!}.$$

This is uniform in spacing and volume. Divide $|t| \le T$ into $K = \max(1, \lceil 8U_{\max}T \rceil)$ intervals and choose $N = m + \ell$ at each step. The resulting approximation has degree at most

$$M = 3^K(m_{\text{initial}} + \ell - 1) - (\ell - 1)$$

and relative norm error at most $(1 + 2^{-\ell})^K - 1$. The quadratic coefficient bound then proves $\|[Y_a, X_a]\| \to 0$ for every finite-time field polynomial X_a; choosing $2^{-\ell} \le a^3/(2K)$ gives an explicit $O(a^3 \log(1/a))$ bound at fixed time. Therefore

$$\|Y_a X_a \Omega_a\| \le \|[Y_a, X_a]\| + \|X_a\| \, \|Y_a \Omega_a\| \longrightarrow 0.$$

The same argument covers initial composite fields, whose degree is three.

Adjoining their equal-time smearings to the common subsequence constructs operators $D_b(f)$ alongside the original $C_b(f)$, with

$$\{D_b(f), D_e(g)^*\} = \delta_{be}\langle f, g\rangle I, \qquad \{D_b(f), C_e(g)^*\} = \{D_b(f), C_e(g)\} = 0.$$

Together with the original CAR and vanishing same-family anomalous anticommutators, these are two copies of the eight-valley CAR algebra. This is algebraic independence; the state is not asserted to factorize or obey Wick's rule. The original correlations and their cyclic positive-energy representation are preserved. Its continuous time action is not automatically extended to the entire enlarged GNS space: the composite derivative contains kinetic terms of order $1/a$. Thus the result resolves the contact-force channel but does not determine whether its bounded-frequency weight survives, escapes to infinite energy, or produces an interacting continuum. It supplies no physical species, Lorentz stress or geometric modular identification. A positive quadratic two-mode control makes this boundary explicit: for $h = \left(\begin{smallmatrix} E_0 & \kappa \\ \kappa & \Lambda \end{smallmatrix}\right)$, with $E_0\Lambda > \kappa^2$ and $\Lambda > E_0 > 0$, the memory Gram stays κ^2 but $\Sigma(z) = \kappa^2/(z - \Lambda) \to 0$. The field error from free energy E_0 is at most $(2\kappa + \kappa^2|t|)/(\Lambda - E_0)$. The full proof and exact controls are in `model/continuum_composite_channel`.

9.71 Regional modular data on the full particle ground.

The invariant finite reference $\rho_G = P_G/5$ now supplies regional modular data on the same ground used for the particle spectrum. In the hidden spin frame, its highest-weight vector is the Slater determinant of four negative spatial orbitals, with projector $P_- = (I - A/\sqrt{3})/2$. Averaging a common spin rotation yields ρ_G. For a region R, let λ_j be the eigenvalues of the spatial compression $(P_-)_R$. The Slater Schmidt decomposition and the spin average give, for every occupied natural-orbital subset I,

$$w_I = \prod_{j \in I} \lambda_j \prod_{j \notin I} (1 - \lambda_j), \qquad \rho_{R,I} = \frac{w_I}{|I| + 1} P_{\text{sym},I}. \tag{13}$$

Here $P_{\text{sym},I}$ projects onto the symmetric spin space of dimension $|I| + 1$; all other states have zero weight. Indeed phase integration deletes unequal spin counts, and the remaining diagonal integral is $\binom{k}{l} \int_0^1 t^{k-l}(1 - t)^l \, dt = 1/(k + 1)$. Thus, retaining the complete kernel,

$$S(\rho_R) = \sum_j h_2(\lambda_j) + \sum_I w_I \log(|I| + 1). \tag{14}$$

For $0 < \lambda_j < 1$ the modular generator on the positive support is

$$K_R = -\sum_j \log(1 - \lambda_j) I + \sum_j \log \frac{1 - \lambda_j}{\lambda_j} N_j + \log(N_R + 1). \tag{15}$$

Endpoint occupations are handled as fixed blocks, without taking logarithms of zero. This is a support identity, not a full-CAR faithfulness claim.

Every adjacent two-site region has positive spectrum

$$\left\{\frac{1}{6}\ [1],\quad \frac{2-\sqrt{3}}{12}\ [2],\quad \frac{2+\sqrt{3}}{12}\ [2],\quad \frac{1}{18}\ [3]\right\}, \tag{16}$$

where brackets denote multiplicities. Its eight other eigenvalues are zero. The support consists of the vacuum, the four one-particle states, and the three two-particle triplet states. With H_R the actual internal regional Hamiltonian, whose contact term vanishes on this support,

$$K_R = \log 6\, I + \beta_R H_R + \log(N_R + 1), \qquad \beta_R = 16 \log(2+\sqrt{3}). \tag{17}$$

Exact signed CAR traces of all five source ground vectors verify this identity on all twelve cube bonds; the sixteen nonadjacent pairs give $K_R = \log 4\, I + \log(N_R + 1)$ instead. The same local spin-frame change conjugates the density, support and Hamiltonian back to the original source coordinates.

For every Hermitian trace-zero variation δ within the positive support, including noncommuting variations,

$$DS_{\rho_R}(\delta) = \beta_R \operatorname{Tr}(\delta H_R) + \operatorname{Tr}\big(\delta \log(N_R + 1)\big). \tag{18}$$

The second term is central on the supported number-preserving algebra, not on the full CAR algebra. Since $\lambda^+_{\min} = (2-\sqrt{3})/12 > 1/48$, for $\delta = \sigma - \rho_R$ with σ supported there and $0 \le \epsilon < 1$,

$$0 \le D(\rho_R + \epsilon\delta \| \rho_R) \le \frac{24\epsilon^2 \|\delta\|^2_{\mathrm{HS}}}{1-\epsilon}. \tag{19}$$

In contrast, a density $\sigma_\perp$ in the orthogonal kernel obeys the full-state identity $S((1-\epsilon)\rho_R + \epsilon\sigma_\perp) = (1-\epsilon)S(\rho_R) + \epsilon S(\sigma_\perp) + h_2(\epsilon)$, whose derivative diverges at zero. The kernel is therefore retained in the operational domain; no supported first law is applied to such a disturbance. These are same-ground finite modular and entropy inputs. Identifying their central term with a geometric edge, their energy pairing with a diamond charge, or their coefficient with a physical clock still requires the existing Einstein theorem's common geometric and scaling premises.

9.72 Deriving a centered clock and constructing its new ground.

The CPT obstruction motivates a separate, explicitly changed charge clock. On a bipartite Weyl lattice with V sites, the one-particle antiunitary $J = \eta(i\sigma_y)\mathcal{K}$, where $\eta_x = (-1)^{x_1+x_2+x_3}$ and $\mathcal{K}$ is conjugation, anticommutes with the hopping. Its canonical particle-hole transform satisfies

$$\mathcal{P}(N) = 2V - N, \qquad \mathcal{P}(H) = H + U(V-N), \qquad \mathcal{P}(H-\kappa N) - (H-\kappa N) = (U-2\kappa)(V-N).$$

Thus $K = H - UN/2$ is the unique scalar charge centering invariant on the full Fock algebra. This changes charged dynamics; the original Hamiltonian and its earlier ground are not silently replaced.

The actual cube gives a rigorous changed-reference test. Its retained all-number theorem states $H \ge -\sqrt{3}/4$. The old $N = 4$ ground has centered energy $-2-\sqrt{3}/4 > -4$, while a singly occupied $N = 8$ product state has expectation -4. Moreover, for $N \le 7$, $K \ge -7/2 - \sqrt{3}/4 > -4$, and particle-hole symmetry transfers this bound to $N \ge 9$. Every new global centered ground therefore has exactly eight particles. A product of four exact dimer grounds on a perfect matching improves

the variational bound: inter-dimer hopping expectations vanish because each factor has fixed charge. Hence

$$-4-\frac{\sqrt{3}}{4}\le E_K\le -2-\frac{\sqrt{17}}{2},\qquad \inf\operatorname{spec}(K|_{N\neq 8})-E_K\ge \frac{\sqrt{17}}{2}-\frac{\sqrt{3}}{4}-\frac{3}{2}>\frac{1}{8}.$$

These are exact full-sector bounds, not relativistic rest masses.

We additionally import Lieb's repulsive Hubbard ground-spin theorem, in the form of Theorem 1, printed p. 8, of his 1993 review; see also the 1989 result and its erratum. The actual hidden-frame hopping is real, spin independent and connected, with a balanced bipartition of four plus four sites and strictly positive onsite repulsion. These finite hypotheses are checked from the source; the theorem allows mixed real hopping signs. Together with the separately proved $N=8$ ground sector, it gives a unique global centered cube ground with hidden spin zero. The review's centered interaction convention differs from K only by the constant 2. No quantitative neutral gap, physical spin dictionary or thermodynamic uniqueness follows here.

The actual cube also has a faithful full-CAR one-site state. Write $K=T+D-4$ on $N=8$. The source identity $A^2=3I$ gives $\|T\|=\sqrt{3}/2$. If P_0 projects onto the 256 singly occupied configurations, every nonzero hopping leaves this subspace, so $P_0TP_0=0$. The source's signed vertex flips are transitive. Set $p=\omega(1-P_0)\le\omega(D)$ and $\delta=\sqrt{17}/2-2>3/50$. The block Cauchy–Schwarz estimate gives

$$|\omega(T)|\le 2\|T\|\sqrt{p(1-p)}+\|T\|p\le 3\|T\|\sqrt{\omega(D)},\qquad \omega(T)+\omega(D)\le-\delta.$$

Thus the common site doublon expectation $d=\omega(D)/8$ satisfies $d\ge\delta^2/54>1/15000$, while $d\le(\|T\|-\delta)/8<1/8$. Gauge, hidden-spin and particle-hole invariance determine the complete one-site state and its modular operator:

$$\rho_x=\operatorname{diag}(d,\tfrac12-d,\tfrac12-d,d),\qquad K_x=-\log(\tfrac12-d)I+\log\left(\frac{\frac12-d}{d}\right)(N_x-1)^2.$$

In particular, $\lambda_{\min}(\rho_x)>1/15000$ and $\|K_x\|<\log 15000$. For every same-site density σ, put $\Delta=\sigma-\rho_x$ and $\rho_\epsilon=\rho_x+\epsilon\Delta$. The full-site entropy first law and remainder are

$$\left.\frac{dS(\rho_\epsilon)}{d\epsilon}\right|_0=\operatorname{Tr}(\Delta K_x),\qquad 0\le D(\rho_\epsilon\|\rho_x)\le\frac{7500\epsilon^2}{1-\epsilon}\|\Delta\|_{\mathrm{HS}}^2,\quad 0\le\epsilon<1.$$

This finite result uses no fitted ground vector and covers noncommuting variations. It is not an infinite-ground faithfulness theorem or an identification of the modular operator with a geometric charge.

For a dimer with hopping $t>0$, let $s=\sqrt{U^2+16t^2}$. Its unique full-Fock centered ground is a singlet/doublon superposition with energy $-(U+s)/2$. Its one-site density is

$$\rho_{\mathrm{site}}=\operatorname{diag}(d,\tfrac12-d,\tfrac12-d,d),\qquad d=(1-U/s)/4>0.$$

Thus this changed reference is faithful on the full local CAR algebra. For a fixed local spin, both addition and removal have energies and weights

$$g_\pm=s/2\pm t,\qquad w_\pm=(1\mp 4t/s)/4.$$

Each charge branch has total weight $1/2$; zero weights are retained. At the actual bond parameters $U=1,t=1/16$, the positive energies are $(2\sqrt{17}\pm1)/16$, and the signed second moment is $65/256$.

Normalized full centered-ground projectors, with a tracial exterior, are particle-hole, gauge and hidden-spin invariant. Translation averaging and the existing boundary/volume and local-dynamics estimates give a new infinite-volume ground with positive centered GNS energy and exactly one particle per site in expectation. The old unshifted state is not reused. The exact canonical composite $d = (2n_{\text{opposite}} - 1)c$ gives

$$[K, c(f)] = -c(hf) - (U/2)d(f), \qquad \|d(f)\| = \|f\|.$$

Consequently the same all-word compactness construction supplies a centered eight-valley continuum net. Its moment identities are $M_1 = h$, $M_2 = h^2 + U^2/4$; only the corresponding second-moment upper bound is passed to the continuum. Particle-hole symmetry pairs the complementary valleys and equates averaged addition/removal measures. This resolves the earlier first-moment obstruction, without establishing full spacetime CPT or Lorentz covariance.

There is also a stronger microscopic occupation bound. On a positive kinetic band with $\epsilon \geq \delta > 0$, the GNS maps $Cf = c(P_+ f)\Omega$, $Df = d(P_+ f)\Omega$ obey $LC + C\epsilon = -(U/2)D$. Integrating $e^{-tL}Ce^{-t\epsilon}$ and using $L \geq 0$, $\|D\| \leq 1$ gives $\|C\| \leq U/(2\delta)$. Translation invariance, arbitrary measurable band windows and rational thresholds yield $w(k) \leq U^2/(2c^2 \sum_j \sin^2 k_j)$, including the negative-band hole estimate. Folding into the node cells and integrating over a circumscribed ball gives

$$\int \frac{d^3k}{(2\pi)^3 \sum_j \sin^2 k_j} \leq \frac{\pi\sqrt{3}}{2} < \frac{11}{4}, \qquad D \leq \min\{2, 11(U/c)^2/8\}.$$

The same argument on the original uncentered ground uses $\|B\Omega\| \leq \sqrt{r}$ and $r \leq 1/2$, giving the separate bound $D \leq \min\{2, 11(U/c)^2/4\}$. Neither bound is uniform on a normalized packet in a shrinking momentum region. Native preparation of the new ground, surviving continuum interactions, measured particle identities and the common geometric Einstein premises remain further requirements. The exact source checks and proofs are in `model/centered_source_ground`.

9.73 The centered bounded-contact limit is a free field theory.

The preceding centered construction admits a stronger conclusion. For $c(a) = v/a$, fixed $v > 0$ and $0 \leq U(a) \leq U_{\max} < \infty$, every retained continuum net of the canonical charged fields is the free eight-valley Weyl vacuum net. This determines the full field hierarchy, not only a two-point approximation. Its particles and holes have $E(p) = v|p|$ and zero rest masses. These modes are not identified with measured species; the bounded-contact branch does not supply the required interacting theory or its massive sectors.

The proof begins with the exact canonical composite $d = (2n_{\text{opposite}} - 1)c$ and the centered occupation bound $\Delta \leq 11u^2/8$, where $u = U/c$. At microscopic time s/c, write $F_c(s; f) = \omega(\{\tau_{s/c}(d(f)), d(f)^*\})$, and let F_0 denote the free filled-sea response at normalized hopping speed one. A degree-m CAR polynomial has centered onsite interaction frequencies in $[-m/2, m/2]$. Bernstein's bound and Duhamel give the degree-three dynamical error. For the state error, the cubic/cubic anticommutator has nine single contractions leaving degree-four words; the two cubic/linear terms leave degree-two words. Their local Wick errors are at most $8\sqrt{\Delta}$ and $2\sqrt{\Delta}$. Absolute row and column sums of the free propagator are at most $e^{3|s|}$, and the contracted factor retains the Schur kernel joining the two smearing indices. Consequently, uniformly over all lattice smearings,

$$|F_c(s; f) - F_0(s; f)| \leq \left(3u|s| + 312\sqrt{\Delta}\, e^{9|s|}\right)\|f\|^2.$$

No divergent $\|f\|_1^2$ estimate is used.

The free onsite spin covariance is $I_2/2$, so normal ordering gives $d = 2{:}n_{\text{opposite}}c{:}$ with no linear component. It creates three positive-energy excitations, whose total energy is the sum of three values $e(k) = (\sum_j \sin^2 k_j)^{1/2}$. The spin and antisymmetrization bounds give coefficient sixteen per charge branch. Two independent internal momenta must have $e(k) \le \epsilon$; each such set has normalized volume at most ϵ^3. Thus the complete signed free measure obeys

$$\mu_{0,f}([-\epsilon, \epsilon]) \le 32\epsilon^6 \|f\|^2.$$

All eight nodes and both positive physical excitation branches are retained. Negative response frequency labels removal, not a negative ground excitation energy.

Gaussian smoothing now bounds the actual centered composite weight in any fixed physical energy window:

$$\mu_{c,f}([-R, R]) \le \eta_c(R)^2 \|f\|^2,$$

$$\eta_c(R)^2 \le \min\left\{1, e^{R^2/(2c^2\epsilon^2)} \left(1536\epsilon^6 + \frac{3u}{\epsilon} + 624\sqrt{\Delta}\, e^{81/(2\epsilon^2)}\right)\right\}.$$

First send $a \to 0$ at fixed ϵ, then $\epsilon \to 0$. This gives $\eta_c(R) \to 0$ uniformly in unit smearings. The composite retains its equal-time CAR norm while its ground spectral weight escapes every finite energy interval.

The next steps use both charge branches in the positive ground GNS space. For the annihilation map $Af = c(f)\Omega$ and composite map $Df = d(f)\Omega$, the source equation is $LA + Ah = -(U/2)D$, with conjugate input space understood. On $h \ge \delta > 0$, integration against the two contraction semigroups gives, after splitting output energies at R,

$$\|\mathbf{1}_{[0,R]}(L)A\| \le \frac{U_{\max}\eta_c(R)}{2\delta}, \qquad \|\mathbf{1}_{(R,\infty)}(L)A\| \le \frac{U_{\max}}{2(R+\delta)}.$$

The negative-band creator has the same bound. Taking the two limits and using the CAR determines the entire initial-time state as the pure free filled sea. On a general bounded kinetic window $\|h\| \le P$, Duhamel followed by integration by parts above R gives

$$\|[C_a(f,t) - C_a(e^{ith_a}f, 0)]\Omega_a\| \le \frac{U_{\max}}{2}\left(|t|\eta_c(R) + \frac{2 + |t|P}{R}\right)\|f\|.$$

The adjoint difference satisfies the same estimate. Passing to the continuum therefore gives both vacuum identities for the free Weyl generator $h_b = iv\sum_j (-1)^{b_j}\sigma_j\partial_j$.

Vacuum identities alone would not prove a free hierarchy. Here spatial locality supplies the missing implication. The free filled-sea vacuum is cyclic for the initial CAR algebra outside any bounded spatial region. Indeed, a vector in either Weyl energy band orthogonal to all outside test functions would have compact support. Its Fourier transform would be entire, but the band equation at fixed nonzero transverse momentum contains $\sqrt{z^2 + r^2}$. Continuation around one of its simple branch points reverses that square root while the entire vector and polynomial Weyl matrix remain single-valued, forcing the vector to vanish. This proves density of the projected outside test functions in both bands, separately in every valley. Their particle and hole creation vectors span the one-particle space; Wick recursion then spans all finite exterior products.

For compact smooth f, let $A_b(t, f) = C_b(f, t) - C_b(e^{ith_b}f, 0)$. The retained finite propagation cone and free Weyl finite propagation localize both terms inside one bounded enlargement of the support. Graded locality and $A_b(t, f)\Omega = A_b(t, f)^*\Omega = 0$ make both differences vanish on the dense outside-polynomial vacuum vectors. The initial CAR cyclic space therefore reduces every time field, with free action. Full all-time GNS cyclicity makes it the entire Hilbert space. Thus

$A_b(t, f) = 0$ as an operator and the complete canonical charged-field hierarchy is free. No Lorentz axiom or external renormalization theorem was assumed in this argument.

This excludes a surviving interaction and positive rest masses on the stated centered bounded-U branch. It does not decide unbounded coupling trajectories, different field scalings, physical species identifications or the common gauge and Einstein requirements. The full proof, exact bound calculators and countercontrols are in `model/continuum_free_limit`.

9.74 Other finite interfaces.

The full-CAR locality bounds used above control dynamics at fixed lattice spacing; a chiral spatial continuum remains a further limit. For partition expectations E_c and positive rates,

$$L = \sum_c \gamma_c (E_c - I), \qquad \operatorname{Tr}(X^* L X) = -\sum_c \gamma_c \|X - E_c X\|_{\mathrm{HS}}^2$$

identifies $\ker L$ with the intersection of the partition commutants, without assuming commuting pinchings or independent no-cancellation. Applying this theorem requires the stated generator. Equivariant rational chain maps and homotopies on the supplied finite support tower are homology maps, not quantum channels or a dynamical refinement limit.

9.75 What the negative controls exclude.

Several finite benchmarks constrain possible identifications. A supplied Weyl/contact action has an isolated moving bound-pair band, complete contact residues and an optical identity, but its positive contact section makes the band topologically trivial. Its certified curvature fails the rest/kinetic mass relation with the same speed parameter. Contact interactions also couple proper Weyl momentum pockets to their complements. The lattice scattering theorem, finite thermal positivity and KMS identities do not remove those obstructions or supply physical asymptotic detectors, a chiral continuum or calibrated cross sections.

The centered repulsive dimer gives a sharper family-wide exclusion. For $t = c/2 > 0$, $U \geq 0$, $\mu = U/2$ and $d = \sqrt{U^2 + 16t^2}$, its two charge-one doublets have gaps $g_\pm = d/2 \pm t$, hence

$$1 < g_+/g_- \leq 3.$$

No common positive linear conversion yields the charged-lepton hierarchy or three distinct same-charge families there. Allowing another chemical potential can evade the ratio bound but does not create a third doublet. The fixed complete charged-frequency catalogue also fails its broader lepton-ratio comparison. The periodic four-site extension fails its declared $E(k)^2 = m^2 + c^2 \sin^2 k$ benchmark at $c = 1/8$; a two-parameter fit to two distinct momentum values does not rescue that test. These exclude stated finite dictionaries, not every larger action or a renormalized continuum limit. Electron/positron masses set by one common measured calibration, and retrospective parameter comparisons, provide no independent mass prediction.

Finite current response also has a precise scope. An open-bond phase is pure gauge and measures no magnetic moment; the contact kernel is not a Coulomb kernel. On the complete periodic ring, a genuine loop phase has gauge-invariant current and total curvature,

$$E_0' = \langle \partial_\phi H \rangle, \qquad E_0'' = \langle \partial_\phi^2 H \rangle - 2 \sum_{n \neq 0} \frac{|\langle n | \partial_\phi H | 0 \rangle|^2}{E_n - E_0},$$

under the isolated-ground hypothesis. Degenerate free and atomic controls lie outside that formula. The calculation retains all 256 Fock states but does not identify a photon or physical Lorentz spin.

Clock and memory exclusions are similarly conditional. A frozen real-operation grammar has a normalized trace-distance lower bound $1/2$ for a specified nontrivial modular action of its nontracial reference. Finite independent copies of the original native reference also fail the required field modular-gap ratios. These do not exclude the explicitly enlarged clock and control laws above. The earlier pair/dark-reference encoding covers a fixed corner, not the whole spatial field. Finite stationary observer references concern source/buffer marginals at the stated repair rate and on their recurrent support; delayed variants need their own construction, and growing exported histories are not made stationary by these marginal references. Actively corrected coherent memories can have flat logical energy and a fixed core, so memory lifetime alone does not establish a moving massive particle. Properties proved on distinct carriers require an explicit intertwiner before they can be combined.

9.76 Physical target and remaining obligations.

The target is an interacting quantum field theory with gapless photons and experimentally matched massive sectors, not a positive global mass gap. A gapless photon rules out such a global gap. An exact nontrivial positive-energy conformal theory in a flat-space vacuum representation with unbroken unitary dilations also cannot have a positive flat-space vacuum mass gap: for a spectral value $E > 0$ and proposed gap $g > 0$, scale covariance also puts $g/2$ in the spectrum. This does not exclude a gap in scaling dimensions or discrete finite-radius cylinder levels; the cylinder radius supplies an energy scale. Conformal behavior may describe a critical or short-distance regime, but a deformation or symmetry-breaking mechanism responsible for massive flat-space sectors must then be specified and implemented.

The finite results leave concrete physical joins. A localized defect must carry the protected charge and the same measured field observables. Spatial refinement must control the additional lattice modes and yield the required Lorentz propagation, while keeping the interactions and reference state. The electromagnetic field and gauge–Higgs interactions must be dynamical, with charge constraints, detector coupling and their infrared behavior treated on that same theory. Massive excitations need a common energy and clock normalization, including the relation $E''(0) = c^2/E_0$ for a relativistic branch with $E_0 > 0$, together with the appropriate stress and proper-time relations. The neutrino sector and other measured particle properties require their own matching tests. Neither a successful finite source compilation nor a calibrated spectrum alone discharges these obligations.

The next section uses the resulting energy, faithful-reference and observation interfaces to identify what a gravitational reconstruction requires. Its geometric hypotheses are stated independently of the finite entropy identities just established.

10 The Einstein equation from a realized null balance

The gravitational hypothesis identifies Einstein dynamics with the coarse response of the same process.

Theorem 10.1 (Null tomography and metric ambiguity)**.** *Let V be four-dimensional with Lorentz form η of signature $(-,+,+,+)$, the opposite convention from Q in Section 7.2; nothing below depends on the choice. If a symmetric form X satisfies $X(k,k) = 0$ for every null vector k, then $X = \phi\eta$. Null-null values therefore determine a symmetric tensor modulo the metric line, and nine explicit null directions, the six axis rays $(1, \pm e_i)$ and $(1, s, s, \pm s)$, $(1, s, -s, s)$ with $s = \sqrt{3}/3$,*

reconstruct the quotient $\mathrm{Sym}^2(V^*)/\mathbb{R}\eta$ *with design determinant* $8192/27$ *in the certificate's basis and a decoder obeying* $\|x\|_\infty \le (2+\sqrt{3})\|q\|_\infty$.

Proof. On future null rays $(1, n)$ the identity $X_{00} + 2X_{0i}n_i + X_{ij}n_in_j = 0$ on the unit sphere forces $X_{0i} = 0$, $X_{ij} = \phi\delta_{ij}$ and $X_{00} = -\phi$. The determinant and decoder bound are exact calculations, and the design, injectivity and ambiguity are formalized in Lean [1, 9]. □

Theorem 10.2 (Tensor completion of a realized null balance)**.** *Let* (M, g) *be a connected, time-oriented,* C^3 *Lorentzian four-manifold and* T_{ab} *a symmetric* C^1 *tensor with* $\nabla^a T_{ab} = 0$. *Suppose a common realized tower supplies* $\kappa = 8\pi G_N > 0$ *with*

$$G^{(g)}_{ab}k^ak^b = \kappa T_{ab}k^ak^b \tag{20}$$

for every null vector k *at every point. Then there is a constant* Λ *with*

$$G^{(g)}_{ab} + \Lambda g_{ab} = 8\pi G_N T_{ab}, \tag{21}$$

and a reference event at which the difference equals $-\Lambda_0 g_{ab}$ *fixes* $\Lambda = \Lambda_0$.

Proof. Theorem 10.1 applied pointwise to $G^{(g)} - \kappa T$ gives $G^{(g)}_{ab} - \kappa T_{ab} = \phi g_{ab}$. The contracted Bianchi identity, the Ward premise and metric compatibility give $\nabla_b\phi = 0$, connectedness makes ϕ constant, and $\Lambda = -\phi$. □

The structure of the argument is simple. Light rays cannot see a stress term proportional to the metric, so balance between curvature and stress on every null direction fixes Einstein's equation up to such a term, and the Bianchi identity with stress conservation makes its coefficient constant. That constant is the cosmological constant. The coefficients of entanglement-equilibrium arguments [54] enter through one machine-checked identity: scalars with $\delta S_{\mathrm{bulk}} = \frac{8\pi^2\ell^4}{15}t$, $\delta A = -\frac{4\pi\ell^4}{15}f$ and $\delta S_{\mathrm{bulk}} + \delta A/(4G_N) = 0$ obey $f = 8\pi G_N t$. For OPH to supply (20), one common refinement tower must realize (G1) the effective manifold of Proposition 7.4; (G2) geometrically normalized cap modular flow whose directional charges define one stress tensor; (G3) its Ward identity, independently of that reconstruction; (G4) the edge-center entropy split, the normalization $\delta\langle Z_L\rangle = \delta A/(4G_N)$, the modular first law and generalized-entropy stationarity; (G5) a small-ball family with controlled remainders and enough observer directions; and (G6) universal coupling, a vacuum reference and a scale calibration fixing $\kappa = 8\pi G_N$. The finite layer contains null tomography, the edge-entropy split and the coefficient identity, and the machine-checked composition proves the tensor step from an explicit null-balance premise; the bridge between them is (M4).

Section 9 supplies finite local-energy balance and faithful-reference entropy and observation maps on the same field algebra. Conditions (G2)–(G4) require identifying these objects with one Lorentz stress tensor, geometrically normalized cap modular generators and the corresponding area variation along a common refinement. The finite interfaces provide inputs to that construction; they do not establish its geometric identifications or stationarity.

11 Cosmology and the closure coordinates

The cosmological hypothesis asks the same process to fix the vacuum, the dark sector and the primordial response. In this paper the layer consists of conditional branches and diagnostics of declared maps.

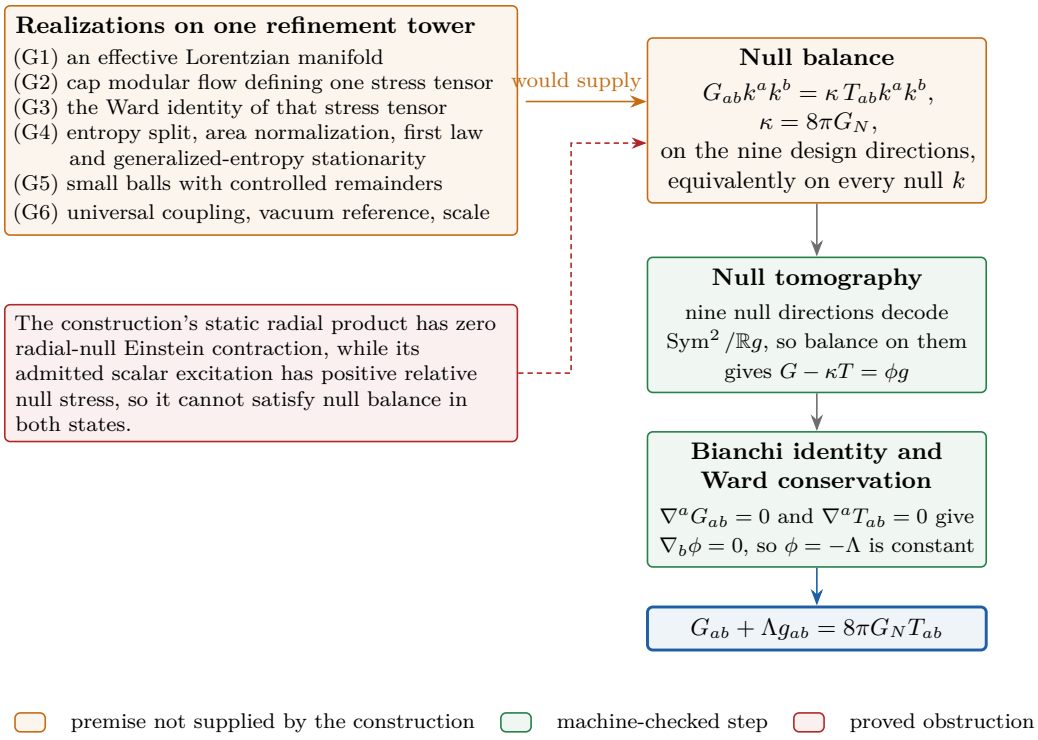


Figure 5: The Einstein equation as tensor completion of a realized null balance (Theorem 10.2). The tensor step is machine-checked from an explicit null-balance premise. Realizations (G1) to (G6) on one refinement tower would supply that premise, and no complete realization of them on one common refinement tower is established; the tested static radial product fails the balance for its own scalar excitation.

11.1 The dark sector: non-luminous charge, the deep galaxy law, and the vacuum coordinate

Under supplied universal-coupling, physical-stress and source-localization premises, sourcing gravity by modular charge classifies any geometric excess over the luminous term as non-luminous modular charge, with anomalous modular energy on overlap collars as one candidate carrier [14]. Conditional on two named premises the deep-regime profile is characterized rather than supplied: an enclosed-mass law that is degree-one scale covariant and composes independent sources in quadrature carries a unique constant $a_0 > 0$ with $M_A(r) = r\sqrt{M_b a_0/G}$, which forces flat rotation curves and gives $v^4 = GM_b a_0$, the baryonic Tully–Fisher relation, each premise load-bearing by an explicit counterexample. A per-cut model gives $a_0 = Gn^2 c$, with only the product $n^2 c$ identifiable, and the value of a_0 is fixed by no source law. Reading the dark-energy density as $\rho_{\mathrm{DE}} = \kappa/N$ at constant κ gives $w = -1 + \frac{1}{3}\,\mathrm{d}\ln N/\mathrm{d}\ln a$, so fixed capacity gives the exact point $(w_0, w_a) = (-1, 0)$ and nondecreasing capacity gives $w \geq -1$; exclusion of that point fails the fixed branch, while agreement supports nothing, since it is also the ΛCDM null [15].

11.2 What closure fixes, and what observation reads off

The architecture carries two dimensionless closure coordinates: the screen grain P and the record capacity N. Under the closure hypothesis these coordinates are fixed points of the closure map, and the loop also licenses reading them off: observers locate a constant by measurement, declare

a map, and certify a fixed point of that map inside the measured basin. Whether a certified fixed point coincides with the measured value is a diagnostic in the sense of Table 2, not a consequence of the hypothesis. The theory stands on its recovered specification whether or not a first-principles evaluation of every coordinate is found.

The screen-grain loop

$$P = \varphi + \sqrt{\pi}/A_T(P),$$

with $A_T(P)$ the inverse coupling a declared map returns at trial grain P and $\varphi = (1+\sqrt{5})/2$, has one interval-certified fixed point per declared map. The bare source map settles at $\alpha^{-1} = 136.994835177413$, a relative distance of 3.0×10^{-4} from the 2022 CODATA value 137.035999177(21) [65]; the self-consistent map, which recomputes the finite-screen unified gauge width inside the loop, settles at $P = 1.6309682414$ and $\alpha^{-1} = 137.035660136947$, a relative distance of 2.5×10^{-6}, the gauge width supplying 99.2 percent of the gap. Both maps carry the hadronic transport by a declared quark-screening ansatz. Replacing the ansatz by the measured hadronic payload gives $\alpha^{-1} = 136.3828$ on the certified interval $[136.3670, 136.3985]$, which excludes the measured value. The certified numerical mismatch is distinct from arithmetic error, but its removal by the omitted physical matching is unproved [10]. Reading the capacity as de Sitter horizon entropy in nats gives

$$N = \frac{3\pi}{\Lambda\,\ell_P^2}, \tag{22}$$

with $\ell_P^2 = \hbar G/c^3$, and the capacity branch gives $N = 3.2921\times10^{122}$ and 3.3001×10^{122} against the coordinate 3.3129×10^{122} read back from the weighted Planck base-ΛCDM chain; in the constant they read $\Lambda = 1.0959\times10^{-52}$ and $1.0933\times10^{-52}\,\mathrm{m}^{-2}$ against $3\Omega_\Lambda H_0^2/c^2 = 1.0891\times10^{-52}\,\mathrm{m}^{-2}$ at the Planck base-ΛCDM centrals, distances of +0.62 and +0.38 percent. Equation (22) is the dictionary that produces the comparison coordinate, so this restates the capacity diagnostic in the constant, and it consumes the measured $\hbar$, G and c through the calibration map (M2). Neither clears the source-only selection and same-quantity gates. Both comparisons are diagnostics of declared maps in the sense of Table 2: the maps were exposed before the comparison, the choice of map is itself declared, and neither comparison carries predictive weight. The executable detector of Section 6 consumes P through the coupling $g(P) = \sqrt{4\pi\alpha(P)}$, and N bounds the correctable records of a declared capacity carrier by $\log M_0 \le N$.

An independent entangled-pair candidate. Wissner-Gross's Chain Theory preprint gives a separate finite spectral construction [105]. Its commuting signed exchanges have gains $1, 2, 3, 6$ with multiplicities $79, 48, 42, 3$, and its stated normalization gives

$$\alpha_{\mathrm{pair}} = \frac{79 + 48\sqrt{2} + 42\sqrt{3} + 3\sqrt{6}}{31104}, \qquad \alpha_{\mathrm{pair}}^{-1} = 137.035999189566\ldots.$$

The construction and expression are due to Wissner-Gross. The OPH comparison independently reconstructs this arithmetic, retains complementary Bell outcomes, and tests support alignment and normalized-readout choices [10]. Identifying the scalar with the physical Thomson coupling is a separate conjecture. Granting it and the same-cell equation gives $P = \varphi + \sqrt{\pi}\,\alpha_{\mathrm{pair}} = 1.630968209402773\ldots$. That substitution does not solve either of the different trial maps above. It selects no family interaction matrix, Higgs vacuum or physical particle mass. The finite comparison imports no claimed lepton mass ratio from the Chain Theory construction.

12 Realized results and physical obligations

A statement about nature needs one physical realization carrying compatible events, fields, action, states and readouts across refinement. This section gives, for each layer, what is established on its stated domain, what is a conditional implication, which join the executable model must realize, and which result would stop the layer or its branch. Exact equality of limiting observable laws is the mathematical target; finite executions and experiments are judged with declared approximation and measurement errors.

12.1 Realized finite domains and physical extension

The finite construction has substantial content before its physical identification. The current attachment map [67] contains 39 explicit finite theorem or model domains, of which 34 have their stated obligations realized. This count concerns those finite domains, not the 84 canonical premise rows in the physical realization register; the two inventories have different domains. This finite attachment tally discharges none of the canonical physical premises as a whole.

The common-process section makes the strongest joins explicit: protected transport sectors and colored CAR exchange; a calibrated field action with all occupation sectors; spatial source factors, a common clock and Higgs particle creation; complete coherent and flagged repair extensions; and energy, faithful-reference and compatible-observation interfaces on the same selected field algebra. These results supply those finite constructions. Their domain restrictions identify what must be extended: a localized physical core, a controlled chiral spatial limit, dynamical gauge fields and a common geometric stress and entropy response.

The table below combines each layer's established result with its remaining physical obligation and decisive negative controls. Detailed statements and proofs occur in the preceding sections, rather than being repeated as a second catalogue here.

12.2 Results and obligations by layer

Table 1 gives the result and obligation of each layer. "Established" means proved on its stated domain or executed with retained receipts; "conditional" means that the conclusion depends on displayed antecedents beyond its realized finite instances; the open join is what the executable model must supply on the same process; and the stopping test is the result that would remove the layer, its branch or the tested law. Negative results are findings about the tested law, stated with their exact scope.

Table 1: Established constructions, physical obligations and negative controls.

Layer	**Established and conditional**	**Physical obligation**	**Negative results and stopping test**
0. Closure and agreement	Established: on the finite consensus branch the closure map is an affine idempotent fixing exactly the packets on consensus normal forms; the record-closure equation is posed exactly. Interpretation and acceptance criterion, not a theorem	A history domain and closure operator for the executable model; interpretation maps fixed independently of validation outcomes, compatible across observers and refinement	The bounded completion class does not identify the solution N; no theorem selects a unique universe. A contradiction among indispensable premises stops the proposal

Table 1 continued

Layer	Established and conditional	Physical obligation	Negative results and stopping test
1. Repair mathematics	Established: canonical observable normalizer; consensus normal form under the transactional diamond; adaptive-run, capacity and fixed-federation bounds; Born weights without continuity axiom, Lüders selection, Tsirelson bound, no-copying; positive residual mismatch from fixed holonomy. Conditional: the finite algebra-state representation of completed records is an input	A source-produced complex-tomographically-complete effect and instrument web with full coexistent-effect additivity; an implemented scheduler within the specified process	Scheduler choices remain explicit inputs; deciding consistent extension is NP-complete for succinct Boolean systems; retained routing logs do not identify federation gluing. As theorems they are refuted only by a counterexample
2. Carrier and repair law	Established: the seam rule from four requirements; the classified convex record-law family; the gap $(5-\sqrt{5})/60$ for every admissible table; the rank-three slow-response completion; one seam per native repair event. Conditional: physical position consumes (M1)	One specified record law and federation gluing validated against the physical process; a physical length	At full-port energy scaling the refining screen loses the rank-three band. The C/E instrument test separates C, E and the current-only law; a resolved dispersion outside the band excludes the whole carrier class
3. Executable observer	Established: native observer with record-caused feedback; electromagnetic and scalar detectors; one interacting process with certified infinite-occupation contrast; overlapping observers exact on 5,768 output words; local event clocks; identification of the record law; coherent observers on matched screens; complete $O(1/d)$ error for the smooth four-read protocol; a continuing C/E finite-buffer observer with counted exports and an exact stationary source/complete-record marginal	One continuing autonomous observer with every service, memory, reset and resource accounted; the declared apparatus, schedules and preparations implemented on that same process	Outcome accuracy does not certify the output state; committed native records form a classical process, so correlations beyond $S = 2$ need declared coherent preparation services

Table 1 continued

Layer	Established and conditional	Physical obligation	Negative results and stopping test
4. Fields and clocks	Established: the native continuum operator of spectral dimension two with Gaussian fields, decay and finite-speed waves; the finite $(1+3)$ causal precursor and count clock; complete spatial CAR source factors, a common calibrated field/repair clock, local energy readouts and faithful Gibbs/Petz observations on the same field. Conditional: the smooth static $(1+3)$ scalar description on a declared radial product	A physically validated clock and energy normalization; an interacting chiral $3+1$ field description from native records; a realized radial observation algebra	Passive repair has no wave sector and implements no nontrivial modular automorphism; the native reference identifies no quantum vacuum; the eigenvalue count excludes an elliptic three-manifold identification of the screen operator
5. Gauge and matter	Established: the forced Lie type; the finite response witness; the exterior scan and $\mathbb{Z}_6$ quotient; protected coherent exchange-history sectors, colored CAR exchange, a full occupation-space repair law and a complete finite repair/field/Higgs/clock experiment. Conditional: the reversible response D and laboratory currents consume (M3); matter identification consumes (M3) and (M5)	A localized native defect sector with its invariant, maintenance energy, kinetics, charge, spin and statistics in the common clock and energy dictionary; the measured spectrum and mixing	The double-load candidate is transient, surviving with probability $(5/6)^n$; the unextended binary class has no nonconstant configuration invariant; the fixed mass laws give $1 : 1.19 : 1.50$ against $1 : 206.8 : 3477$. An exhaustive no-go over a frozen defect class stops that mechanism
6. Gravity	Established: null tomography and the tensor completion from an explicit null-balance premise; same-field finite modular, energy and quantum entropy identities, with a proved regional-retraction obstruction. Conditional: the Einstein equation up to Λ under (G1) to (G6)	(G1) to (G6) realized on one refinement tower with the same matter, modular flow and clocks; defect maintenance energy in the same stress	The static radial product fails null balance for its own scalar excitation; stationary repair entropy is distinct from fixed-volume generalized entropy. Failure of (M4) removes the spacetime and Einstein conclusions together
7. Cosmology and tests	Established: the deep-law uniqueness under two premises; $(w_0, w_a) = (-1, 0)$ at fixed capacity; the de Sitter sign; the dispersion band. Diagnostics: capacity and grain roots	Source-only selection of P, N and a_0; physical encoding and calibration for the C/E test	The measured hadronic payload gives a grain root excluding the measured coupling. Exclusion of the band, a C/E rejection, a charged-lepton average beyond three standard uncertainties from the balanced center, or exclusion of $(-1, 0)$ fails its branch

12.3 The realization maps

Five maps name what a physical realization must supply, and each boundary statement in this paper cites them by label.

(M1) Position and scale. An identification of the intrinsic carrier completion with physical position, with a physical length, cofinal overlap gluing between carriers, and a refinement limit (Section 5).

(M2) Clock and energy calibration. A common comparison between internal evolution, laboratory duration and energy in measured units. The source-executed reference clock measures the same controlled intervals as the field, and the corresponding energy observables have complete detectors. Extending and validating that calibration across the physical process remains necessary. Native event counts are a different clock; the rejected direct count-to-optical-phase attachment in Section 13.3 does not reject the calibrated reference clock.

(M3) Laboratory current. An identification of the finite port response with a physical gauge current, with a physical matter action and a continuum operator limit.

(M4) One common physical tower. A single physical refinement family realizing the event, modular, stress, entropy, continuum, vacuum, and scale premises together, rather than several independently declared islands.

(M5) An implemented and matched action. A specified field and interaction law realized by the same native services, with its complete state space, relative normalizations and operational readouts. Parameters may be declared or calibrated; which predictions follow without consuming their measured targets must be stated. Physical identification requires compatible propagation, currents and interactions across refinement.

The finite constructions realize restricted instances of these maps: a calibrated common clock, internal current operators, a full field action, energy readouts and compatible reference observations. The remaining task is their common physical extension and validation, as specified in Table 1; none of the five full physical maps is yet closed. A theorem that consumes none of them stands on its own finite domain. Section 13.3 tests two simple attachments of native records to published data. The construction paper gives the detailed premise map, including the retained negative controls that bound how the carrier, the response triplets, the feedback class and the kinetic term may be attached [66].

13 Empirical meaning and falsifiability

The custody-bound register fixes target definitions, exclusion thresholds, precision floors, exposure classes and decision rules before eligible comparison [3]. Two kinds of test follow from the construction: a propagation anisotropy fixed by the carrier's symmetry, and an instrument comparison fixed by the executable observer. Published records also test precise attachments of native records retrospectively.

13.1 Propagation anisotropy at angular rank six

Angular rank means spherical-harmonic rank j, not a particle spin. The dimension of the invariant subspace of the sixty proper icosahedral rotations in angular rank j is

$$m_j = \frac{1}{60}\Big[(2j+1) + 15\,U_{2j}(0) + 20\,U_{2j}(\tfrac{1}{2}) + 12\,U_{2j}(\tfrac{\varphi}{2}) + 12\,U_{2j}(\tfrac{\varphi-1}{2})\Big],$$

with U the Chebyshev polynomials of the second kind, and the exact table gives $m_1 = \cdots = m_5 = 0$ and $m_6 = 1$ [49].

Theorem 13.1 (Icosahedral angular-rank universality)**.** *Let the carrier kinetics be any operator* $\lambda_a(k) = a^{-2}\sum_d w(d)\,[1-\cos(a\,k\cdot d)]$ *with a finite direction multiset and weights invariant under the proper icosahedral rotation group. Then the dispersion is exactly isotropic through angular rank five at every order in a, every directional term below angular rank ten is one multiple of the unique normalized invariant* $\mathcal{I}_6$*, and whenever the weighted sixth-moment coefficient is nonzero the first directional artifact appears at order* a^4k^6*, suppressed by exactly* $1/16$ *under one binary refinement at that order. Rank six is the least symmetry-allowed nonzero anisotropic rank in the class.*

Proof. The order-$a^{2m-2}k^{2m}$ angular content is an invariant polynomial of degree $2m$ with harmonic components at even ranks $j \le 2m$, and the invariant table forces zero at $L = 2, 4, 8$ and a one-dimensional space at $L = 6$. The table, the 62-direction stationary census, the orbit verification and the nonvanishing sixth-moment coefficients are certified in exact arithmetic [1]. □

Proposition 13.2 (Primitive twelve-port branch prediction)**.** *Suppose the intrinsic spatial kinetic symbol lies in the cosine class of Theorem 13.1 with scale-independent coefficients, the complete orbit of the twelve port directions* u_i *is its sole hop support through order* k^6*, the carrier scale a is finite and positive with the quadratic term normalized to* k^2*, one carrier rest frame is transported coherently over the experiment, and the tested scalar or polarization-independent sector realizes the operator. Then*

$$\omega^2 = \frac{1}{2a^2}\sum_{i=1}^{12}[1-\cos(ak\,u_i\cdot\widehat{k})] = k^2 - \frac{a^2}{20}k^4 + \frac{a^4}{840}k^6 + \frac{2a^4}{7875}k^6\mathcal{I}_6(\widehat{k}) + O(a^6k^8), \tag{23}$$

with $\mathcal{I}_6(n) = \frac{25}{132}\sum_i P_6(u_i\cdot n)$*, so in a transported frame the branch predicts the scale-free relations*

$$\frac{B_6}{C_4^2} = \frac{32}{315}, \qquad \frac{B_0}{C_4^2} = \frac{10}{21}, \qquad \frac{B_6}{B_0} = \frac{16}{75}, \tag{24}$$

and every intrinsic anisotropic coefficient at $1 \le j \le 5$ *vanishes.*

Proof. The exact carrier moments $\sum_i(u_i\cdot n)^2 = 4$, $\sum_i(u_i\cdot n)^4 = 12/5$ and $\sum_i(u_i\cdot n)^6 = 12/7 + (64/175)\,\mathcal{I}_6(n)$ substituted into the cosine series give the expansion; eliminating a gives the ratios. □

A second frozen ray follows from the source seams.

Proposition 13.3 (Source-seam edge ray and conditional propagation branch)**.** *The boundary current of the complete thirty-seam source packet maps through the antipodal-odd load readout onto the even-sum module* D_6 *of Theorem 5.1, with Smith invariants* $(1,1,1,1,1,2)$*, and every directed seam defines one translation of the internal record carrier whose complete normalized direction multiset is the thirty-direction edge orbit. If the feasible move laws and their objective are natural under the proper carrier action and A3 supplies a unique normalized minimizer, transitivity forces weight* $1/60$ *on every directed seam, and the selected internal operator is the source-counting homogeneous convolution.*

Suppose these record translations are physical displacements under one homogeneous position action, that the complete edge orbit is the sole direct support through the displayed order, that proper-carrier covariance fixes one common weight, that the same operator acts on one scalar or polarization-independent physical sector with quadratic term normalized to k^2*, and that the action carries the stated gluing, scale, frame, readout, nuisance, and exclusivity data. Its spatial kinetic symbol is then*

$$\Lambda_a(k,\widehat{k}) = \frac{1}{5a^2}\sum_{j=1}^{30}\Big[1-\cos\Big(ak\,w_j\cdot\widehat{k}\Big)\Big], \tag{25}$$

with $C_4 = -a^2/20$, $B_0 = a^4/840$ *and* $B_6 = -a^4/12600$, *hence* $B_0/C_4^2 = 10/21$, $B_6/C_4^2 = -2/63$ *and* $B_6/B_0 = -1/15$. *The intrinsic anisotropic coefficients at angular ranks one through five vanish, and the rank-six vector lies on the rotated* $\mathcal{I}_6$ *orbit.*

Proof. Exact integer incidence gives the displayed image and Smith invariants, and the response-selected Gram pullback gives the completion of Theorem 5.1. The sixty directed labels map two-to-one onto the thirty signed edge directions, whose unit-normalized moments are $\sum_j (w_j \cdot n)^2 = 10$, $\sum_j (w_j \cdot n)^4 = 6$ and $\sum_j (w_j \cdot n)^6 = 30/7 - (2/7)\,\mathcal{I}_6(n)$; substitution in (25) gives the coefficients and ratios. The incidence, quotient, orbit binding and moment identities are machine-checked in exact arithmetic [1]. □

The proposition contains an exact finite source theorem and a conditional physical branch. The naturality and minimizer clause supplies a homogeneous internal action whose generator satisfies the positive maximum principle and an exact Dirichlet identity, and whose plane-wave spectrum is the symbol above. The declared symbol also carries an exact remainder certificate with no physical premise: with $q = ak$, $|\widehat{\Lambda} - P_6| \le \frac{7}{388800} q^8$ and $\frac{19}{20} q^2 \le \widehat{\Lambda} \le q^2$ uniformly on $0 \le q \le 1$, where $P_6 = q^2 - q^4/20 + (1/840 - \mathcal{I}_6/12600) q^6$; the certificate turns a into no SI length. The edge branch and the primitive-port branch have opposite rank-six signs, and their coefficient rays, custody packages, exposure exclusions and comparison budgets are separate.

Theorem 13.4 (Carrier-class dispersion band)**.** *For the declared positive-weight scalar cosine class over orbits* O_s *of the sixty proper carrier rotations with radii* $r_s > 0$, *weights* $w_s > 0$ *and the* k^2 *coefficient normalized to one, write* $\Lambda_{\mathrm{norm}} = k^2 + C_4 k^4 + (B_0 + B_6 \mathcal{I}_6(\widehat{k})) k^6 + O(k^8)$ *and* $\mu_m = \sum_s w_s |O_s| r_s^m$. *Then* $C_4 = -(a^2/20)\mu_4/\mu_2 < 0$ *for every member;* $B_0/C_4^2 = (10/21)\mu_2\mu_6/\mu_4^2 \ge 10/21$ *with equality exactly on single-radius members; the anisotropic ranks one through five vanish; and* B_6/B_0 *lies in* $[-16/135, 16/75]$, *the face and vertex orbits attaining the endpoints and the edge orbit sitting at* $-1/15$.

Proof. Summing over the sixty rotations gives $\sum_g ((gu)\cdot\widehat{k})^6 = \frac{60}{7} + \frac{64}{35}\mathcal{I}_6(\widehat{u})\mathcal{I}_6(\widehat{k})$ on unit vectors. The floor is the Lagrange identity $\mu_2\mu_6 - \mu_4^2 = \sum_{i<j} W_i W_j r_i^2 r_j^2 (r_i^2 - r_j^2)^2 \ge 0$ with $W_s = w_s|O_s|$, and the band holds because the rank-six numerator is a positive-weighted mean of seed values of $\mathcal{I}_6$ in $[-5/9, 1]$ [1]. □

A resolved intrinsic dispersion with B_0/C_4^2 below $10/21$, or B_6/B_0 outside $[-16/135, 16/75]$, or a residue at ranks one through five, excludes every member of the declared class at once. The primitive-port relation is a frozen prospective conditional physical-branch prediction: its target statement, coefficient snapshot, registration record, public commit custody and detached calendar attestations fix the prospective content, the dated template search whose icosahedral templates returned a family-wide p-value of 0.64 is ineligible, and a verdict requires a dataset-specific registration with a joint likelihood, frames, orientation profile, nuisance model and calibrated threshold. The branch fails at five standard deviations or more if an isolated intrinsic C_4 is positive, a lower-rank coefficient is nonzero, or the calibrated profile likelihood excludes the complete manifold $C_4 < 0$, $B_0 = \frac{10}{21} C_4^2$, $B_6 = \frac{32}{315} C_4^2$. A null is inconclusive, since the branch supplies no positive lower bound on a, and a failure rejects the primitive twelve-port propagation branch; it reaches the framework as a whole only after a stronger source law makes that branch forced and exclusive [3].

13.2 An instrument test from the executable observer

Under a common current-record interface every admissible record law has the classical statistics of C and E. The multiplier g, the coherence between an empty seam and a waiting single load,

separates C, E and the law that records only the current. On a fixed coherent two-state source preparation the readout probability is $3/8 + \frac{1}{2}\,\mathrm{Re}(g\,e^{-i\theta})$: reversing the phase moves C from 5/8 to 1/8, E predicts 3/8 for both, and the law that records only the current predicts $3/8 \pm \sqrt{2}/4$ [69]. Four fixed settings include common-phase and incoherent controls. The complete-count evaluator uses 8192 trials per setting, an independently established probability-error budget of 1/64 and an exact sampling certificate below 1/1000; it rejects malformed or incomplete records and reports *no data* when no independent counts are supplied. A source-executed implementation runs both complete 78-state instruments, the four preparations and a complete Lüders readout from twelve-port control words, retaining every record and attempted trial [66]. A sequential design on one continuing carrier needs neither exact resets nor independent trials: 379 discarded native attempts before each of 3,927 trials replace the reset, failed heralds count as trials, and conditional-law bounds on the dependent transcript give a wrong-candidate probability of at most 3/400 under the exact-source model [88]. Synthetic controls exercise the decision rules and establish no physical verdict; physical encoding, independent calibration and comparison data are the open requirements.

13.3 Native records against public data

Published records can test precise attachments of native records without a new laboratory realization; each test below is retrospective, states its readout hypotheses, and assigns no prospective verdict.

The diagonal native branch is a classical stochastic process, so under independently chosen settings and local factorized responses it obeys the CHSH bound $S \le 2$. The deposited records of the loophole-free Hensen experiment [91] reproduce 245 eligible trials, 196 wins and $S = 2.42250$, and the published fixed-count analysis gives a tail bound near 0.03917 with its setting-calibration allowance [89]. This rejects the local classical reading of the diagonal branch under that experimental contract. Committed native records form a classical process: the native grammar keeps diagonal inputs diagonal under every classical feedback history, and two implementations with the same committed branches, which give 3/4 and 1/4 under a read before commitment, give 1/2 for both once the record is committed [99]. Correlations beyond $S = 2$ therefore need uncommitted coherence from declared preparation services, and every admissible record law attenuates it [69]. The replay leaves the completed record surface untouched, on which a declared entangled state reaches the Tsirelson bound of Theorem 4.5, and it does not distinguish C from E.

Event clocks need no external validation, but identifying their readings with a published measurement is a testable hypothesis. Let two native ports, or two distinct regions of one carrier, count incident repairs, and attach the published 300-second blocks of the Ar^{13+}/Yb^{+} optical frequency ratios of King et al. [92] to the normalized count ratio through a gain of at least one, a declared bracket on the Yb^{+} frequency and an absolute error of 10^{-12} per block, with direct phase-count readback. The complete count law, after charging every choice of 2021 subwindow, gain and retained blocks, bounds the probability of the observed block ranges below 10^{-104} for two native ports and below 10^{-22} for every pair of the 3,433 distinct regional event sets [90]. The direct attachment is rejected together with its readout assumptions; the experiment's interrogations and servos lie outside that hypothesis, and the native event clocks are unaffected as clocks of the model.

Published trapped-ion records test a supplied experimental model executed through the source interface. Two calibrated preparation parameters predict 100 reserved complex readouts with per-spin-mean root-mean-square error 0.06647, and a changed-potential transfer predicts 125 derived observations with root-mean-square error 0.2876 in position-estimator units [68]. These comparisons test the supplied oscillator model and derive neither its action nor its apparatus from native repair.

13.4 Diagnostics and what fails together

The other quantitative comparisons carry their own decision shapes and do not count as predictions. The closure roots of Section 11.2 are diagnostics of declared maps. The registered conditional primordial branch $n_s = 1-P/48 = 0.96602$, at the calibrated grain P, lies at the lower 0.612 percent position of the official ACT DR6, Planck and DESI joint posterior chains, whose mean is 0.9743 ± 0.0034 [93, 90]. Those data precede the registration, so this is a retrospective tension under baseline ΛCDM, not a likelihood ratio, and the registered comparison with a later dataset is unchanged. A separate ACT DR6 likelihood slice on the measured bandpowers, with the background and absolute calibration held fixed, prefers a free tilt, fitted at 0.9773, by $\Delta\chi^2 = 4.07$; no significance or framework-wide verdict is assigned to it [90]. A charged-lepton world average more than three standard uncertainties from the balanced-circulant center refutes the equal-rank-two tracial reading of the face-circulant sector; a resolved deep-regime galaxy sample violating degree-one radial scale covariance or quadrature composition removes the uniqueness of the deep law; and exclusion of $(w_0, w_a) = (-1, 0)$ fails the fixed-capacity branch. The conditional implications share premises, so they fail in groups: failure of (M4) removes the effective-spacetime and Einstein conclusions together, failure of (M3) removes the gauge and matter interpretations together, and failure of (M2) removes every statement in laboratory units. The finite normal-form, invariant-theory and algebraic results retain their stated domains under all three.

14 Machine verification and reproducibility

The results have different logical types, and Table 2 gives the classes used throughout; each result throughout the paper carries one of them. Each step is identified by its artifact as a Lean theorem, exact executable certificate, conditional paper argument, numerical receipt, or explicit unproved premise.

Each result maps to an artifact class: a Lean library with no admitted propositions in the dependency closure reported here, including formal premise boundaries and countermodels; exact-arithmetic code receipts, integer and rational computations and outward-rounded interval certificates, each emitting a canonical hashed payload; executable observer receipts that replay every retained result and check its artifact hashes; simulation receipt bundles pinning the source revision, configuration, seed where stochastic, tolerances and output hashes; verifiers that recompute verdicts from clause vectors and reject receipts whose stored verdict disagrees; and adversarial negative controls, so that every certificate ships with mutations that must be detected. Machine-readable records classify every quantitative row as a source result, reconstruction implication, diagnostic, prospectively fixed test or rejected candidate, and reject any stronger classification unsupported by the recorded dependencies.

Table 3 names the artifact family behind each group of results. The repository, receipts, schemas and rebuild instructions are public [1, 2, 68]. The formal and executable stack checks the claims assigned to it and enforces their stated boundaries; analytic arguments in this manuscript retain their displayed proofs, and no part of the stack establishes physical realization.

Table 2: Epistemic claim classes used in the manuscript.

Finite theorem	A deductive result proved from displayed premises on a stated finite domain. An exhaustive computation counts as part of the proof only with a specified domain and a soundness argument.
Certified finite computation	An executable witness whose inputs, controls, and output are fixed by a reproducible certificate; execution alone does not promote its interpretation to a theorem.
Conditional analytic theorem	A theorem whose hypotheses include a supplied continuum action, state space or limit; its conclusion holds on that supplied object.
Source limit theorem	A theorem about the limit of a declared family of finite source executions along a stated subsequence, with its normalizations and order of limits displayed; the limit is derived, and its physical identification is a separate claim.
Reconstruction implication	An exact implication whose antecedents include displayed geometric or physical premises.
Unsupplied realization map	A required map from a finite object to a physical event, current, field, clock, scale, or continuum structure for which no construction is asserted.
Diagnostic	A certified computation compared against measured values that it consumes; never counted as a prediction.
Prospectively fixed conditional test	A diagnostic with a custody-bound target definition, kill band, precision floor, and decision rule fixed before comparison.
Frozen prospective branch prediction	A source calculation and decision rule fixed before any eligible comparison, conditional on a named physical branch. Failure rejects that branch; its scope extends to the whole framework only if the branch is proved forced and exclusive.

Table 3: Principal verification artifacts and their scope.

Results	**Principal artifacts**
Executable observer, detectors, interacting process and C/E instruments	The observer receipts with exhaustive path replays, seeded executions, feedback controls, exact and interval certificates, and a single verification entry point that replays every receipt and checks its hashes [68, 66]
Overlapping observers, local clocks, screens and the continuum operator	The model packages for the observer correspondence, the compiled clock history, joint and active clocks, repair identification, federated screens and the coherent screen observer, each with exact or interval receipts, independent oracles and replay; the analytic limit arguments with their constants and order of limits [77, 84]
Public-data comparisons	Hash-pinned public records (Bell trials, optical ratio traces, CMB posterior chains) with exact rational reducers, their stated readout hypotheses and replay checks [89, 90]
Normal forms, stability, refinement, complexity and the transactional diamond	The observable-normal-forms library and component-paper reductions; the local-diamond and Newman arguments of this manuscript; the finite transaction-model verifier
Born weights, Lüders selection, Tsirelson saturation, no-cloning	The event-algebra modules for the interface, conditioning, the bound and the public record algebra, with per-module axiom audits
Rank-three completion	The port-Gram repair band, covariance and frame-quotient modules with a source-pinned receipt and independent verifier
Gauge Lie type, matrix witness, matter image and exterior selection	The screen modules for the carrier action, commutant and incidence response; the exact-arithmetic port current certificate; the trace-balanced kernel and exterior scan modules
Causal precursor, count limit and proper-time readout	The provenance and geometry modules; the source-record family receipts with exact all-pairs counts through $q = 89$ and the sampled $q = 144$ level, the carrier realization receipts and event logs, and the public archive
Null tomography, small-ball arithmetic and tensor completion	The Einstein-branch tensor module for null directions, design, decoder, injectivity and metric ambiguity; explicit null-balance, Ward, Bianchi and scale premises
Angular-rank theorem, frozen branches and the dispersion band	The primitive-port prediction and carrier-class band modules with mutation tests; the custody-bound prediction register
Exact federation and the closure loop	The level-six configuration, seeds, seam arrays, loads, per-schedule ledgers, quotient hashes, standalone verifier and public archive; the level-ten settlement receipts; the event logs and recovered specifications of the canonical and control sources
Dark sector, vacuum coordinate and closure roots	The dark-sector, deep-profile, per-cut and fixed-capacity modules with the matched seen-data diagnostic; the capacity closure and nonidentifiability modules; the interval-certified grain root

15 Discussion and comparison

Operational reconstructions derive quantum structure from information and composition principles [17, 18, 19]. OPH makes a narrower statement at the quantum stage: it takes the finite algebra-state representation as an input and contributes the completed-record surface on which the event, conditioning and correlation identities operate, together with the consensus theorem that defines completion. Quantum Darwinism locates classical objectivity in redundant environmental records [44], and the completed-consensus record plays the same public role with redundancy replaced by

protected overlap agreement and an explicit repair dynamics. The cellular-automaton programme shares the finite substrate [45], while OPH locates the quantum identities on the completed record surface. Wigner's-friend extensions constrain agents reasoning about one another's unfinished measurements [46], whose premises concern pre-consensus perspectives that the completed surface does not represent. Relational quantum mechanics treats physical facts as relational [42] and the thermal-time programme reads time from modular structure [43]; OPH attaches public facts to protected overlap records, and its executable observer reads modular flow from the actual observation state of a running process and lets native incident events drive a modular step whose generator is the native transfer, with a faithful thermal reference derived from closed native histories; passive repair alone implements no nontrivial modular automorphism. The stationary native state of a fixed-number sector is tracial, so its modular flow is trivial, while conditioning on a native current event gives a Palm reference whose coherent modes carry modular frequency $\log(5/4)$, the ratio of the ten moving branches of a non-adjacent occupied pair to the eight of an adjacent one, a degree-census identity of the carrier: modular time in this construction is relative to an event [68]. The local-diamond proof sits beside asynchronous agreement [38, 39], and Dijkstra's self-stabilization separates recovery from continued legitimate operation [40].

The gravitational branch belongs to the thermodynamic and entanglement-based routes to geometry [53, 54, 57], with Bisognano–Wichmann modular flow and half-sided modular inclusions as established analytic ingredients [50, 51, 52]. The route from a null-projected balance and the Bianchi identity to the full equation, with Λ as an integration constant, is familiar from trace-free and unimodular formulations [55, 56]; what is added is the finite nine-direction design with exact constants and the premise contract (G1)–(G6) stated as one implication. The gauge section uses Doplicher–Roberts reconstruction for the sector group [59, 60] and contributes the forcing of the local Lie type from complete response and rechartings inner on that response, a stronger premise than mere internal availability of their implementers.

The closure hypothesis has precedents of a different kind. Wheeler pictured a participatory universe as a self-excited circuit [101]. Tegmark's mathematical-universe hypothesis identifies physical existence with mathematical structure [102], and Nozick proposed that a fundamental principle might subsume itself [103]. OPH adds two demands: the self-subsumption must be inhabited, carried out through records and agents inside the structure, and it must be testable in part, through an executable specification whose observers can be run and whose instruments make predictions. Lawvere's fixed-point theorem shows why unrestricted self-representation fails [104]; the closure condition is restricted to the observable and interventional structure that observers can recover.

A useful comparison of unification proposals asks how much physical structure follows from each declared input and which observations distinguish the resulting models; OPH offers a concrete case with proofs and executable checks attached to individual implications. Formal verification measures deductive reliability, predictive success requires physical realization and eligible measurements, and neither theorem counts nor numerical coincidences define a probability that an architecture describes nature.

16 Conclusion

The goal is a consistent and complete theory: the closure hypothesis realized as one executable process whose observers measure what we measure, with every layer of the tower carried by the same law, state, clocks and refinement, or else proved obstructions that remove OPH or a stated branch of it. Completeness here means a complete operational construction with its services, domains and claimed physical identifications accounted for. It is not a claim that a formal axiom system

decides every sentence of its language. This paper places the evidence on that tower. Layer by layer the finite theorems turn readouts of the executable model into familiar structure: public records independent of the repair schedule, the Born rule and the Tsirelson bound on completed records, a carrier whose slow response has exactly three directions, the Standard Model gauge Lie algebra with a conditional anomaly-free generation, a source-derived $(1+3)$ causal precursor with proper-time ratios from record counts, and the Einstein equation conditional on independent null balance and conservation. The model runs: its observer acts on retained records, one interacting quantum process carries flux, charged matter and a scalar field through complete measurements, overlapping observers read one complete law, and joined carriers on a refining screen carry one continuum operator whose effective description on a declared radial product is a smooth $(1+3)$ Lorentzian field theory.

Specified coherent memories protect native exchange-history sectors, and a complete finite source experiment joins fermionic field operations, particle creation, repair and one calibrated clock. Energy readouts and faithful-reference observations act on that same selected field algebra. The original Weyl/contact action now also has a translation-invariant infinite-lattice ground and positive GNS energy at fixed spacing. Its energy density is bracketed by finite-volume calculations, and the grounded cube's local charged response is completely resolved into 78 positive spectral lines. Exact retarded response shows how matter changes propagation on the unchanged microscopic graph. The finite ground is prepared and its charged response read by a complete native observer experiment. The infinite ground has a quantitatively nonzero connected four-point witness, an ordered Euclidean hierarchy with time-reflection positivity, a nonzero memory kernel and a controlled spatial inverse-response expansion away from real frequencies. The bounded-contact refinement now constructs a positive-energy continuum field net with every Weyl valley and a conservative propagation cone. Its controlled zero-coupling limit is free; persistence of interaction at nonzero limiting coupling remains to be proved. Ground occupation and first-energy-moment constraints distinguish microscopic convergence from macroscopic particle claims. Exact spin selection covers every finite ground preparation, and the complete visible spectrum rejects its direct identification with the observed charged-lepton hierarchy under any common positive unit factor. A native four-bin phase clock now retains every charge branch, clock outcome and first failure while executing coherent histories through model time three. Its resolution does not separate all seventy-eight lines. Regional modular generators on the same invariant ground have an exact central correction and a controlled entropy remainder; their kernels forbid applying the supported first law to arbitrary full-CAR disturbances.

The signed-energy symmetry test also forces an explicit reference choice. At positive limiting coupling, the original canonical fields and clock cannot have the stated CPT vacuum action. Requiring particle-hole invariance fixes the alternative charge clock $H-(U/2)N$. Its new finite ground is unique with hidden spin zero at half filling, a proved charged-sector gap and faithful one-site states. Its thermodynamic and continuum ground constructions restore the averaged addition/removal symmetry. The new occupation-defect bound is quadratic in U/c. These are separate references and spectra, rather than a redefinition of an old finite gap as a physical mass. Uniform spectral estimates now prove that the centered composite force escapes every fixed energy window. Its entire bounded-coupling continuum hierarchy is free, with zero rest mass in all eight Weyl valleys. This rules out that trajectory as the required interacting massive endpoint, while leaving other justified coupling trajectories and actions to be examined. The physical extension must retain these joins while supplying a validated normalization across an interacting chiral $3+1$ field description of native records, a localized defect sector whose maintenance energy is its rest energy, and a responsive geometry realizing the Einstein premises. The controlled massive relativistic calculation currently uses a specified one-dimensional coupling continuation; different continuations through the same

bare point can select different masses. It therefore does not determine the measured spectrum. The negative controls retain their stated carrier and law domains. The five physical maps have not been realized jointly on this continuum construction, so the architecture is not yet established as a description of our universe. The prospective tests are the frozen dispersion band of Theorem 13.4 and Proposition 13.2, under which a resolved intrinsic dispersion with $C_4 \geq 0$, with $B_0/C_4^2 < 10/21$, with B_6/B_0 outside $[-16/135, 16/75]$ or with a nonzero coefficient at angular ranks one through five excludes every member of the declared carrier class at once, and the C/E instrument test, whose decision rule is fixed before any independent count.

Reproducibility statement

All theorems, certificates, simulation configurations, receipts, and the formal library are public in the project repositories [1, 2]. The paper release manifest records the release identifier (r2056) with PDF hashes and sizes; computational receipt bundles pin the applicable source revision, configuration, seed and output hashes, with clean-checkout rebuild instructions and negative controls; and machine-readable claim and prediction provenance accompanies the artifacts [3]. This manuscript synthesizes and strengthens results whose longer proofs also appear in public component preprints on observable normal forms, finite event algebras, consensus, Einstein reconstruction, gauge structure and the executable observer [4, 5, 6, 9, 11, 66], a relationship disclosed here so that overlap is visible.

Statements and Declarations

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Author contributions. Bernhard Mueller developed the OPH concept and mathematical synthesis, wrote the main manuscript, and coordinated the formal and computational evidence. Mario Poneder developed the executable observer construction, including the native observer, the interacting source process and their certificates, and contributed the finite port-current construction and verifier, gauge and matter certificates, and proof review of the edge-entropy and Einstein-normalization branches. Alexander Osika proposed that the screen pixel ratio and the correctable-record capacity are derivable rather than fitted, supplied the detuning fixed-point equation that determines the pixel ratio, and originated the echosahedral patch-hardware design on which the screen-microphysics branch is built, in addition to the physical-realization and hardware boundary, prototype framing, and manuscript review. Kai Xue contributed consistency checks, simulator review, visualization review, and prototype-engineering review. Ben Cassie contributed implementation-claim and physical-evidence review. Peter Nguyen contributed proof auditing, branch-boundary analysis, compact-gauge and global-quotient analysis, and manuscript consistency review. Jinwook Kim developed the constructions, proofs, and counterexamples for the profile partition and structural characterizations, exact behavior-cut recovery, exact finite-state profile transport, adaptive repair stratification, the cumulative attempt-capacity classification, fixed computation-federation progress, and the fixed-federation execution classification, together with the mechanism-variant comparison kernel and conditional example and the source-derived phase-effect selection. He constructed a provably fair emission-order round-robin scheduler with a linear exact-output attempt bound, established order-sharp quadratic accepted-step bounds for the actual compiler with the exponential fanout control, and proved the historical weak-relation fair-stuttering no-go and the weak-fair delay versus bounded-waste horizon separation. He formalized these results in Lean, con-

nected the repair and execution results to the Tower library, and contributed their manuscript, claim registry, ledger and build records. David Matscheko contributed proof review of consensus repair, observable normal forms, modular and Einstein algebra, edge entropy, hypercharge, and the $\mathbb{Z}_6$ quotient. Daniel Ari Friedman contributed independent red-team audits of the papers, the book, the claim and premise ledgers, the Lean library, continuous integration and the custody records, which led to scoped headline claims, displayed values matched to their certified receipts and repaired falsification rules, and the independently reviewed programme-obstruction audit whose controls bound how the carrier, the response triplets, the feedback class and the kinetic term may be attached. Jonathan Hill contributed Lean formalization and proof auditing of observable normal forms, refinement and repair results, complexity classifications, and artifact coverage. William T. Glynn contributed simulator development, computational implementation, reproducible-build infrastructure, and release-validation work in the main OPH research repository. Maarten Antonie Visser contributed consensus-protocol review, physics feedback on the foundations and emergent-spacetime framing, and the source-derived causal-poset reading of authenticated provenance that the event-precedence layer implements. Kale Arnav Anirudha contributed consensus-paper material, asynchronous-agreement analysis, repair-map definitions, and formal review of the protocol surface. Brieuc de La Fournière contributed finite Lean constructions on the source-derived causal-order and carrier-symmetry surfaces, including the cover-relation provenance compiler and the antipodal port-action identification, and external comparison controls for the charged-lepton branch. All authors reviewed and approved the manuscript. All authors agree to be accountable for all aspects of the work.

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Data availability. The finite witnesses, simulation configurations, receipts, scientific claim registries, and release manifests supporting this study are available in the public OPH and simulation repositories [1, 2]. Individual receipts and their bundle documentation provide the source and output provenance for computational claims. The artifact map identifies the level-six exact-federation archive and its level-five to level-ten extension, the closure-loop archive, and the causal-poset archive.

Code availability. The Lean sources, exact-arithmetic verifiers, finite scans, simulation code, tests, and rebuild instructions are available in the same public repositories [1, 2]. The computational claims in this manuscript identify their principal files in Section 14.

Ethics approval and consent to participate. Not applicable. The study involved no human participants, human data, or animals.

Consent for publication. Not applicable.

Competing interests. The authors participate in OPH-related research, software, simulation, or prototype-development programs and may receive professional or reputational benefit from this work. These relationships are disclosed. The authors declare no other competing interests.

Credited prior work

The results of this paper build on the following established work. Each entry names an established result as it is used here and cites its sources.

- Icosahedral harmonics, finite-difference dispersion, and nonminimal Standard-Model Extension coefficient searches [113, 115, 111, 112].
- Root lattices, graph incidence, transverse projectors, free-vector oscillators, icosahedral edge harmonics, finite-difference symbols, Lorentz-violating threshold kinematics, and nonminimal Standard-Model Extension coefficient searches [107, 109, 116, 114, 110, 111, 112].

- Spherical-design moment theory, lattice dispersion expansions, and Lorentz-violation coefficient parametrizations [108, 114, 111, 106].

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